02-07-24

081-06-04

06-04/06

AI

Artificial Intelligence

* Jupiter Notebook
* Python

numpy → for python learning.

pandas.

matplotlib

kaggle → data set

datasets

train   test

→ features
→ Algorithm choose
→ Divide data into ~~trainspeed~~ train and test.
→ Model training.
→ defining confusion ~~pa~~ matrix
→ prediction

081-06-06

Chapter-1:

* what is AI?

↳ AI is the ability for a computer to think, learn, and simulate human mental processes such as reasoning, percieving, and learning.

↳ It can also independently perform complex tasks that once required human input.

↳ AI refers to the development of computer systems that are capable of performing data that would typically require human intelligence.

06-06

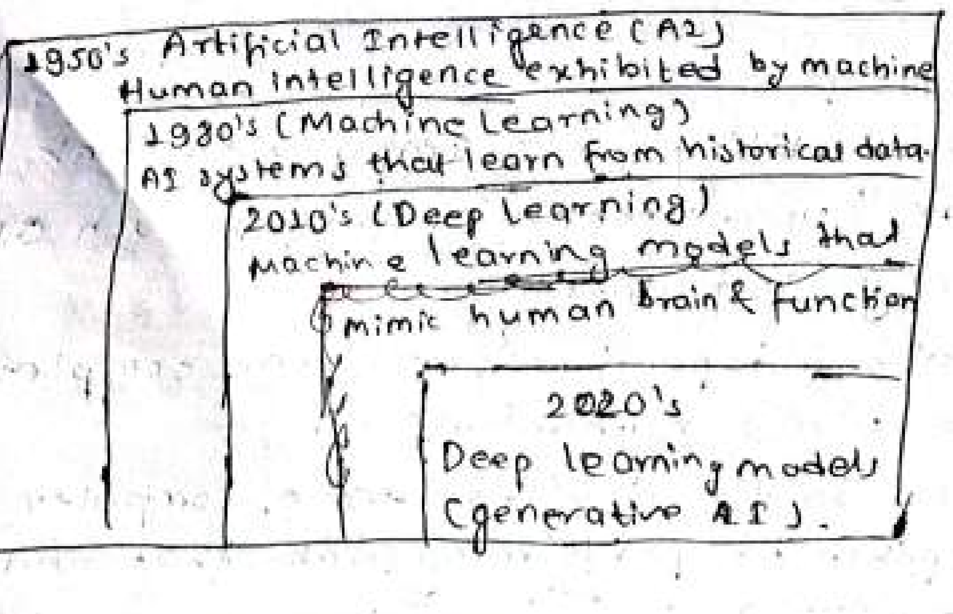
AI

↳ It involves the creation of intelligent machines that can analyze, reason, learn, and make decisions similar to humans.

↳ AI encompasses a wide range of techniques and approaches, including machine learning, natural language processing, computer vision, and robotics.

↳ Applications and devices equipped with AI can see and identify objects. They can understand and respond to human language. They can learn from new information and experience. They can make detailed recommendations to users and experts. They can act independently, replacing the need for human intelligence or interventions (classic example can be a self-driving car).

↳ But in 2024, most AI researchers and practitioners and most AI related headlines are focused on breakthrough in generative AI, a technology that can create original text, images, videos and other contents.



* Types of Intelligence: 06-06

↳ According to the theory of multiple intelligence, AI human intelligence can be categorized into seven types.

1) Linguistic Intelligence:
↳ linguistic intelligence refers to the ability to effectively express one's thoughts in spoken or written language, understand others words or texts, flexibly master the phonology, semantics, and grammar of a language, manage verbal thinking, and convey or decode the linguistic expression through the verbal thinking.

2) Logical mathematical intelligence
↳ It designates the capability to calculate, quantify, reason, summarize, and classify effectively to carry out complicated mathematical operations.

3) Spatial intelligence:
↳ It features the potential to accurately recognize the visual space and things around it and to represent what they perceived visually on painting and graphs.

4) Bodily-kinesthetic intelligence:
↳ It integrates the capacity to express thoughts and emotions and to use hands and other tools to fashion products or manipulate objects.

5) Musical Intelligence:
↳ It is the ability to discern pitch, tone, melody, rhythm, and timbre.

6) Interpersonal Intelligence:
↳ It is the capability to understand and interact effectively with others

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AI

7) Intrapersonal Intelligence:
↳ It is about self-recognition which means the capability to understand oneself and then act accordingly based on such knowledge.

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*Components of Intelligence:

1) Cognition:
↳ The ability to process information, think critically, and solve problems.

2) Memory:
↳ The capacity to store, retain, and recall information.

3) Perception:
↳ Sensory processing to interpret the environment.

4) Language:
↳ Communication through words and symbols.

5) Reasoning:
↳ Logical thinking and deduction.

6) Creativity:
↳ Generating noble ideas and solutions.

7) Emotional Intelligence:
↳ Understanding and managing emotions.

8) Social Intelligence:
↳ Navigating social interactions and relationships.

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AI

9) Adaptability:

↳ Adjusting to new situation and learning from experience,

10) Practical Knowledge:

↳ Applying acquired knowledge effectively in real-life situations.

* Approaches of AI

Imp 1) Acting Humanly : Turing Test Approach

2) Thinking Humanly : Cognitive Modelling Approach

3) Thinking Rationally : "The laws of thought" Approach

4) Acting Rationally : The Rational Agent Approach.

* Acting Humanly : Turing Test Approach: W1

↳ The art of creating machines that perform functions that require intelligence when performed by people.

↳ The study of how to make computer do things at which, at the moment people are better.

↳ The turing test proposed by Alan Turing [1950], was designed to provide a satisfactory operational definition of intelligence.

↳ Imp The computer passes the test if human interrogator after passing some written question cannot tells whether the response were made by a human or not.

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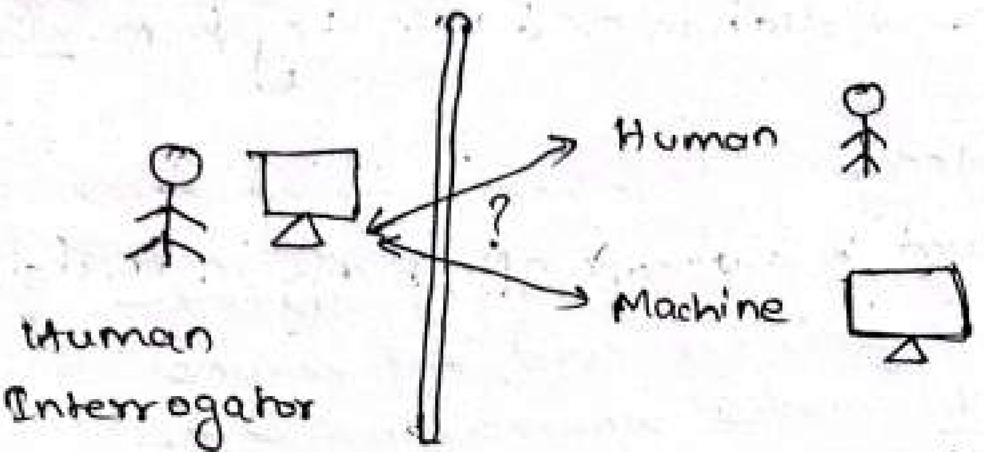
AI



↳ Capabilities need of the computer for the test are:-
1) Natural language Processing (Ability to communicate successfully.)
2) Knowledge Representation (store what it knows or hear)
3) Automated Reasoning (Use the stored information to answer question and draw new conclusions).
4) Machine Learning (Adopt to new circumstances and to detect an extra plot patterns).

and for total turing test,
5) Computer Vision (To percieve objects)
6) Robotics (To manipulate objects and move about).

2) Thinking Humanly : ~~Creative~~ cognitive Modelling Approach

↳ The exciting new effort to make computer think... Machines with minds in the full and literal sense.
↳ The automation of activities that we associate with human thinking, activities such as decision making, problem solving, learning...

↳ It is based on Cognitivie Modelling Approach

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AI

(Cognitive Science).

* Cognitive Science:

↳ Cognitive science ~~meads~~ brings together compute models from AI and experimental techniques from psychology to try to construct precise and testable theories of the workings of human mind.

↳ Needs understanding of how human think.

Eg:- general problem server (GPS) [1961].

3) Thinking Rationally: "The laws of thought" Approach

↳ The study of mental faculties through the use of computational models.

↳ The study of computations that make it possible to percieve, reason, and act.

↳ Based on rational thinking.

1) Irrefeutable reasoning process

2) logistic tradition within AI hopes to build on such programs to create intelligent systems.

4) Acting Rationally: The Rational Agent Approach.

↳ Computational intelligence is the study of the design of intelligent agents.

↳ AI is concerned with intelligent behaviour in artifacts

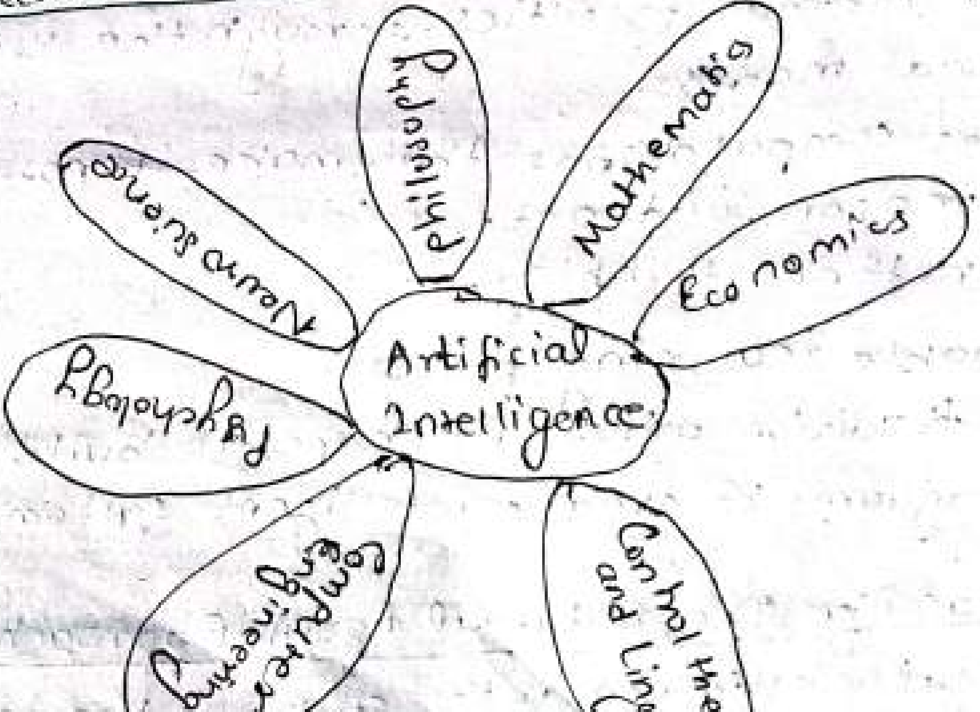
↳ Based on intelligent agents.

→ Agents are the things that act. Computer agent are expected to have other attributes that distinguish them from ~~rear~~ mere program such as operating under autonomous control, perceiving their environment, ~~p~~ persisting over a prolong time period, adopting to change and being capable of taking on anothers goal.

→ Rational agents are those who act so as to achieve the best outcome or the best expected outcome whenever there is uncertainity.

081-07-29

* foundation of AI:



Q.1) What is AI? Explain the importance and application of AI.

07-27
AI

↳ Artificial Intelligences (AI) refers to the simulation of human intelligence in machines that are programmed to think, learn, and perform tasks typically requiring human cognition. These tasks include reasoning, problem-solving, understanding natural language, and visual perception.

The importance of AI can be explained as below:

a) Automation and Efficiency: AI automates repetitive tasks, reducing errors and increasing productivity.

b) Decision Support: AI processes vast amounts of data quickly, aiding decision-making in fields like medicine, finance, and business.

c) Innovation: AI enables the creation of new products and services, such as self-driving cars or virtual assistants.

d) Economic Growth: AI drives industrial transformation and opens new avenues for entrepreneurship.

e) Problem-solving: AI tackles complex challenges, such as climate modelling, drug discovery, and disaster management.

The application of AI can be explained as below:

a) AI powered systems assist in diagnosing diseases, personalizing treatment plans, and accelerating drug development.

b) Used for fraud detection, algorithmic trading, and risk assessment.

c) AI enhances gaming, music recommendation, and content creation.

d) Chatbots and virtual assistants like Siri and Alexa streamline ~~applica~~ interactions.

e) Autonomous vehicles and traffic management systems improve safety and efficiency.

f) Optimizes crop management, pest control, and resource allocation.

2) AI can change the future of the technology. Give your opinion on this statement.

↳ I strongly agree with the statement that AI can change the future of technology. AI is not just a technological advancement; it is a foundational tool shaping how future innovations will unfold. The supporting reasons for it are :-

→ It enhances existing systems by making them smarter and more efficient.

→ AI drives innovation in robotics, space exploration, and augmented reality.

→ It boosts automation and productivity across various industries.

→ AI enables personalized experiences in healthcare, education, and entertainment.

→ ~~&~~ It addresses global challenges like climate change, food security, and disaster management.

However, its integration comes with ethical consideration, including data privacy, bias, and its impacts on jobs. Despite these challenges, AI remains a cornerstone of technological progress, with the capacity to revolutionize industries and improve the quality of life worldwide. Proper development and responsible we will ensure its benefits are maximized for future generations.

AI will undoubtedly shape the future of technology, driving innovation and progress across industries. However, its development and integration must be managed responsibly to ensure its benefits humanity as a whole.

3) Have you ever used any technology with AI? Why do you think it has intelligence? Justify.

↳ Yes, most people, including myself, have used technologies powered by AI.

Examples include virtual assistants like Siri or Alexa, recommendation systems on platforms like Netflix and YouTube, and AI chatbots used for customer service.

These technologies exhibit intelligence because they can learn, analyze, and adapt based on input and data.

The reasons for thinking that they have intelligence are explained below:-

a) Learning and Adaptation: AI systems use data to improve over time. For example, a recommendation system learns your preferences by analyzing our viewing history.

b) Problem-Solving:

07-27
AI

Virtual assistants understand natural language, process queries, and provide relevant answers, demonstrating reasoning skills.

c) Automation of Complex Tasks:

AI can perform tasks like facial recognition or fraud detection, which mimic human cognitive abilities.

d) Decision-Making:

AI systems, such as self-driving cars, make real-time decisions based on their environment, showcasing advanced computational logic.

HW

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AI

1) What is AI? Explain the importance and application of AI.

2) AI can change the future of the technology. Give your opinion on this statement.

3) Have you ever used any technology with AI? Why do you think it has intelligence? Justify.

4) Explain turing test.

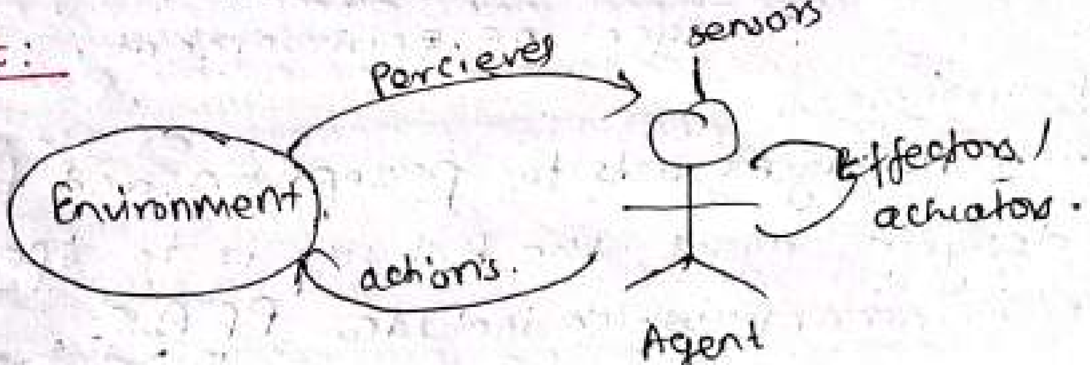
Russel and Peter
⇓
text book of AI

081-07-29

Chapter - 2

Intelligent Agents

*** Agent:**



↳ Agent is anything that percieves from the environment through sensors and acts upon that environment through effectors / actuators.

- Human agent : eyes, ears and other organs for sensers and hands, legs, mouths and other body parts for actuators.
- Robotic agent : might have cameras and infrared range finders for sensors and various motors for actuators.

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AI

- **Software Agent**: recieves keystroke, file contents and network packets as sensory inputs and acts on the environment by displaying on the screen, writing files, and sending network packets.

Agent Terminologies:

1) Percept: the agent's perceptual input at any instant
2) Percept sequence: the history of perceptual i/p complete
3) Agent function: the action performed after perceptual i/p.
4) Agent Program: the code implemented for agent to function
5) Performance measures: to used to measure the success of agent on environment.

* Task Environment
↳ To which AI agent acts for perception and fxn.
↳ While designing agent, the first step is to specify task environment which include PEAS

PEAS → Performance measures, Environment, Actuators, Sensors

Performance measures → criteria for the success in of agents behavior

Environment → that where agent interact

Actuators → required to perform some action

Sensors → perceiving environment and take percept

* Types of Environment:/Properties of task Environment

07-29
AI

1) Fully observable vs. Partially observable

↳ If the agent can see everything in the environment at any given time, it is fully observable.

Eg:- a chess game, where all the pieces are visible on the board.

↳ If the agent can only see part of the environment at any given time, it is partially observable.

Eg:- Autonomous vehicle.

2) Deterministic vs. Stochastic

↳ In a deterministic environment, everything is predictable. This means if we take the same action in the same situation, it will always lead to the exact same result. There is no randomness or surprise.

Eg:- In puzzle game, solving a piece always leads to specific result.

↳ If the environment has some randomness, meaning the same action can sometimes lead to different results.

Eg:- Rolling a dice in a game where the result is unpredictable.

3) Episodic vs. Sequential

↳ If agents experience is divided into independent episodes (where each action doesnot affect future decisions), it is episodic.

Eg:- A customer service chatbot handling one question at a time with no long term memory or past interactions.

↳ If the agent action depend on previous one (past decision affect future one), it is ~~seq~~ sequential.
Eg:- A chess game, where each move depends on earlier move.

4) Static vs. Dynamic

↳ If the environment doesnot change ~~when~~ while the agent is thinking or making decisions, it is static.
Eg:- A board game, where the game state doesnot change, unless the player takes action.

↳ If the environment changes while the agent is acting, it is dynamic.
Eg:- In real-time video game or driving a car, where things like other players or traffic conditions change as the agent acts.

5) Discrete vs. Continuous

↳ If there are limited number of distinct states or actions, the environment is discrete.
Eg:- A turn based games like checkers, where there are only a few move possible.

↳ If there are infinitely many ~~move~~ possible states or actions, it is continuous.
Eg:- A robot navigation in open space, where it can move in any direction at any speed.

6) Single Agent vs. Multiagent: 07-29/30 AI

- If only one agent is acting in the environment, it is single agent environment.

Eg:- A robot navigating in a room.

- If multiple agents are involved, interacting with each other, it is multi agent environment.

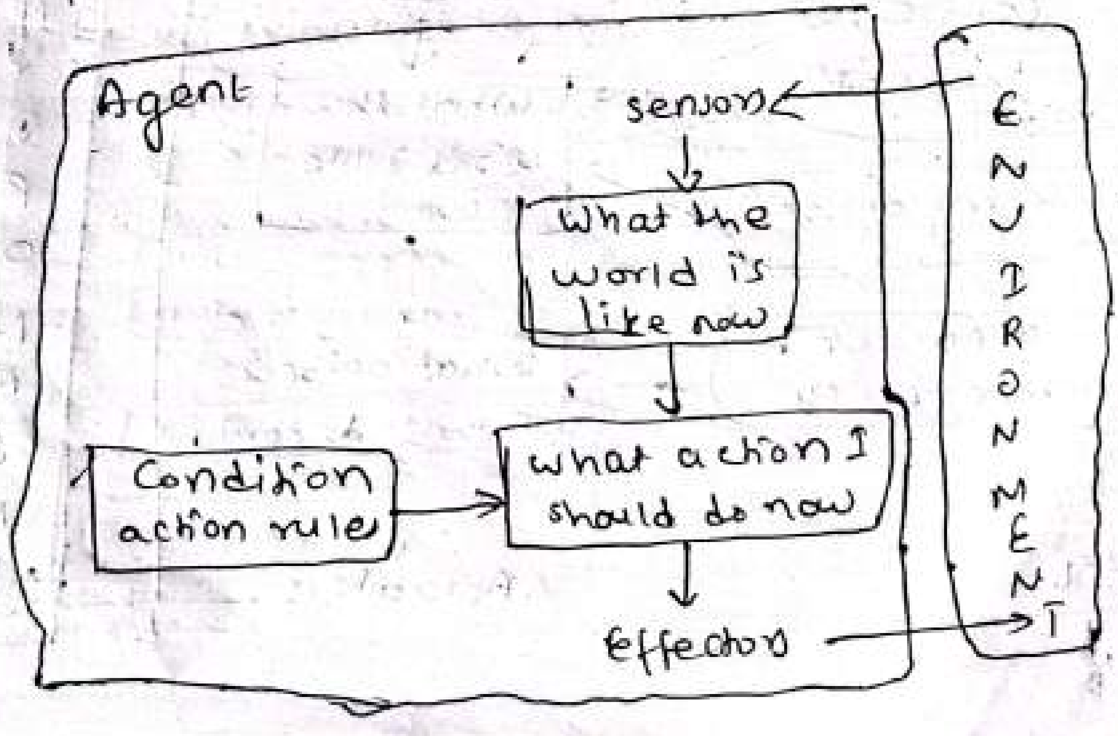
Eg:- In multiplayer game, each player is an agent in the environment.

081-07-30

*Types of Agent: Imp

1) Simple reflex agents
2) Model-based reflex agents.
3) Utility-based agents
4) Goal-based agents
5) Learning agents.

1) Simple Reflex Agent



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AI

↳ Simplest kind of Agent.

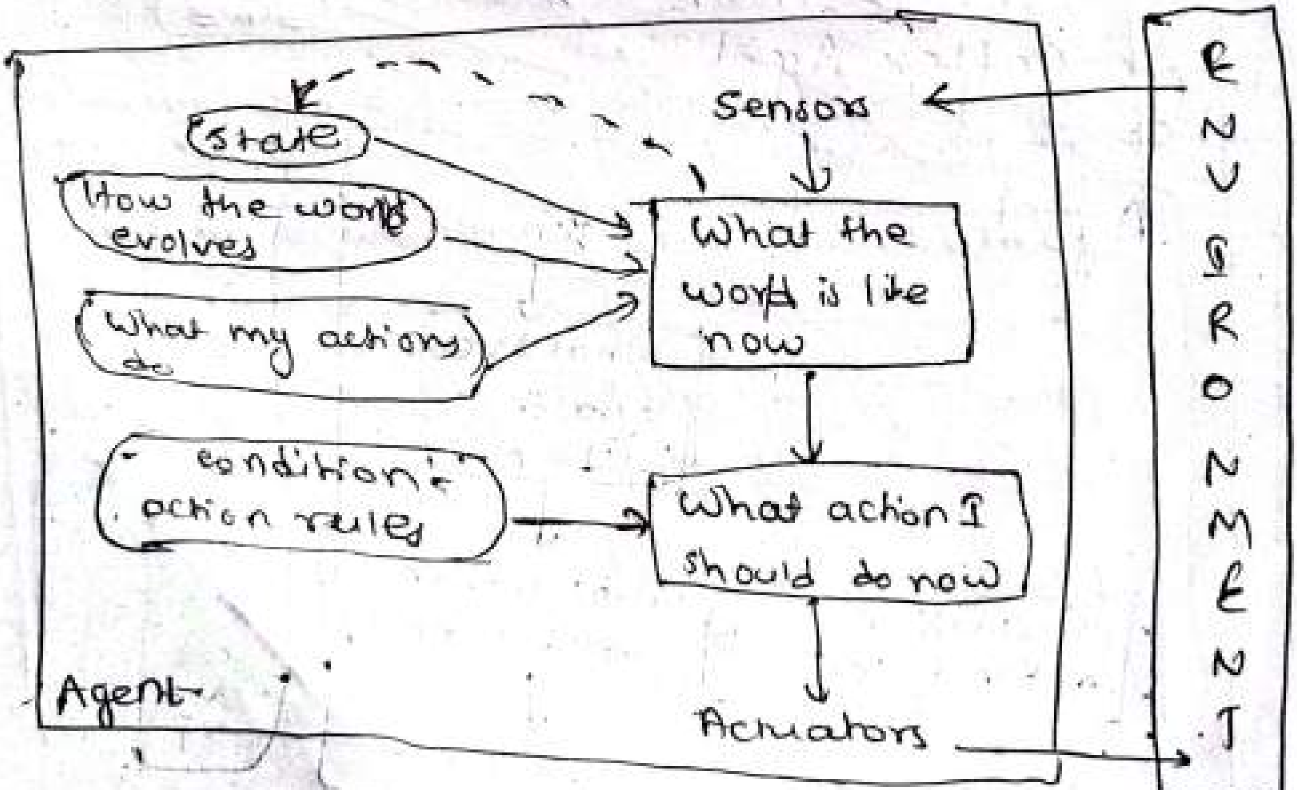
↳ A simple reflex agent reacts to the current situation using predefined rules.
↳ It doesn't think ahead or remember past actions. It only reacts to the current percept.

↳ ** <u>How it works</u> **
→ Based on a condition in the environment, it immediately performs a corresponding action.

↳ ** <u>Example</u> **
→ A robot vacuum that moves forward until it hits an obstacle, then turns and moves in another direction.

↳ ** <u>Rule</u> **
→ If an obstacle is detected, turn left.

2) Model-based Reflex Agent : Imp



↳ A model-based reflex agent remembers past states of the world and uses this memory to make better decisions.

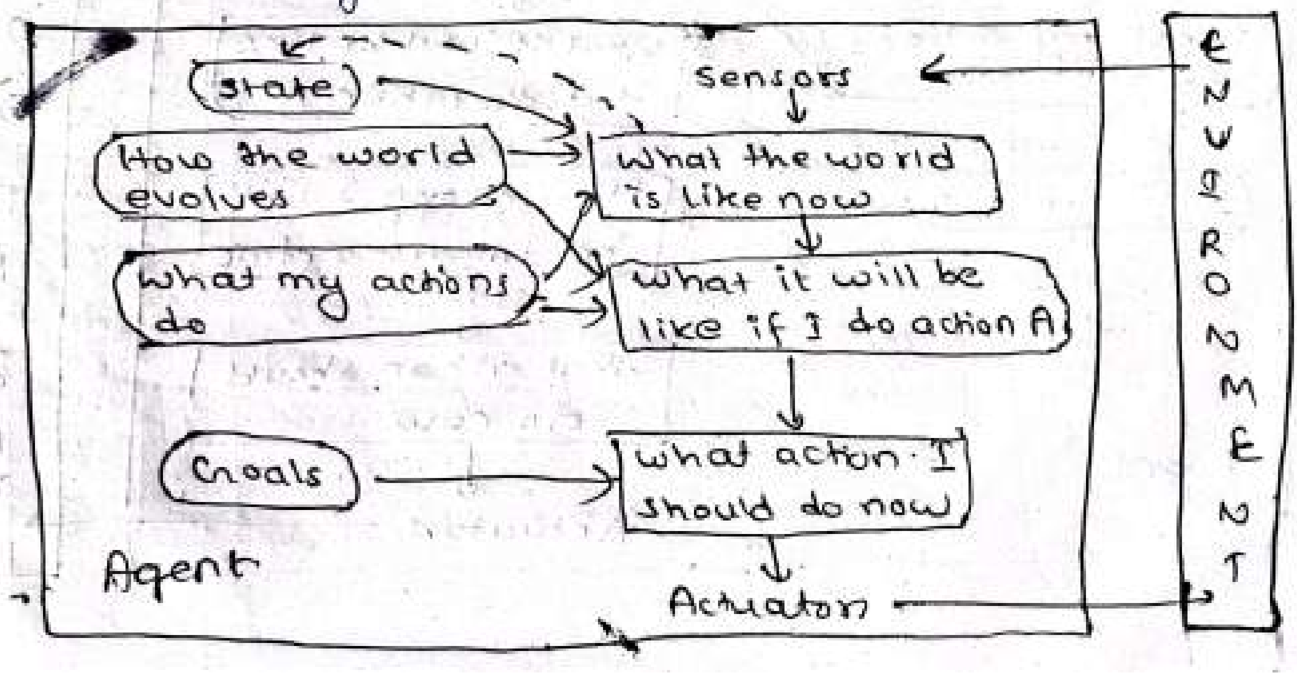
↳ Unlike the simple reflex agent, it doesn't just react to the current situation; it considers how the world has changed.

↳ **How it works**:
→ It keeps track of what has happened before and uses this information to update its model of the world.

↳ **Example**:
→ A thermostat. It doesn't just react to the current tempr but remembers the last tempr and adjusts the heating or cooling accordingly.

↳ **Rule**:
→ If the tempr is too low (and it was also low earlier), turn on the heater.

3) Goal-Based Agent : Imp



↳ A goal-based agent has specific goals it aims to achieve, and it plans actions to reach those goals. 07-30 AS

↳ Unlike reflex agents, it isn't just reacting. It actively thinks about how to achieve a desired outcome.

↳ **How it works**:

→ It evaluates its options and chooses the best sequence of actions to achieve its goal.

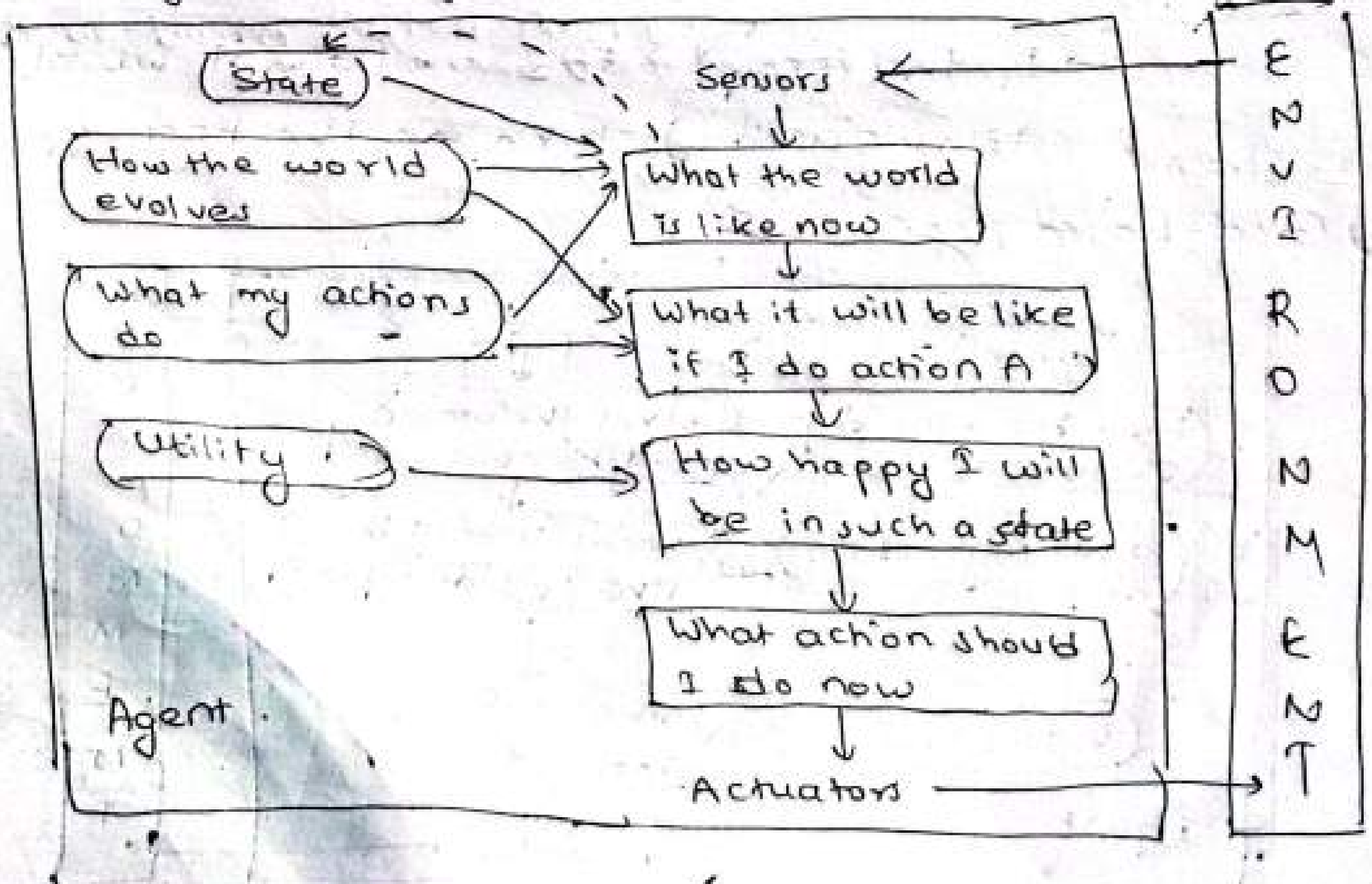
↳ **Example**:

→ A GPS navigation system. The system doesn't just react to roads or traffic; it figures out the best route to a destination.

↳ **Goal**:

→ Find the fastest route from point A to point B.

4) Utility-based Agent:



↳ A utility-based agent chooses actions based on a measure of "utility" or satisfaction.

↳ It doesn't just aim to achieve a goal; it tries to maximize its "happiness" or "satisfaction" from different options.

↳ **How it works**:

→ It evaluates possible actions according to how much utility they provide and selects the one that gives the highest utility.

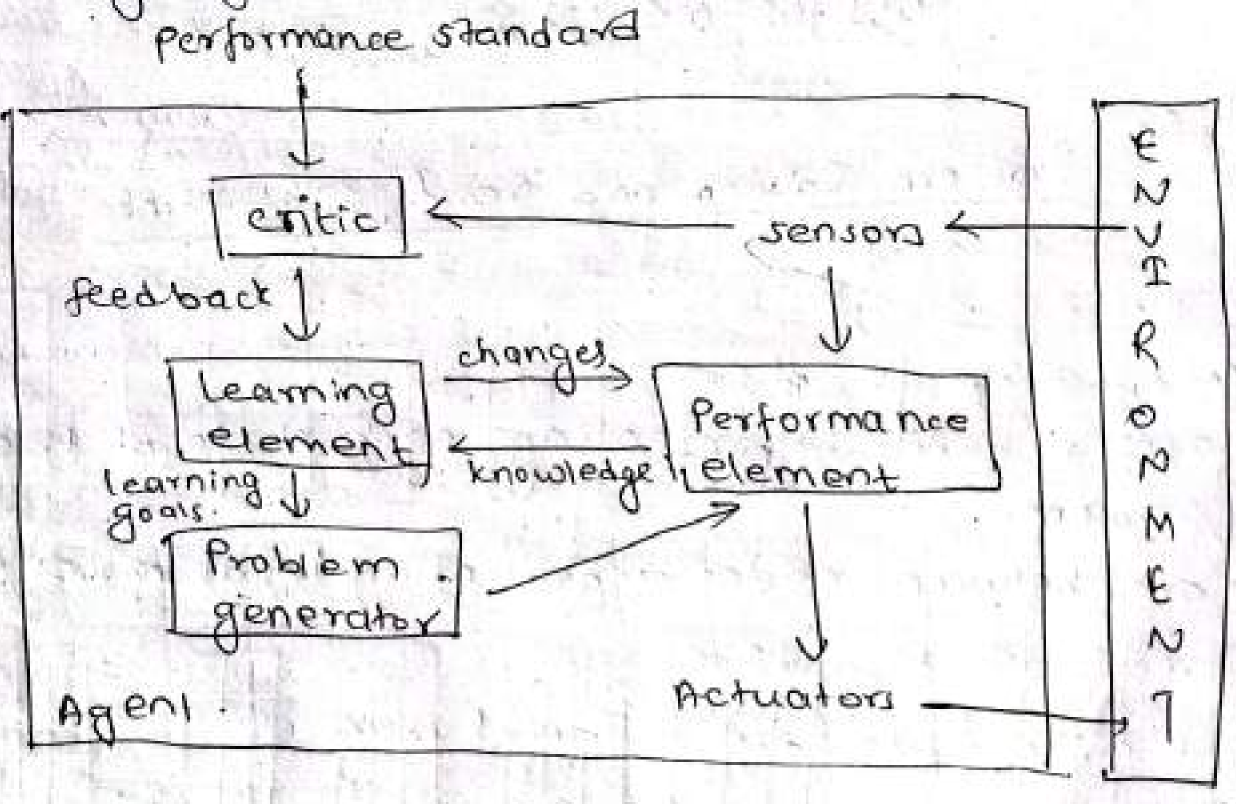
↳ **Example**:

→ A shopping assistant. It helps you choose the best product by evaluating quality, price, reviews, etc., to find the option that maximizes your satisfaction.

↳ **Utility**:

→ Choose the product that gives the most value (quality + price) based on your preferences.

5) Learning Agent vvvI



↳ ↳ A learning agent improves over time, by learning from its experiences. 07-30 AI

↳ It starts with some basic knowledge or rules and then
↳ uses feedback from the environment to improve its actions.

↳ ↳ **How it Works**:

→ It uses past experiences (correct or incorrect actions) to update its knowledge or strategy, improving its future performance.

↳ ↳ **Example**:

→ A game-playing AI that gets better at playing chess the more it plays, learning new strategies from its wins and losses.

↳ ↳ **Learning Process**:

↳ Observe → Learn → Improve → Perform better

4)

081-08-04

Unit-3:

Problem Solving and Search Algorithms

State Space

↳ state space consists of:

→ a set of nodes representing each state of the problem.

→ arcs between nodes representing legal moves from one state to another.

→ an initial state and a goal state.

↳ The state space can take the form of:

→ a tree, or a graph

08-04
AI

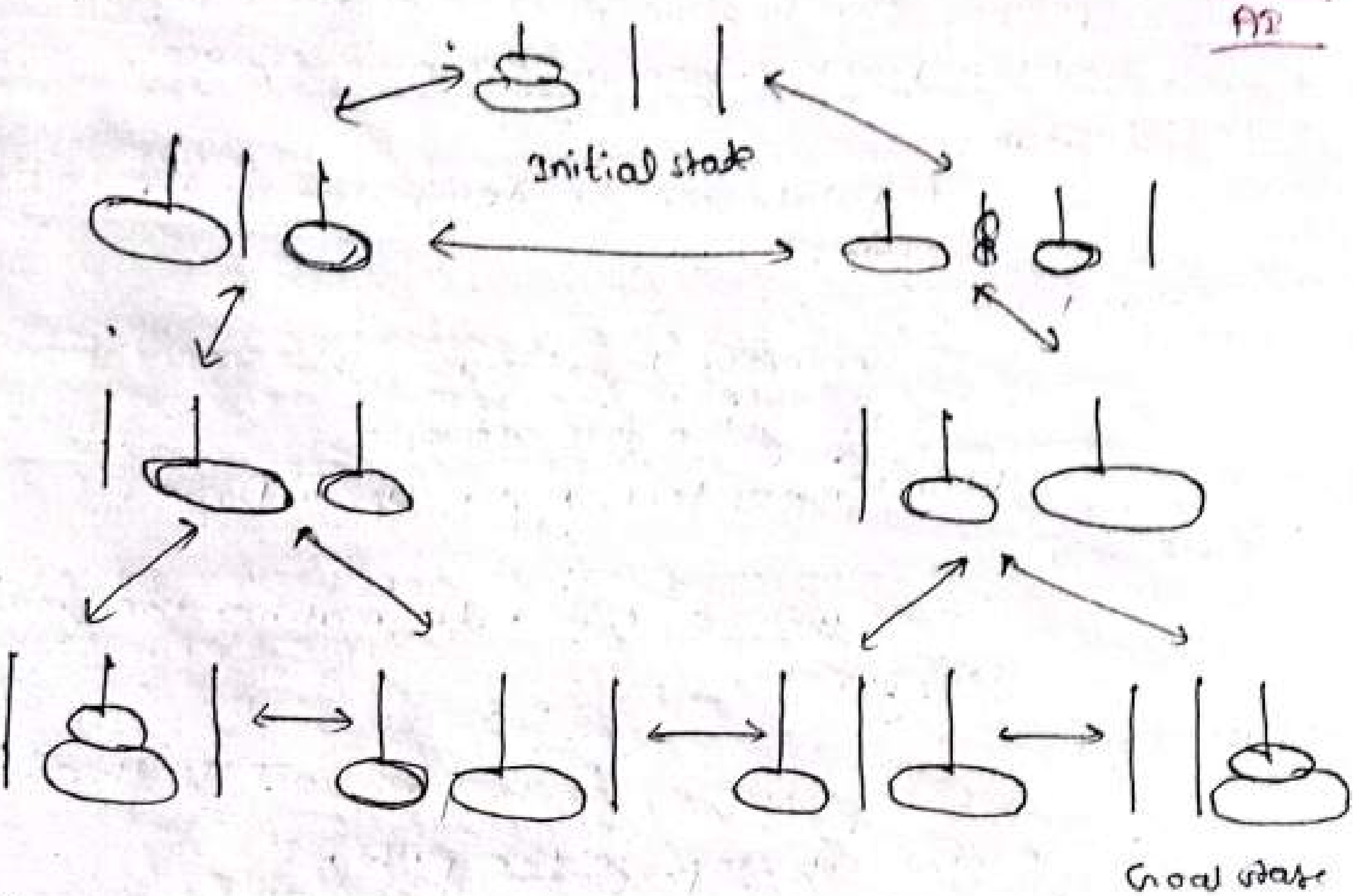


fig:- State space representation of the Tower of Hanoi problem using a graph.

* Problem formulation

Example problem: 8- Puzzle

↳ The 8-puzzle consists of a 3×3 board with eight numbered tiles and a blank space. A tile adjacent to the blank space can slide into the space. The problem definition is as follows:

7	2	4
5		6
8	3	1

Initial state

	1	2
3	4	5
6	7	8

Goal state

fig:- 8-puzzle Example.

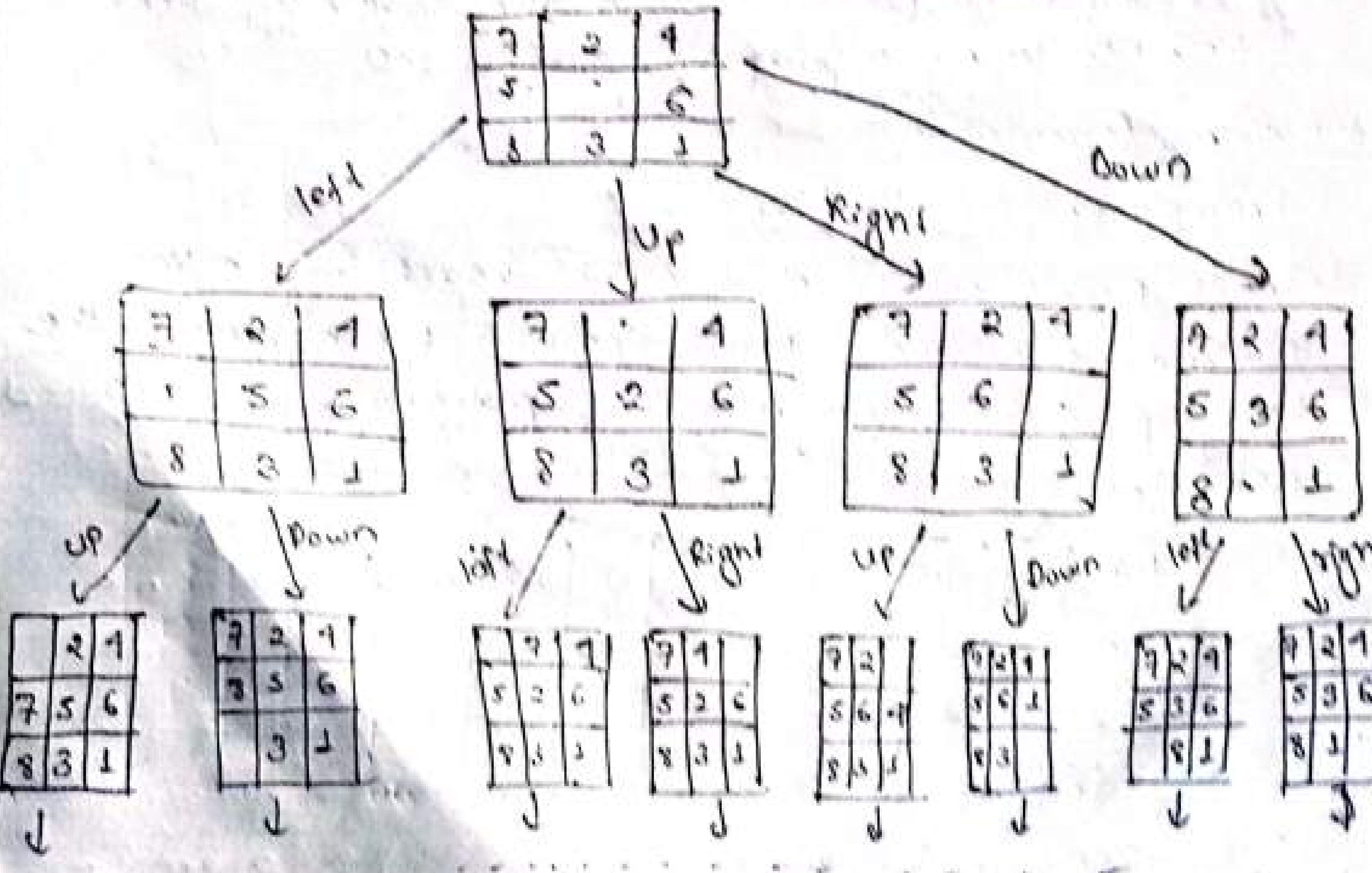
→ States:
Specifies the location of each of the tiles and the blank in one of the nine squares.

→ Initial state:
Any state can be designated as the initial state.

→ Successor function:
Generates legal states that result from trying the four actions:
(Blank moves left, Right, Up, or Down.

→ Goal Test:
Checks whether the state matches the goal configuration.

→ Path cost:
Each step costs 1, so the path cost is the number of steps in the path.



08-04
AI

Q) You are given two jugs, a 4 ltr and a 3 ltr, a pump which has unlimited water which you can use to fill the jug, and the ground on which water may be poured. Neither jug has any measuring markings on it. How can you get exactly 2 ltr of water in 4 ltr jug?

↳ The Water Jug Problem can be formulated as follows:

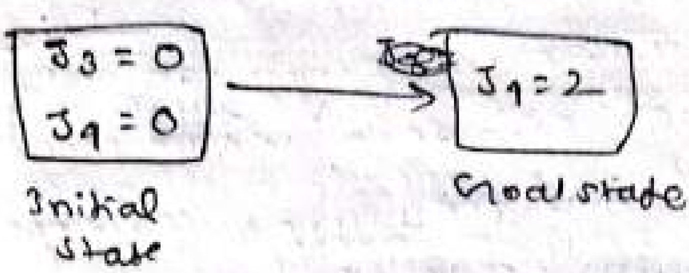
- State:
  Determined by the amount of water in each jug.
- State Representation:
  Two real-valued variables, J_3 and J_4 indicating the amount of water in the two jugs, with the constraints:
  $$0 \le J_3 \le 3, \quad 0 \le J_4 \le 4$$
- Initial state:
  $$J_3 = 0, \ J_4 = 0$$
- Goal state:
  $$J_4 = 2.$$

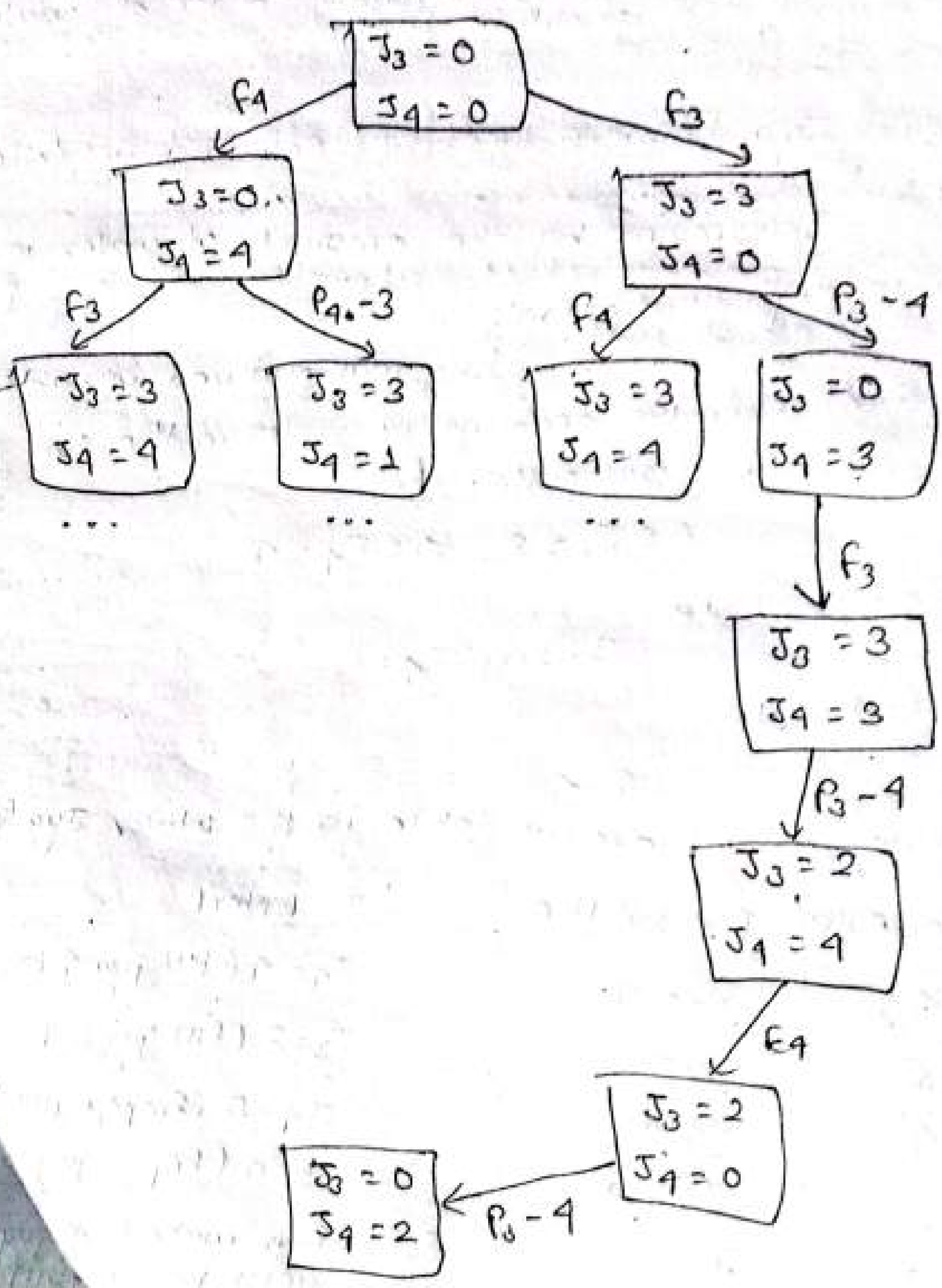
The following actions are available in the Water Jug Problem:

Action/Rule	Precondition	Effect
F4	$J_4 < 4$	$J_4 = 4$ (fill jug 4 to full)
F3	$J_3 < 3$	$J_3 = 3$ (fill jug 3 to full).
E4	$J_4 > 0$	$J_4 = 0$ (Empty jug 4)
E3	$J_3 > 0$	$J_3 = 0$ (Empty jug 3)
P4-3	$J_3 < 3$	$J_3 = 3$ (Pour water from jug 4 into jug 3 until jug 3 is full)
P3-4	$J_4 < 4$	$J_4 = 4$ (Pour water from jug 3 into jug 4 until jug 4 is full)

State Space

08-09
AI





08-04
A2

Q) Farmer, Goat, Wolf, and cabbage Problem

A farmer needs to get a goat, a wolf, and a cabbage across a river. He has a rowboat that can carry himself and, one at a time. The problem condition are:

→ If the wolf and goat are left together, the wolf will eat the goat.

→ If the goat and cabbage are left together, the goat will eat the cabbage.

Formulate this problem as a state-space search problem and find the solution.

↳ The farmer, Goat, Wolf, and Cabbage Problem can be formulated as follows:-

• States:

Determined by the position of the farmer, goat, wolf, and cabbage on each side of the river.

• State Representation:

Four variables F, G, W, and C, representing the positions of the farmer, goat, wolf, and cabbage respectively.

• Constraints:

→ Goat and Cabbage (GC) can not be left together alone.

→ Wolf and goat (WG) cannot be left together alone.

• Initial state:

[West: FGWC, East: -]

• Goal state:

[West: -, East: FGWC]

07-04
AI

The actions that can be performed in the farmer, Goat, Wolf, and Cabbage problem are as follows :-

Action/Rule	Description
F_{w-e}	farmer alone move from west side to east.
F_{e-w}	farmer alone move from east side to west.
FG_{w-e}	farmer and Goat move from west side to east.
FG_{e-w}	farmer and Goat move from east side to west.
FW_{w-e}	farmer & Wolf move from west side to east.
FW_{e-w}	farmer & Wolf move from east side to west.
FC_{w-e}	farmer & Cabbage move from west side to east.
FC_{e-w}	farmer & Cabbage move from east side to west.

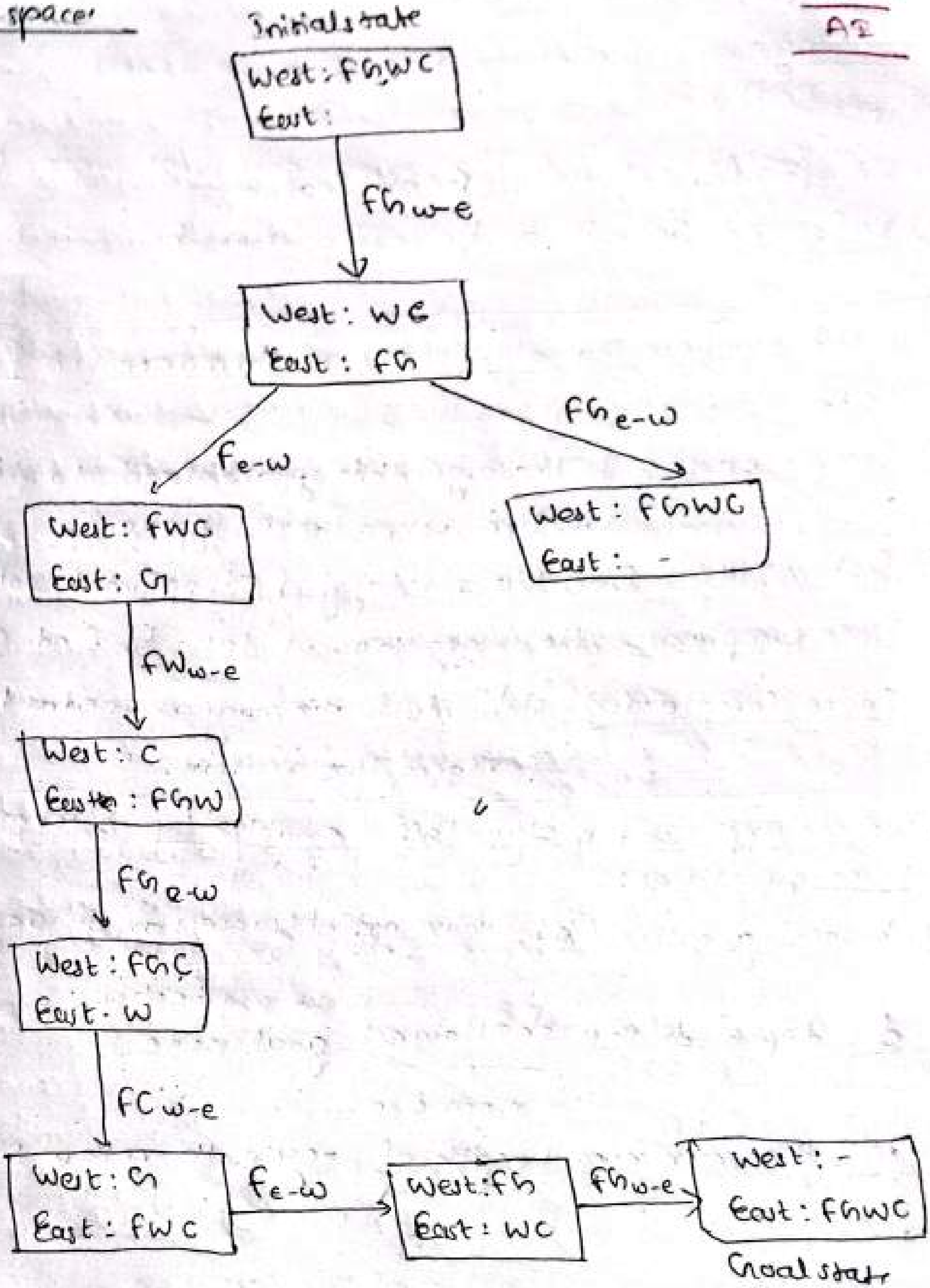
$$\boxed{\begin{array}{l} \text{West : FGWC} \\ \text{East : -} \end{array}} \longrightarrow \boxed{\begin{array}{l} \text{West : -} \\ \text{East : FGWC} \end{array}}$$

Initial state   Goal state.

ctd →

08-04
A2

State space:



* Searching:

Types:

1) Uninformed Search or Blind Search
2) Informed Search or Heuristic Search

↳ We can evaluate the algorithm's performance in four ways:

1) Completeness: Is the algorithm guaranteed to find a solution when there is one?
2) Optimality: Does the strategy find optimal solution?
3) Time Complexity: How long does it take to find a sol^n?
4) Space Complexity: How much memory is needed to perform the search?

↳ We can express the algorithm's complexity in terms of three quantities:

1) branching factor (b), max. no. of $\underbrace{\text{successor}}_{\text{branch.}}$ of any node
2) d, depth of the $\underbrace{\text{shallowest}}_{\text{near one.}}$ goal node
3) m, the maximum length of any path in the state space.

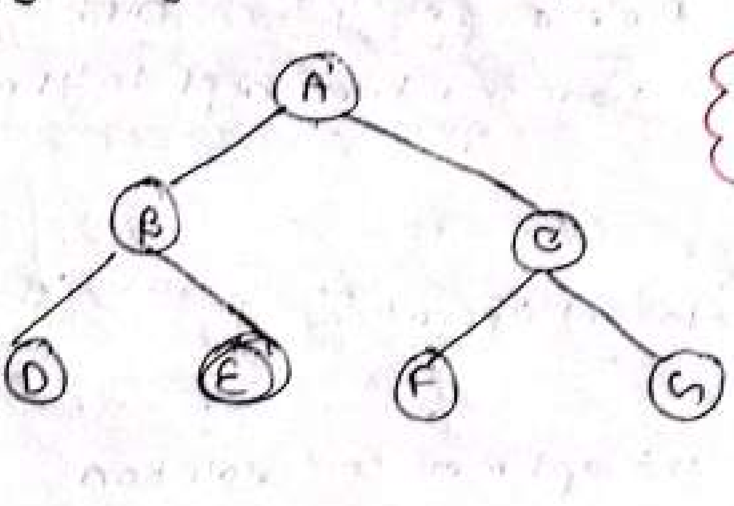
1) Uninformed Search:

08-06
A1

- search provided with problem definition only and no additional information about the state space.
- Various uninformed search techniques / strategies are:
  - a) Breadth-first Search.
  - b) Uniform-cost search
  - c) Depth-first search.
  - d) Depth-limited Search
  - e) Iterative deepening depth-first Search.

a) Breadth-first Search:

- Root node is expanded first.
- Then, all the successors of the root node are expanded.
- Then their successors are expanded and so on.
- Nodes, which are visited first ~~before~~ will be expanded first (FIFO).
- All the nodes of depth 'd' are expanded before expanding any node of depth 'd+1'.



BFS uses Queue Data Structure

Traverse:
A B C D E F G

- **Completeness:** 07-06 AI
  - This search strategy finds the shallowest goal first.
  - Complete, if the shallowest goal is at some finite depth.
- **Optimality**
  - The shallowest goal node is not necessarily the optimal one.
  - Optimal, if the path cost is a non-decreasing function of the path of the node (for example: when all the actions have the same cost).
- **Time Complexity**
  - The time complexity is:

$$O(b^{d+1})$$

where,

b = branching factor

d = level of goal node in the search table.

- **Space Complexity**
  - same as time complexity i.e, $O(b^{d+1})$.
  - since each node has to be kept in the memory.

*** Disadvantages:**

↳ Memory Wastage.

↳ ~~Irrelevant~~ Irrelavant Operations

↳ Time Intensive

↳ Doesn't assure the optimal cost solution.

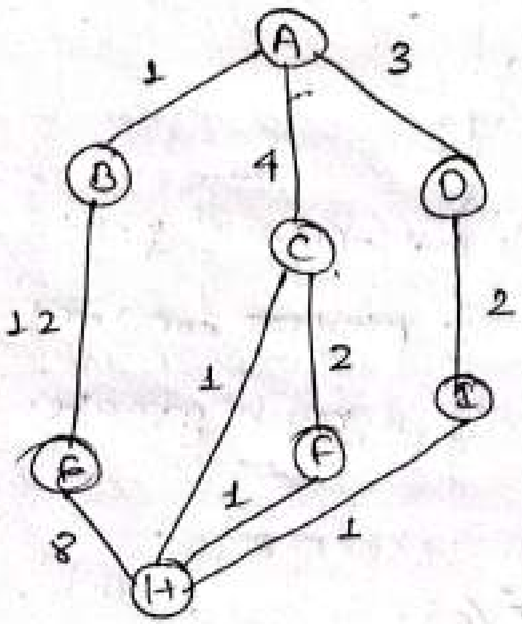
b) Uniform Cost Search

08-06 AI

↳ It expands the lowest cost node on the fringe.

↳ The first solution is guaranteed to be the cheapest one because a cheaper one would have expanded earlier and so would have been found first.

↳ If all step costs are equal, this is identical to BFS.



↳ Required Conditions: A to H

- ABEH = 21, ACH = 5, ACFH = 7, ADIH = 6.

Disadvantages:

↳ Doesn't care about the number of steps a path has but only about their cost.

↳ It might get stuck in an infinite loop if it expands a node that has a zero cost action leading back to same state.

Four Criterial

07-06
AI

- **Completeness**
  - Complete, if the cost of every step is greater than or equal to some small positive constent ϵ.
- **Optimality**
  - The same ensures optimality.
- **Time Complexity**
  - $O(b^{C^*/\epsilon})$

    where,

    $C^* \to$ cost of optimal path.

    $\epsilon \to$ small positive constant
  - This complexity is much greater than that of BFS.
- **Space Complexity**
  - $O(b^{C^*/\epsilon})$

Q Missionaries and Cannibals Problem

Three missionaries and three cannibals must cross a river using a boat which can carry at most two people, under the constraints that, for both banks and boat, if there are missionaries present on the bank (or the boat), they cannot be outnumbered by cannibals (if they were, the cannibals would eat the missionaries). The boat cannot cross the river by itself with no people on board.

- formulate the problem for state space representation
- Construct a complete search tree.

5 Problem formulation

08-07

08-06
A F

States:

Determined by the position of missionaries and cannibals on each side of the river.

State Representation:

Two variables M and C represent the no. of missionaries and cannibals respectively.

Constraints:

→ No. of cannibals should not be more than missionaries

→ Boat cannot cross the river itself.

Initial state:

[West: MMMCCC, East: -]

Goal state:

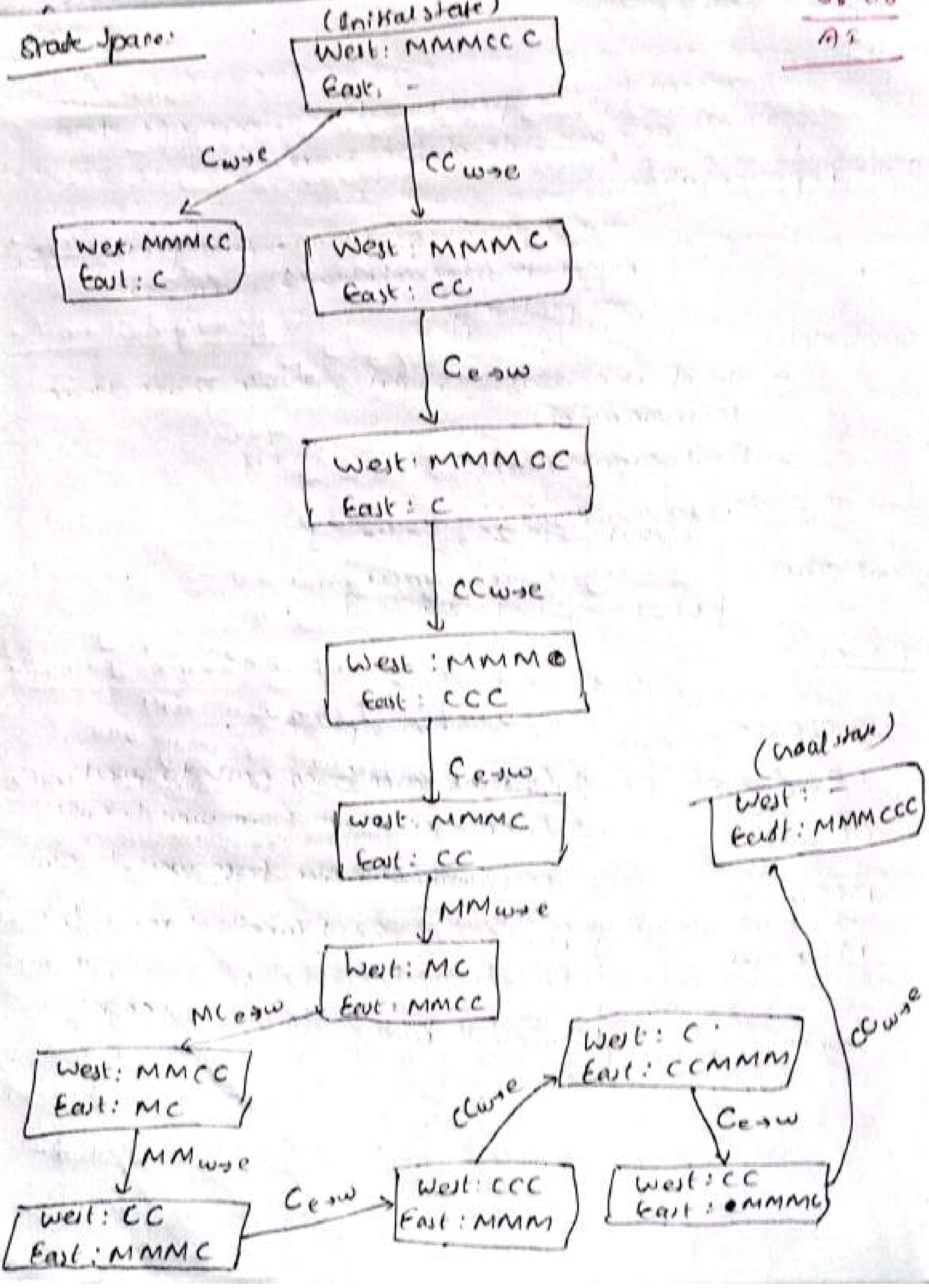
[West: - , East: MMMCCC]

Actions

Action/Rule	Description
$C_{e \to w}$	1 Cannibal move from east to west.
$CC_{w \to e}$	2 Cannibals move from west to east.
$MM_{w \to e}$	2 Missionaries move from west to east
$MC_{e \to w}$	1 Missionary and 1 Cannibal move from east to west.
$C_{w \to e}$	1 cannibal move from west to east.

State space:

08-06
AS



081-08-07

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AI

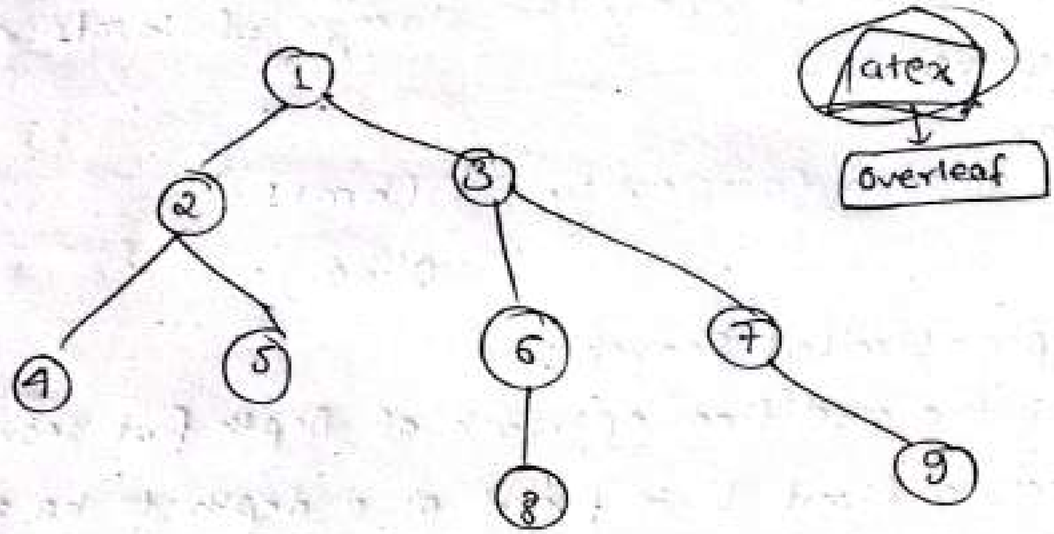
c) Depth-first search

↳ Always expands the deepest node in the current fringe of the search tree.

↳ The search proceeds immediately to the deepest level of the search tree, where the nodes have no successors (dead end).

↳ When a dead end is reached, the search backup to the next shallowest node that still has unexplored successors.

↳ This strategy can be implemented by tree search with a last-in-first-out (LIFO) queue, also known as stack.



[1-2-4-5-3-6-8-7-9] → Traversing

i) Completeness:

↳ Can get stuck going down the wrong path when a different choice would lead to a solution near the root of the search tree.

↳ Not complete.

ii) Optimality:

↳ The strategy might return a solution path that is longer than the optimal solution, if it starts with an unlucky path.

↳ Not optimal.

iii) Time Complexity:

↳ In the worst case all the b^m nodes of the search tree would be generated. Hence,

Time Complexity = $O(b^m)$.

iv) Space Complexity:

↳ For a search tree of branching factor 'b' and maximum tree depth 'm', only the storage of bm+1 node is required.

↳ Hence

$$\text{Space Complexity} = O(bm+1)$$
$$= O(bm)$$

d) Depth-Limited Search:

↳ It is the modified approach of Depth first Search.

↳ Here, a limit 'l' is fixed as a depth to be ~~se~~traversed and going backward after that.

2. Informed (Heuristic) Search strategies:

08-10
A5

Heuristic:

↳ In informed search, a heuristic is a fxn that estimates how close state is to the goal state. for example: man hattan distance, eucligean distance, etc. (Lesser the distance, closer the goal).

↳ Informed search algorithms use additional knowledge (heuristics) to guide the search towards the goal more efficiently.

↳ These algorithms reduce the number of paths explored by using heuristics to estimate the cost to the goal.

↳ Common informed search algorithms include
  i) Greedy Best-first search
  ii) A* Search

a) Greedy Best-First search:

↳ In greedy search, we expand the node closest to the goal node. The closeness is estimated by a heuristic $h(x)$.

↳ Heuristic, 'h' is defined as:

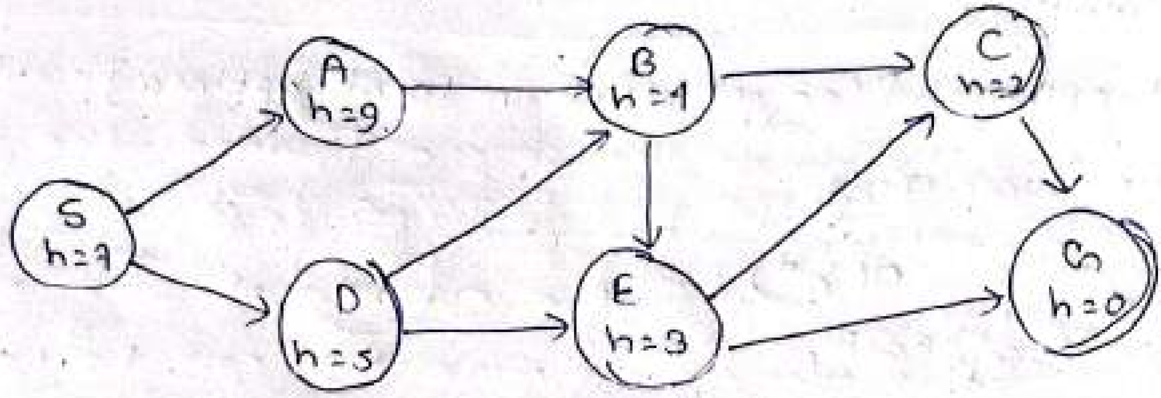
$h(x)$ = estimate of distance of node 'x' from the goal node

{ Lower the value of $h(x)$, closer the node from the goal. }

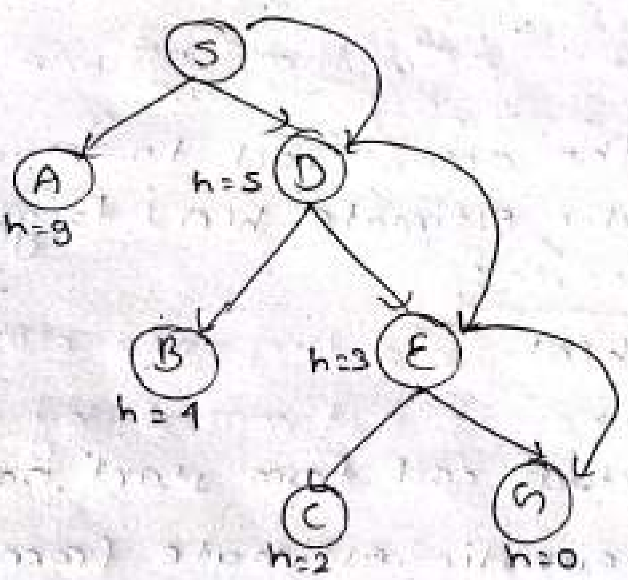
Strategy:

↳ Expand the node closest to the goal state.
(i.e, expand the node with lower 'h' value).

Q. Find the path from S to G using greedy search. The heuristic value, h of each node is given below the name of the node.



↳ soln;



$S \to D \to E \to G$

Advantages:

↳ Works well with informed search problems, with fewer steps to reach a goal.

Disadvantage:

↳ Can turn into unguided DFS in the worst case.

i) Completeness:

↳ Not complete [Can start down with an infinite path and never return to any possibilitgies.]

ii) Optimality:

↳ Not optimal [Can get stuck in local optima].

iii) Time Complexity:

$$O(b^d)$$

iv) Space Complexity:

$$O(b^d)$$

b) A^* Search:

↳ A^* Search combines the actual cost to reach a node $g(n)$ and the heuristic estimate $h(n)$ to guide the search.

$$f(n) = g(n) + h(n)$$

where,

$g(n)$ is the actual cost from start node to node 'n'.

and,

$h(n)$ is the heuristic estimate from 'n' to the goal.

> A^* search is optimal only when for all nodes, the forward cost for $h(x)$ under estimates the actual cost $h^*(x)$ to reach goal. This property of A^* heuristic is called admissibility [$0 \le h(x) \le h^*(x)$].

Q. Find the path to reach from S to G using A* search.

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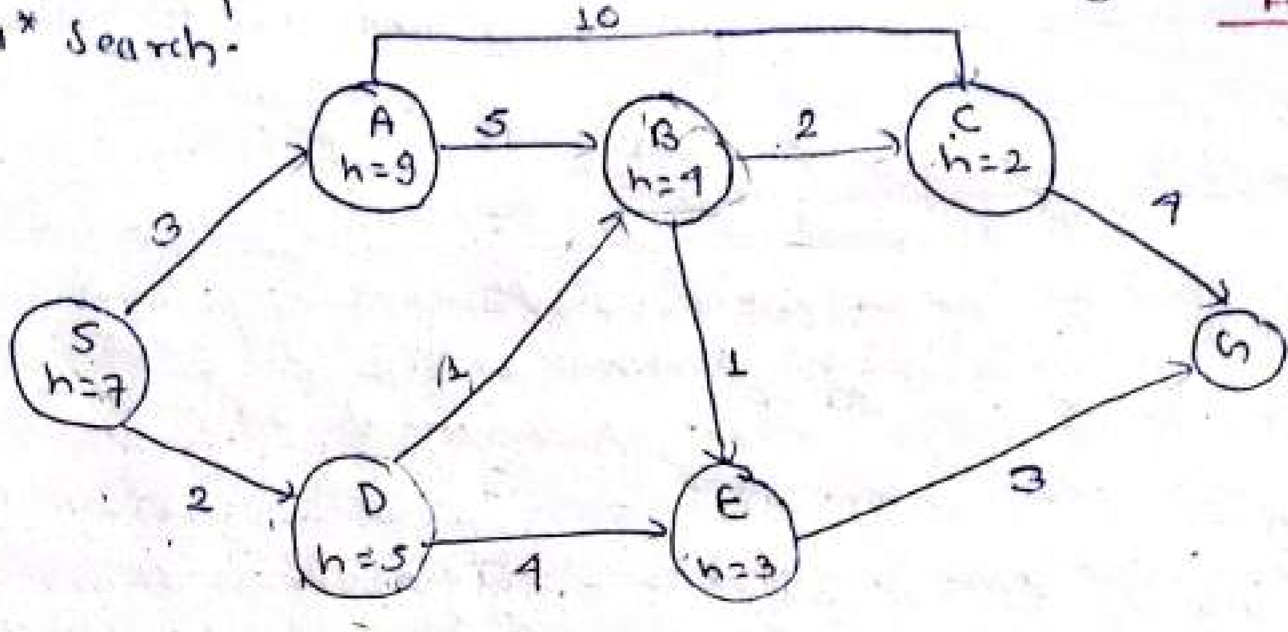
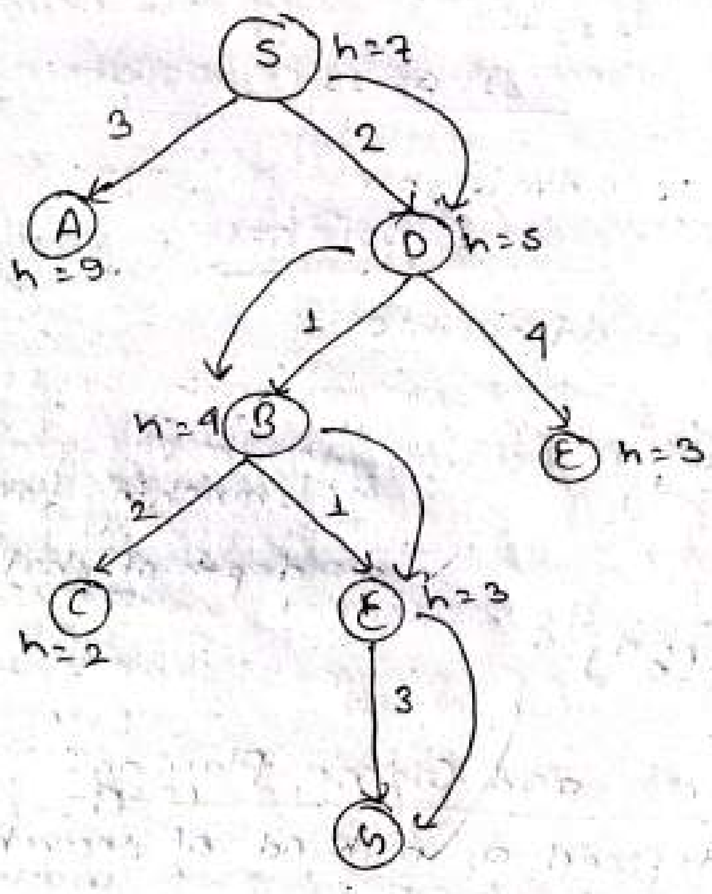
AI



↳



S → D → B → E → G.

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AC

D	Path	h(x)	g(x)	f(x)
↳	S	7	0	7
i)	S→A	9	3	12
↓	S→D	5	2	7
	S→D→B	4	2+1=3	7
ii	S→D→E	3	2+4 = 6	9
↓	S→D→B→C	2	3+2=5	7
iii	S→D→B→E	3	3+1=4	7
	S→D→B→C→G	0	5+4=9	9
iv	S→D→B→E→G	0	4+3 = 7	7

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b) i) Completeness:
↳ Complete if h(n) is admissible.

ii) Optimality:
↳ optimal if h(n) is admissible.

iii) Space Complexity:
$O(b^d)$

b = branching factor

iv) Time Complexity:

d = depth of shallowest goal

$O(b^d)$

*Adversarial Search and Game Playing:

↳ A game can be defined as a kind of search problem (game tree) with the following components.

i) Initial state : Identifying the initial position in the game and the identification of the first player.

Adversarial search is an Artificial Intelligence technique used in competitive environments where two or more agents have conflicting goals

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AI

ii) Actions:

Returns the set of legal moves in a state.

iii) Successor fxn:

Returning a list of move (move, state pairs)

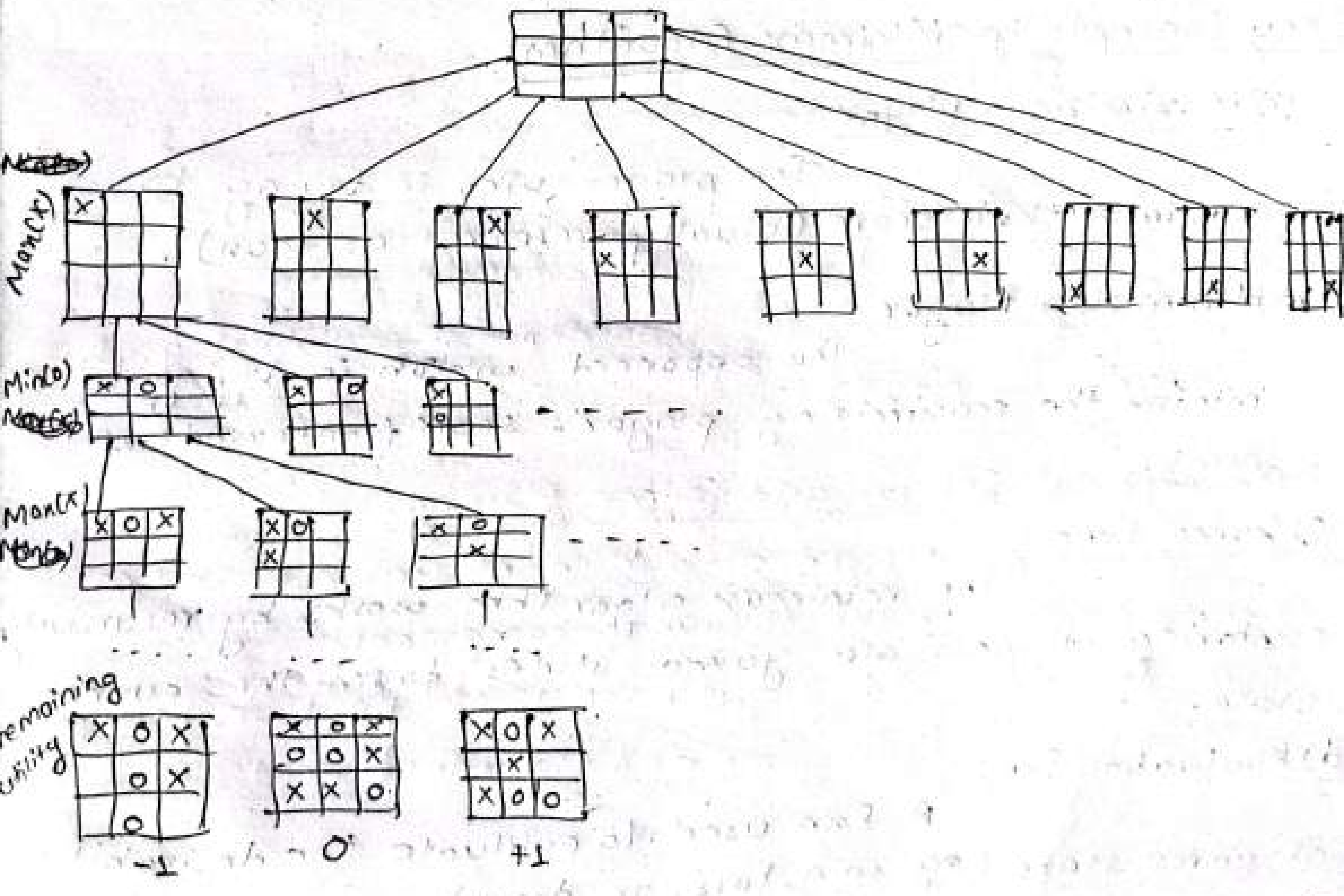
iv) Terminal test:

Which is true, if the game is over and false otherwise. States where the game has ended are called terminal states.

v) Utility fxn:

Which gives a numeric value for the terminal states. Example: In tic tac toe lose, draw, win with -1, 0, +1.

Game tree example for tic tac toe:



↳ There are two adversarial search algorithms.

08-11
AI

1) Minimax algorithm
2) Alpha-beta pruning

1) Minimax Algorithm:

↳ The minimax is classic decision making algorithm used in adversarial search for two player games like tic tac toe, chess, checkers, etc. In these games, two players alternative making moves with the goal of maximizing their own advantage while minimizing their opponent's advantage. The minimax algorithm is used to choose the best move for the current player assuming both players are playing optimally.

Key Concepts in Minimax Algorithm:

a) Maximizing player

The player who is trying to maximize their score (usually referred as max).

b) Minimizing Player

The opponent who is trying to minimize the maximizing player's score (referred as min).

c) Game tree:

The minimax algorithm works by recursively exploring all possible game states from the current state.

d) Evaluation fxn:

A fxn used to evaluate the desirability of game state (eg: win, loss, or draw).

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* Steps of Algorithm:

1) Maximizing player's turn :
The maximizing player tries to choose the move that maximizes the score

2) Minimizing player's turn :
The minimizing player tries to choose the move that minimizes the score.

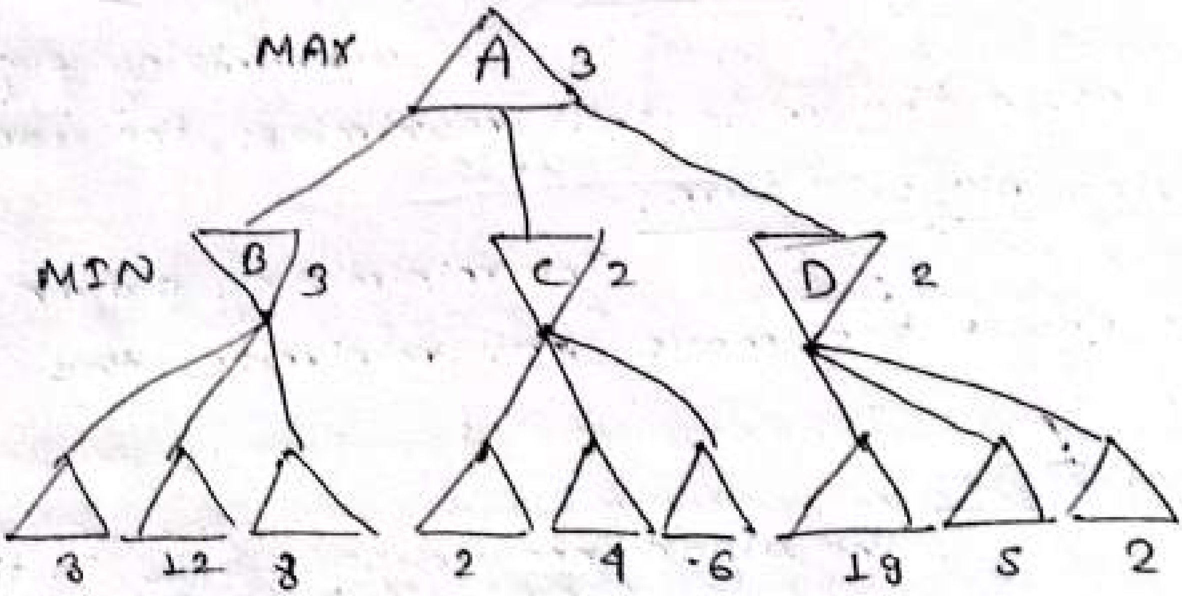
3) Terminal State:
The algorithm assign scores to the terminal state based on the outcome.
[Eg:- In tac tac toe, ~~d~~ loss, ~~win~~ draw, win with -1, 0, 1]

4) Recursion :
The algorithm explores all possible future states (moves) recursively, alternating between maximizing and minimizing until it reaches the terminal states.

5) Back propagation:
The algorithm evaluates the game tree starting from the terminal nodes propagating the values back up to the root, where the current best move is choosen.

Gd
→

Eg.-



↳ In the figure, the algorithm first recurses down to the 3 bottom leaf nodes and uses utility fxn to discover that their values are 3, 12, 8.

↳ Then, it takes minimum of these values, 3 and return it is as backed up value of node B.

↳ A similar process gives backed up value ~~of~~, 2 for C and 2 for D.

↳ Finally, we take maximum of 3, 2, 2 to get backed up value of 3 for the root node A.

2) Alpha-beta Pruning:

↳ The main disadvantage of minimax algorithm is that all the nodes in the game tree cut off to a certain depth are examined.

Four Criteria for minimax algorithm:

~~Comp~~ ↳ It performs the depth first exploration of a game tree in a complete way. So, it is both complete & optimal.

↳ Time Complexity:

$O(b^m)$.

· Space Complexity:

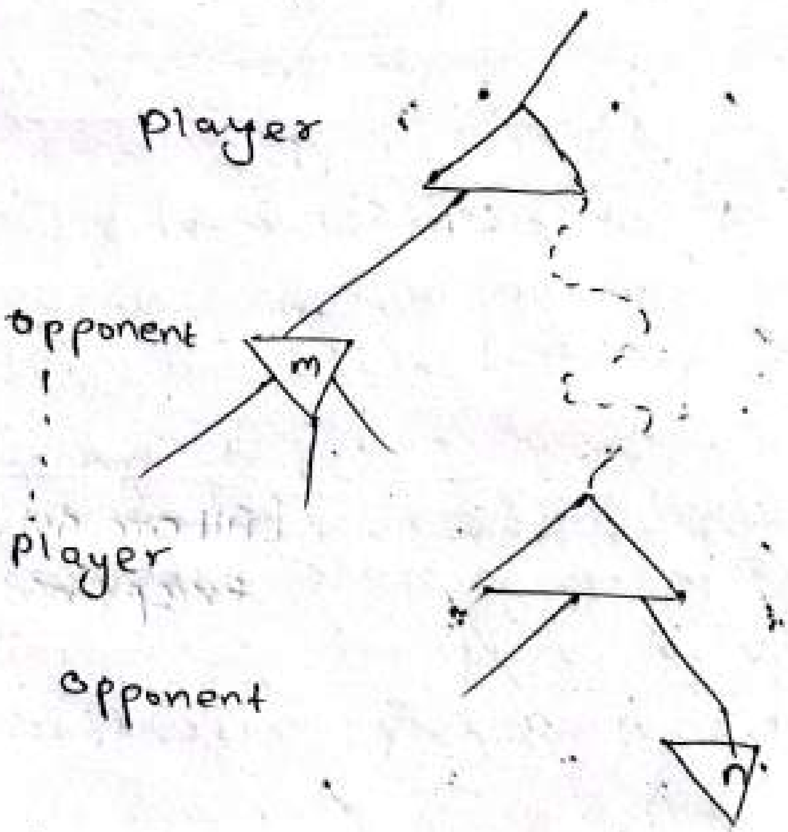
$O(bm)$.

{ b = legal moves at each step
m = maximum depth of the tree.

2) ~~Ad~~

↳ Alpha-beta pruning helps reduce the number of nodes explored.

↳ When applied to standard minimax tree, alpha-beta pruning returns the same move as minimax would but prunes away the branches which could not possibly influence the final decision.



↳ In the figure above if m is better than n for player, we will never get n in play.

↳ Alpha-beta pruning gets its name from the following two terms

i) α = the value of best (i.e. highest value choice), we have found so far at any choice point along the path of max.

ii) β = the value of the best (i.e. lowest value choice) that we have found so far at any choice point along the path of min. 08-11/13 A.I

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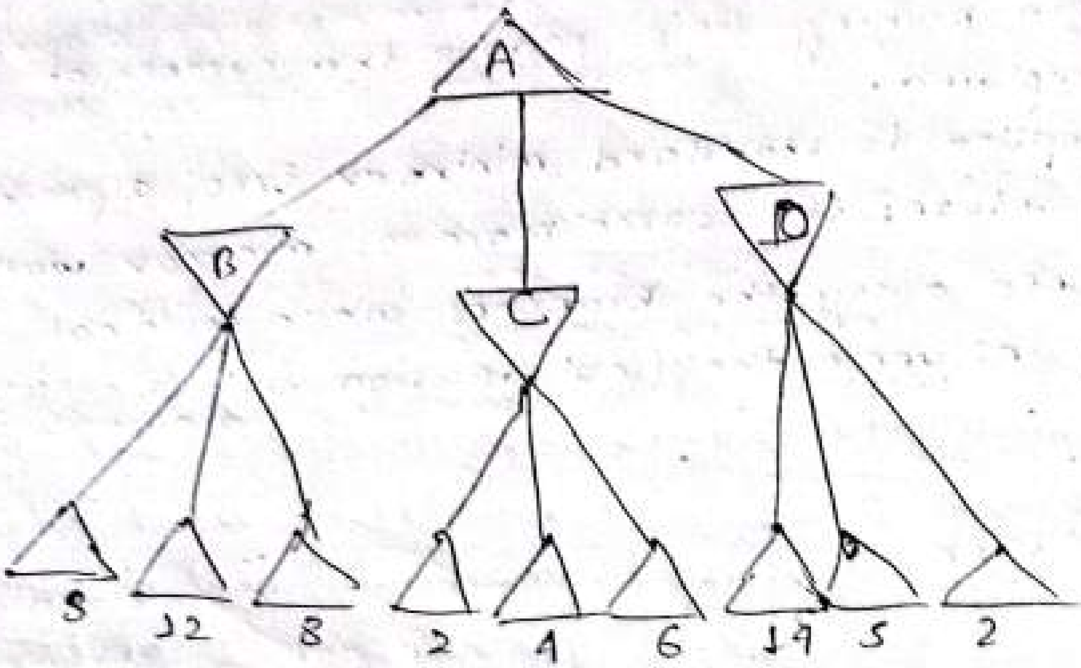
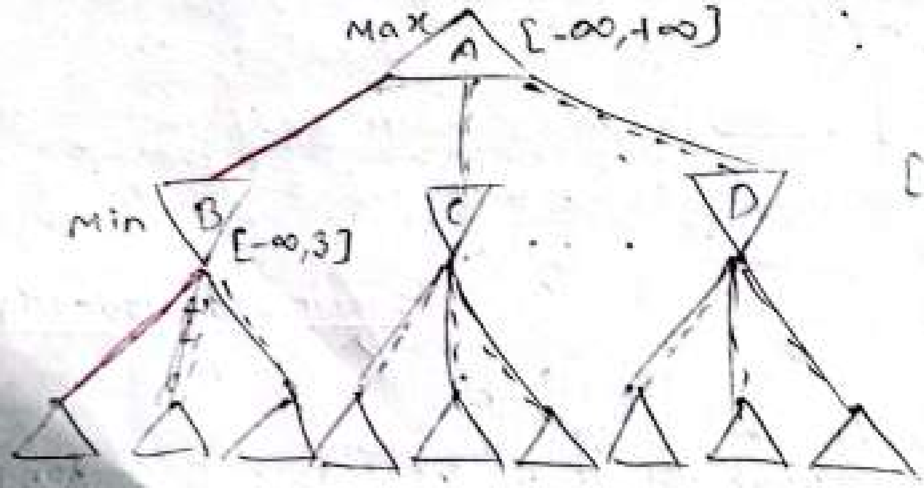
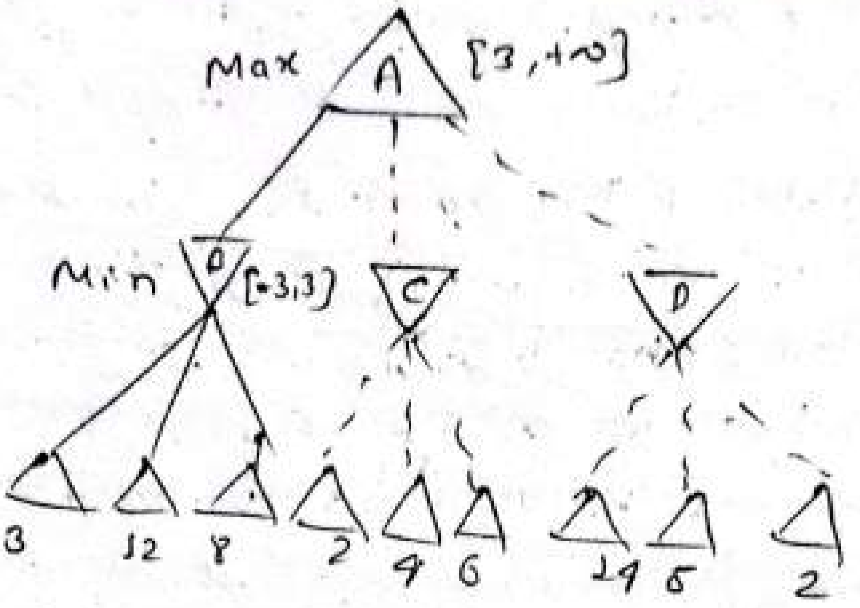
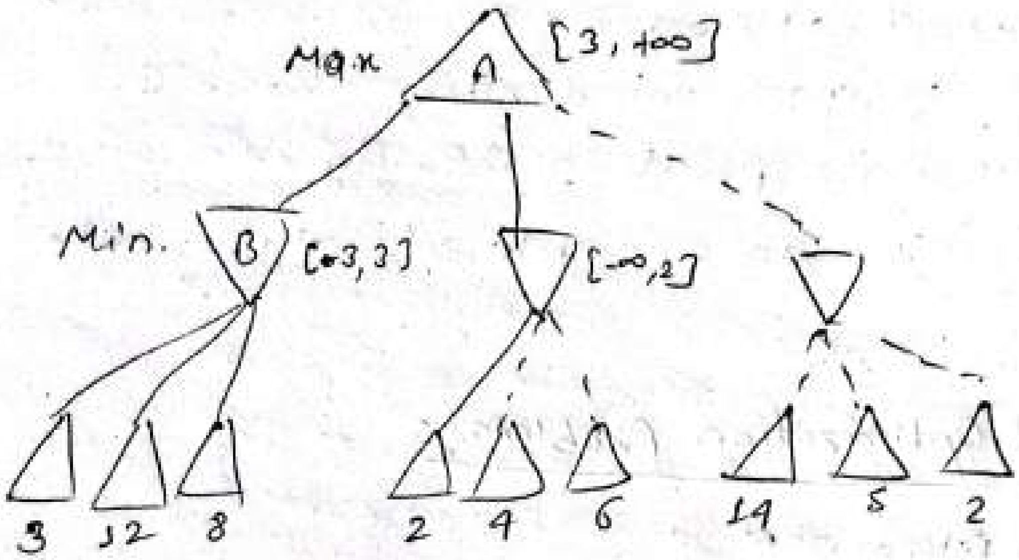


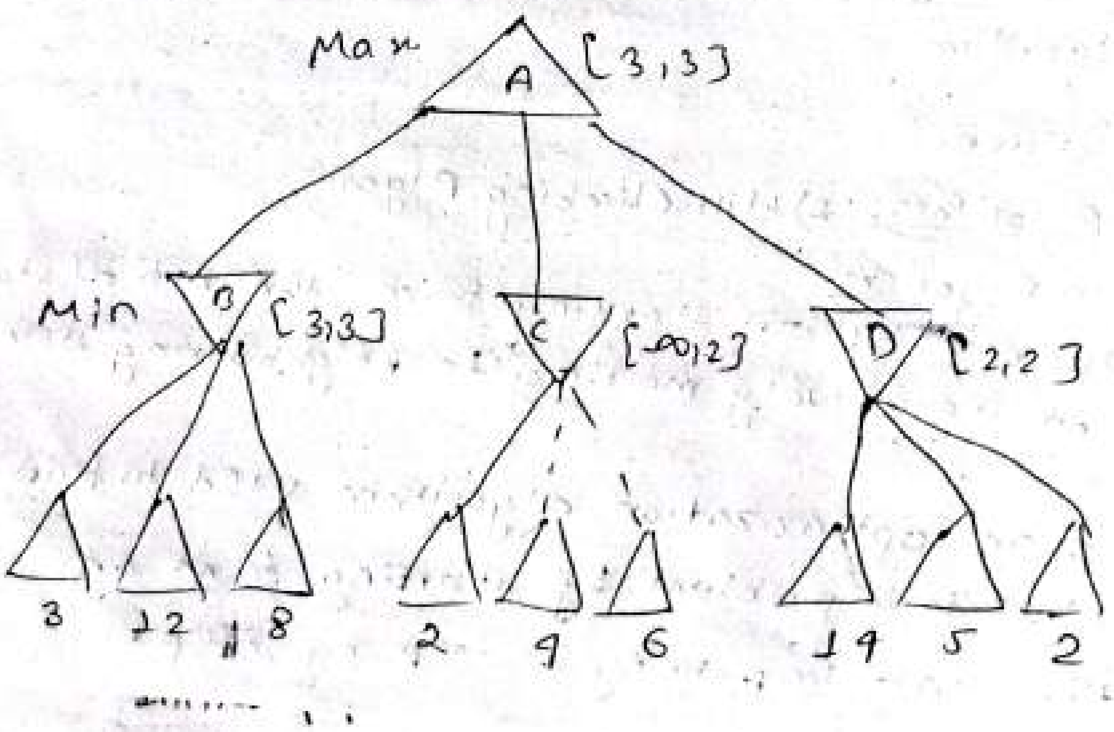


[All are dot line except red]



[two steps are skipped for traversing 12 & 8 of B.]





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08-13 A1

↳ The first leaf below B has the value 3. Hence, B which is min node has a value of atmost 3.

↳ The second leaf below B has value 12, min would avoid this move so the value of B is still atmost 3.

↳ The third leaf below B has a value of 8, we have seen all these successor's states so the value of B is exactly 3. Now, we can save that the value of the root node is at least 3.

↳ The first leaf below C has value 2. So, C which is a min node has value of at most 2. But we know that B ~~is~~ worth 3 so max would never choose C. Therefore, there is no point in looking other successors states of C. This is an example of alpha-beta pruning.

Local Search and Optimization Problems:

1) Hill climbing Algorithm
2) Simulated Annealing
3) Genetic Algorithm
4) Gradient Descent.

~~1. Hill Climbing Algorithm:~~ 1) Hill Climbing Algori

~~Local~~ Local search Algorithm

↳ Operates by ~~init~~ starting from the initial states and iteratively moving to neighbouring state

↳ Hill climbing is an optimazation algorithm used to find the bst solution to a problem by starting from an initial solution and improving it step by step.

↳ It works by always moving to the neighbouring solution that is better.

How it works?

↳ Start with a random solution.

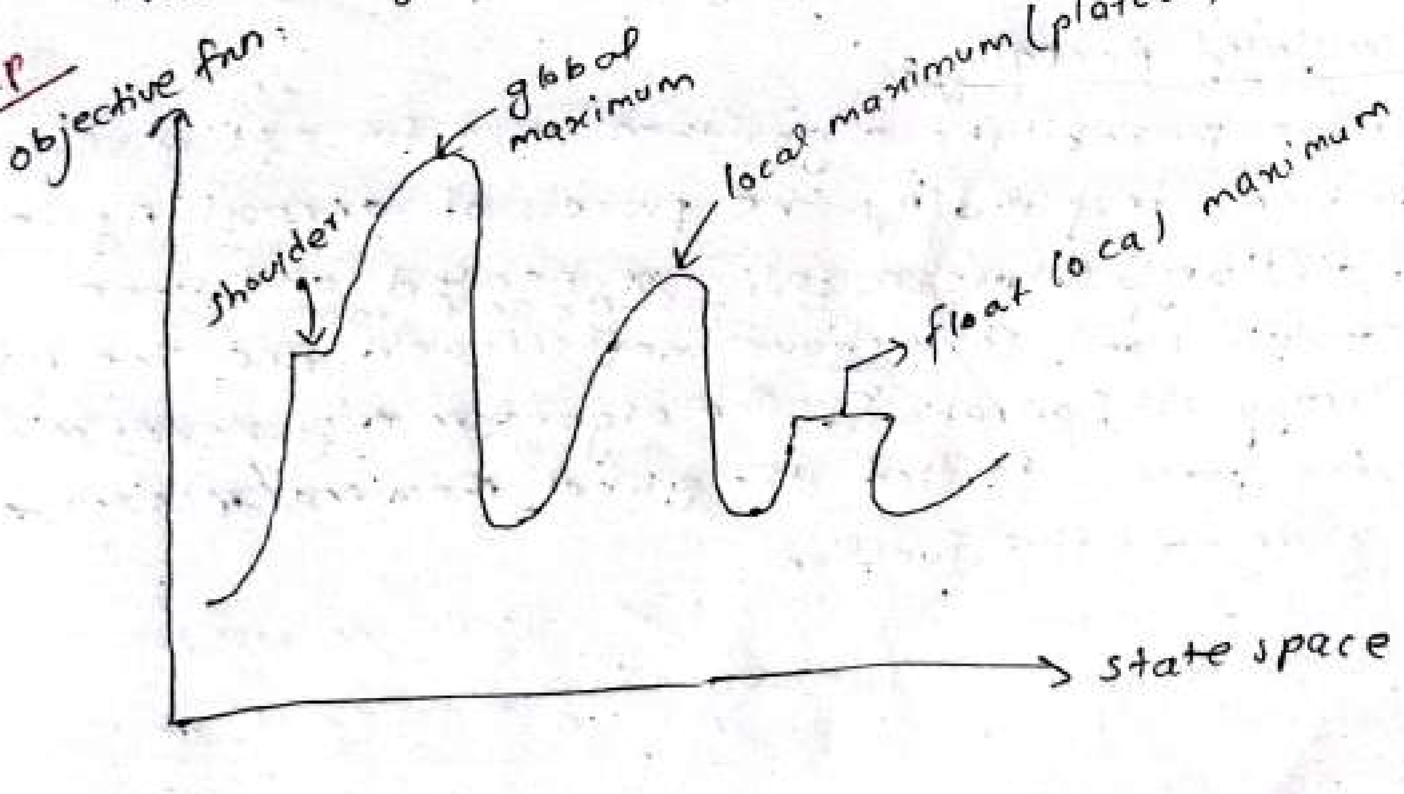
↳ Check neighbouring solution: look at the solution close to the current one.

↳ Move to the best neighbour: If neighbouring solⁿ is better, move to that one.

↳ Repeat: Keep moving to the best neighbour until no ~~better~~ best solution is found.

- Hill climbing always moves to the best possible neighbour at each step.
- It stops when no better solⁿ is available.
- It can get stuck in local optimum [a solⁿ that is better than its neighbour but not the best overall]

Imp



* Problems with hill climbing: imp

03-23
AI

1) local maxima:

↳ The algorithm might get stuck in a solⁿ that is best in its neighbourhood but not the best overall.

↳ Eg:- Imagine climbing a hill and reaching a peak that seems high, but there is an even taller peak nearby. you are stuck in local maxima.

2) Flat local maxima (plateau):

↳ A plateau is a flat area where all neighbouring solⁿ have the same value making it hard to decide which direction to move in. The algorithm might move around without finding an improvement

3) Ridges.

↳ A ridge is a narrow uphill path where the algorithm might not be able to move in the right direction to reach the highest peak. Hill climbing might struggle to follow the ridge to the global optimum

2) Simulated Annealing:

↳ It is a probabilistic algorithm used for optimization problems inspired by the process of annealing in metallurgy where metals are heated and then cooled slowly to remove defects and find the minᵐ energy configuration.. The algorithm ~~[illegible]~~ mimics the process to find the global maxima/minima of given objective function.

08-11 / 08-1
08

1. Objective function:

↳ The functions to be optimized.

2. Temp' (T):

↳ Controls the probability of accepting worse solution. Initially high, gradually decreasing.

3. Cooling schedule:

↳ The process of decreasing the temp over time.

4. Acceptance probability:

↳ Determines whether to accept a worse solution.

$$P(E, E_{new}, T) = \exp\left(\frac{E - E_{new}}{T}\right)$$

↳ If new solⁿ is better (low energy), it is always accepted.
↳ If the new solⁿ is worse, it is accepted with the probability that decreases as the temp' decreases.

031-08-107

* Cooling Schedule:

↳ Temperature should decrease slowly over time to ensure convergence to the global optimum.

Common cooling schedule:

↳ Exponential decay, $T_{new} = \alpha \times T_{current}$ where, $0<\alpha<1$

↳ Linear decay, $T_{new} = T_{current} - \delta$, where δ is constant.

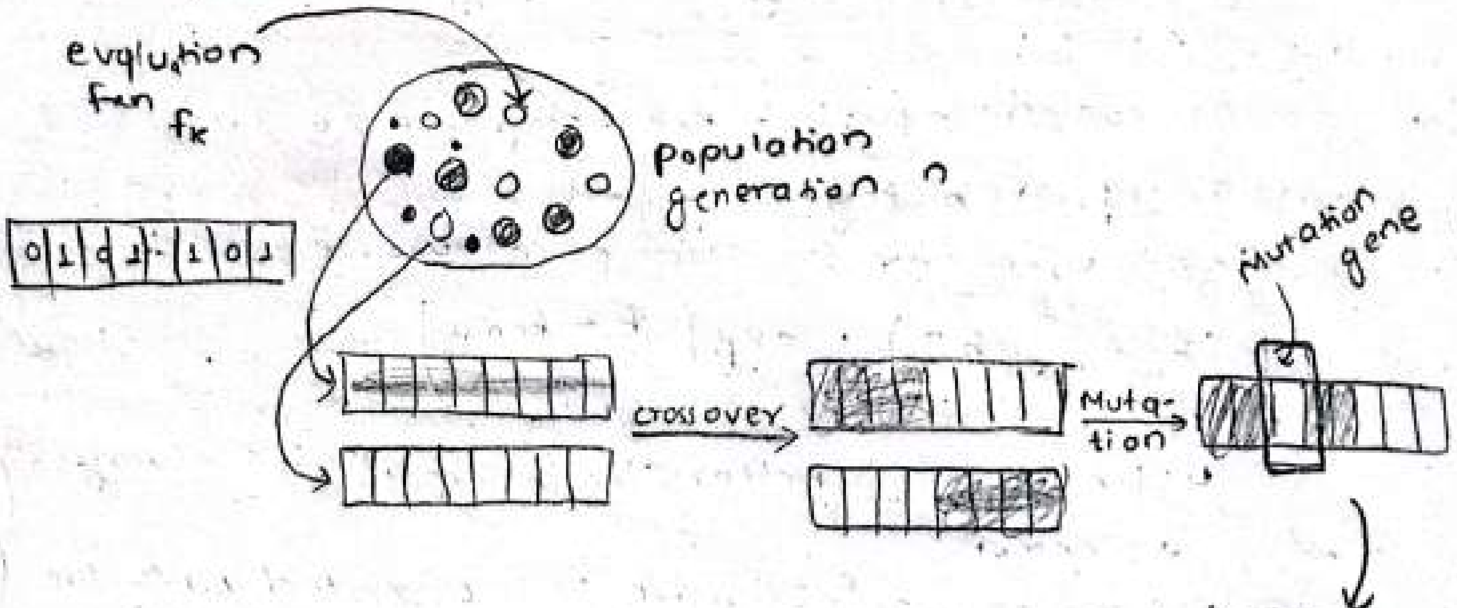
Advantage of simulated annealing

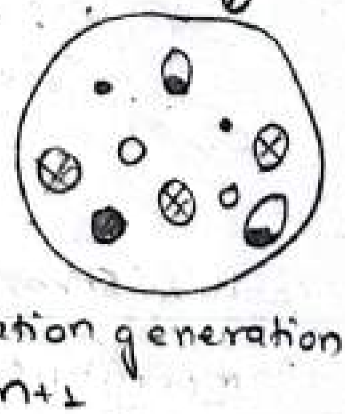
↳ Can escape local maxima / minima.

↳ Suitable for large, complex search space where other method like, hill climbing algorithm and gradient descent stuck.

X 3) Genetic Algorithm: 08-1-07 AI

1) ↳ Genetic algorithm is a method used to solve optimization problem by mimicking the process of natural selection. It is a family of algorithm known as evolutionary algorithm.





key steps of genetic algorithm:

a) Initial Population:

↳ Start with a group of possible soln, called individuals or chromosomes. These are usually generated randomly.

b) Fitness fun:

↳ Evaluate each individual based on how good it is at solving the problem. This is done by using fitness fun.

c) Selection:

↳ Choose the best individuals based on their fitness score to create the next generation. This idea is that the fittest individuals have a higher chance of being selected.

d) Cross over:

↳ Combine two selected individuals to create new one. This is similar to reproduction in nature.

e) Mutation:

↳ Occasionally, make random changes to individual chromosome (like small mutation is genes). This helps introduce diversity and prevent algorithm from getting stuck in local optima.

f) New generation:

↳ Replace the old pop^n with new one, and repeat the process (step b-e) until a stopping condition is met.

Q. Maximize the fxn $f(x) = x^2$ in the range $[0, 31]$.

↳ Encoding Technique:

5 bits are enough to represent the range from 0-31.

0 → 00000
1 → 00001
2 → 00010
3 → 00011
⋮
31 → 11111

Step-1: select Initial Pop^n to start with problem we initially select random no. of pop^n. ~~Here~~

Here,

Initial pop^n ~~size~~ of size 4 is choosen.

ctd →

{Probability = $f(x) / \Sigma f(x)$} {Exp. count = $\frac{f(x)}{Avg. f(x)}$} 08-1-07 AS

String no.	Initial Popn	x value	Fitness fun $f(x^a)=x^2$	Prob	%prob	Expected count	Actual count
1	01100	12	144	0.12467	12.47%	0.4987	1
2	11001	25	625	0.5411	54.11%	2.1645	2
3	00101	5	25	0.0216	2.16	0.0865	0
4	10011	19	361	0.3126	31.26	1.2502	1
Sum.			1155	1	100	3.999	4
Average			288.75	0.25	25	0.999	1
Maxm			625	0.5411	54.11	2.1645	2

String no.	Mating pool	Crossover point	offspring after cross over	x value	fitness $f(x)=x^2$
1	01100	4	0110 1	13	169
2	11001		1100 0	24	576
3	11001	2	11 011	27	729
4	10011		10 001	17	289
Sum					1763
Average					440.75
Maxm					729

String no.	offspring after cross over	Mutation chromosome for flipping	offspring after mutation	x-value	fitness $f(x)=x^2$
1.	01101	10000	11101	29	841
2.	11000	00000	11000	24	576
3.	11011	00000	11011	27	729
4.	10001	00101	10100	20	400
Sum					2546
Avg.					636.5
Max.					841

and so on, we can find the max^m value.

(Q.) Maximize the fxn $f(x) = 15x - x^2$ in the interval [0-15]

↳ Encoding Technique:

4 bits are enough to represent the range from 0-31.

0 → 0000
1 → 0001
⋮
15 → 1111

Step-1: select initial pop^n to start with problem we initially select random no. of pop^n.

Here →
Initial pop^n of size 4 is choosen.

08-18
AS

String no.	Initial popⁿ	x-value	fitness, fx^n $f(x) = 15x - x^2$	Prob	% prob	Expected count	Actual count
1	0101	5	50	0.3165	31.65	1.2658	1
2	1010	10	50	0.3165	31.65	1.2658	2
3	0100	4	44	0.2784	27.84	1.1139	1
4	1110	14	14	0.0886	8.86	0.3544	0
Sum			158	1	100	3.999	4
Average			39.5	0.25	25	0.9999	1
Max			50	0.3165	31.65	1.2658	2

String no.	Mating pool	Crossover point	offspring after crossover	x-value	fitness $f(x) = 15x - x^2$
1	0101	2	0110	6	54
2	1010		1001	9	54
3	1010	3	1010	10	50
4	0100		0100	4	44
Sum					202
Avg					50.5
Max					54

String no.	offspring after cross over	Mutation chromosome for flipping	offspring after mutation	x value	fitness $f(x) = 15x - x^2$
1.	0110	0000	0110	6	54
2.	1001	0100	1101	13	26
3.	1010	0000	1010	10	50
4.	0100	1001	1101	13	26
Sum					156
Avg.					39
Max.					54

and so on we can find the max^m value.

* Constraint Satisfaction Problems:

* Crypto arithmetic:

↳ Types of constraint satisfaction problem.

* Constraint:

↳ No two letters have same value.

↳ Sum of digit must be as given in the problem

↳ There should be only one carry.

- Digit that can be assigned to the letter is in range 0 - 9.

Eg:-

$$\begin{array}{r} T\ O \\ +\ G\ O \\ \hline O\ U\ T \end{array}$$

T = 2, O = 1

\+ G = 8, O = 1

O = 1, U = 0, T = 2

Max^m carry 1 હોઈ શકે
so O ની value 1 હશે

$2 + G \geq 10$ for carry.
so,
G ની value 8, 9 માત્ર
possible છે।
9 લેવાથી U 0 આવશે
જે already O ને assign
થયેલ। So, 8 લેવાથી ઠીક છે.

S E N D
\+ M O R E
M O N E Y

C_1 C_2 C_3 C_4

S = 9, E = 5, N = 6, D = 7

\+ M = 1, O = 0, R = 8, E = 5

M = 1, O = 0, N = 6, E = 5, Y = 2

(5)

C R O S S
+ R O A D S

D A N G E R

	C_0	C_1	C_2	C_3	C_4
	C 9	R 6	O 2	S 3	S 3
+	R 6	O 2	A 5	D 1	S 3
D 1	A 5	N 8	G 7	E 4	R 6

* Constraint Satisfaction Problem (CSP):

↳ A CSP is a type of problem in AI where the goal is to find a solution that satisfies a set of constraints. These problems are about finding values for variables that meet certain conditions or restriction.

Components:-

1) Variables.
2) Domains
3) Constraints:

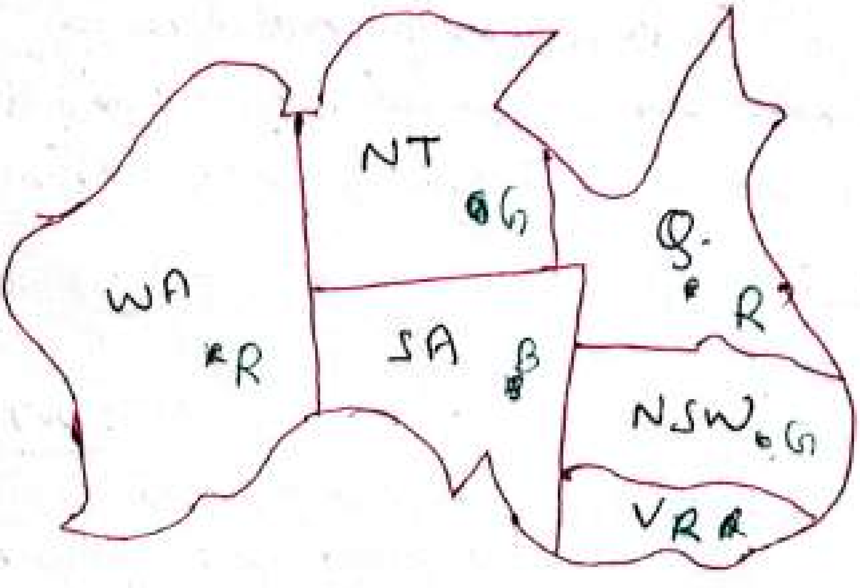
1) variables:

↳ These are the things you are trying to assign values to. for example, in a puzzle, the variables could be the position of pieces or the colors you need to assign to certain objects.

2) ~~Vs~~ Domains:
↳ The possible values that each variable can take. for example, if you are solving a sudoku puzzle, the domain for each cell might be the numbers from 1 to 9.

3) Constraints:
↳ These are the rules or restrictions that limit the values the variables can take. Constraints define which combinations of values are allowed and which are not. In Sudoku, a constraint might be that no number can appear twice in the same row, column, or grid.

Example:

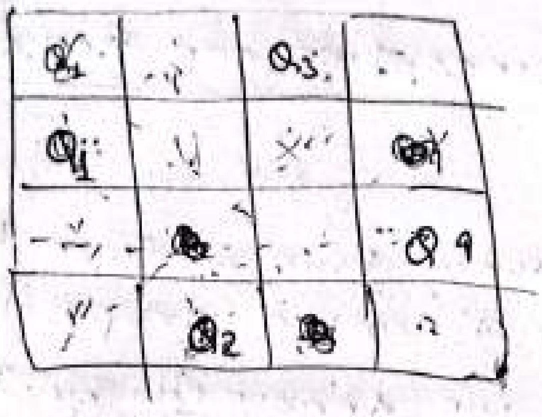


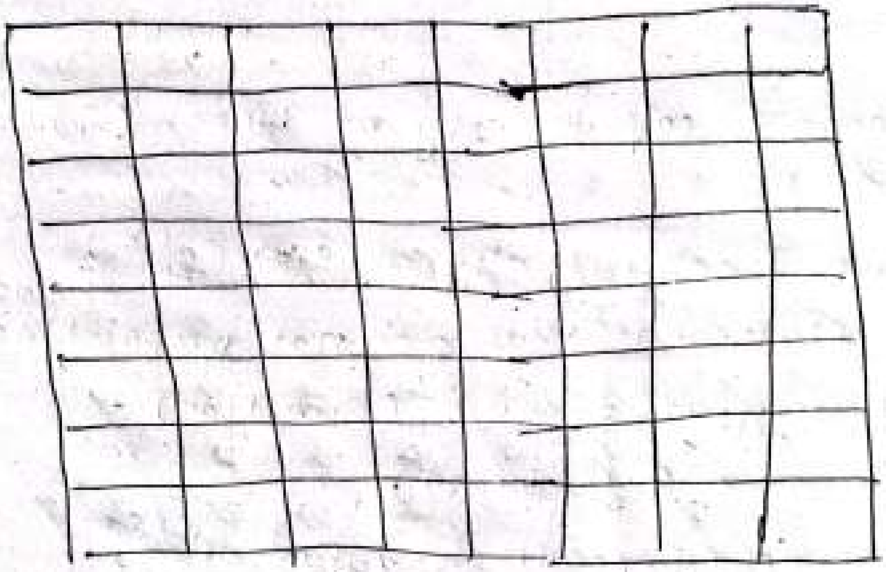
- Variables : WA, NT, SA, Q, NSW, V, T

- Domains D_i = {red, green, blue}
- Constraints : adjacent regions must have different colors

The 4-Queen Problem

08-20
AS





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AI

Unit-4

Knowledge Representation and Reasoning:

* Propositional logics:

↳ A statement that is either true or false but not both at the same time is called proposition. If the proposition is true, we denote it by T and if proposition is false, we denote it by F.

↳ Example:

→ Kathmand is capital city of Nepal (T).

→ $2 < 4$ (T)

→ London is capital city of India (F)

↳ Example of statement that are not propositional

→ what is your name? (Question)

→ Do you work? (Command)

→ x is even (Depends on value of x).

↳ To represent propositions, ~~propositons~~ propositional variables are used (p, q, r, s).

* Logical connectives:

↳ Logical operator are used to construct mathematical statement having 1 or more propositional statement. The combined proposition is called compound proposition.

* Operators:

1) Negation: If p is the proposition, then negation of p is denoted by $\neg p$ which means not p or it ~~to be~~ not case that p.

ctd

Eg:

P: "Today is Monday"

¬P: "Today is not Monday".

Truth table

P	$\neg P$
T	F
F	T

2) Conjunction (AND)

↳ Let P and Q be propositions. The conjunction of P and Q is denoted by $P \wedge Q$. The conjuction $P \wedge Q$ is true when both P and Q are true and is false otherwise. We generally use "AND" for conjunction.

Eg:-

P: It is Monday

Q: It is raining

$P \wedge Q$: It is Monday AND raining.

Truth table

P	Q	$P \wedge Q$
F	F	F
F	T	F
T	F	F
T	T	T

3) Disjunction (OR)

↳ Let P and Q be propositions. The disjunction of P and Q is denoted by $P \vee Q$. The disjunction $P \vee Q$ is false when both P and Q are false and is true otherwise. We generally use "OR" for disjunction.

Eg: P: It is monday
Q: It is raining
P∨Q: It is Monday or it is raining.

Truth table

P	Q	P ∨ Q
F	F	F
F	T	T
T	F	T
T	T	T

4) Implication (if, then)

↳ let P and Q be proposition. The condition statement P→Q is called implication. In implication, P is called hypothesis, and Q is conclusion.

Eg:
P: "Today is saturday".
Q: "college is closed."
P→Q: if today is saturday then college is closed.

Truth table

P	Q	P→Q
F	F	T
F	T	T
T	F	F
T	T	T

* some other terminologies for implication

↳ P only if Q
↳ P is sufficient for Q
↳ Q is necessary for P

5) Bi-implication (if and only if)

$\frac{03\text{-}24}{05}$

↳ If P and Q be propositions, the bi-conditional statement is denoted by $P \leftrightarrow Q$. The bi-conditional statement is true when P and Q have the same truth values and false otherwise.

Eg:

P: I am alive.

Q: I am breathing.

$P \leftrightarrow Q$: I am alive ~~if~~ ~~or~~ if and only if I am breathing.

Truth table

P	Q	$P \leftrightarrow Q$
F	F	T
F	T	F
T	F	F
T	T	T

Q. You can access internet from NCIT only if you are master student or you are newly admitted.

P: You can access internet from NCIT.

Q: You are master student

R: you are newly admitted

$P \rightarrow (Q \lor R)$.

Q. You cannot have voting right if you are mentally unfit and you are not over 18 years.

P: You can have voting right

Q: You are mentally unfit

R: You are ~~not~~ over 18 years

$\neg P \rightarrow (Q \land \neg R)$

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A I

Q The automated reply can not be sent when file system is full.

↳ P: The automated reply can be sent

Q: File system is full

$\neg P \rightarrow Q$

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Q. Leaders will make correct decision only if you choose good leader or you raise your voice against incorrect decision

↳ P: leaders will make correct decision

Q: You choose good leader

R: You raise your voice against incorrect decision

$P \rightarrow (Q \vee R)$

Q. Construct the truth table for the statement.

$(P \vee \neg Q) \rightarrow (P \wedge Q)$

P	Q	$\neg Q$	$P \vee \neg Q$	$P \wedge Q$	$(P \vee \neg Q) \rightarrow (P \wedge Q)$
F	F	T	T	F	F
F	T	F	F	F	T
T	F	T	T	F	F
T	T	F	T	T	T

Q) $(P \to Q) \wedge (Q \to R)$ 08-25 A5

P	Q	R	$P \to Q$	$Q \to R$	$(P \to Q) \wedge (Q \to R)$
F	F	F	T	T	T
F	F	T	T	T	T
F	T	F	T	F	F
F	T	T	T	T	T
T	F	F	F	T	F
T	F	T	F	T	F
T	T	F	T	F	F
T	T	T	T	T	T

* Converse, contrapositive and inverse of conditional statement

Converse

↳ The converse of propositional symbol $P \to Q$ is $Q \to P$

Eg:-

$P \to Q$: If today is saturday then the college is closed.

P : Today is saturday

Q : The college is closed

$Q \to P$: If the college is closed then today is saturday.

Contrapositive

↳ The contrapositive of proposition $P \to Q$ is $\neg Q \to \neg P$

Eg:-

$P \to Q$: If today is saturday then the college is closed.

P : Today is saturday.

Q : The college is closed.

$\neg Q \to \neg P$: If the college is not closed then today is not saturday.

Inverse

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↳ The inverse of proposition $P \to Q$ is $\neg P \to \neg Q$

Eg:-

$P \to Q$: If today is saturday then the college is closed.

P : Today is saturday.

Q : The college is closed.

$\neg P \to \neg Q$: If today is not saturday then the college is not closed.

Q) State the converse, contrapositive and inverse of the following conditional statements.

a) If it snows tonight, then I will stay at home.

b) When I stay up late, it is necessary that I sleep until noon.

c) A positive integer is prime only if it has no divisor other than 1 and itself.

a) soln:

P : It snows tonight

Q : I will stay at home.

Converse : If I will stay at home then it snows tonight
($Q \to P$)

Contrapositive : If I will not stay at home, then it doesnot snow tonight.
($\neg Q \to \neg P$)

Inverse : If it doesnot snow tonight, then I will not stay at home.
($\neg P \to \neg Q$)

b) soln:

P : I stay up late

Q : I sleep until noon.

Converse : When I sleep until noon, it is necessary that I stay up late.
$(Q \to P)$

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Contrapositive : When I donot ~~as~~ sleep until noon, it is necessary that I donot stay up late.
$(\neg Q \to \neg P)$

Inverse : When I donot stay up late, it is necessary that I donot sleep until noon.
$(\neg P \to \neg Q)$

c) soln;

P : A positive integer is prime.

Q : It has no divisor other than 1 and itself.

Converse : It has no divisor other than 1 and itself only if a positive integer is prime.
$(Q \to P)$

Contrapositive : It ~~has~~ doesnot have no divisor other than 1 and itself only if a positive integer is not prime.
$(\neg Q \to \neg P)$

Inverse : A positive integer is not prime only if it doesnot have no divisor other than 1 and itself.
$(\neg P \to \neg Q)$

Propositional Equivalence :

a) Tautology

↳ A compound proposition that is always true no matter what the ~~tr~~ truth values of propositional variables that occurs in it, ~~it~~ is called tautology

Eg: ① $P \vee \neg P$.

P	$\neg P$	$P \vee \neg P$
F	T	T
T	F	T

Eg:- ②

$(P \to Q) \vee (Q \to P)$

P	Q	$P \to Q$	$Q \to P$	$(P \to Q) \vee (Q \to P)$
F	F	T	T	T
F	T	T	F	T
T	F	F	T	T
T	T	T	T	T

b) Contradiction:

↳ A compound proposition that is always false no matter what the truth value of the propositional variables that occur in it, is called contradiction.

Eg: ② $\neg(P \wedge Q) \leftrightarrow (Q \wedge P)$

P	Q	$P \wedge Q$	$\neg(P \wedge Q)$	$Q \wedge P$	$\neg(P \wedge Q) \leftrightarrow (Q \wedge P)$
F	F	F	T	F	F
F	T	F	T	F	F
T	F	F	T	F	F
T	T	T	F	T	F

Eg:- ① $P \wedge \neg P$

P	$\neg P$	$P \wedge \neg P$
F	T	F
T	F	F

c) Contingency:

↳ A compound proposition that is neither tautology nor contradiction is called contingency.

* Logical Equivalence

↳ compound proposition having same truth value in all possible cases are called logically equivalent.

Q. Prove that:

a) $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$

b) $(P \to Q) \equiv \neg P \vee Q$

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AS

* Inference Rule:

Rules of inference	Tautology	Name
① P $P \to Q$ $\therefore Q$	$(P \land (P \to Q)) \to Q$	Modus Ponens
② $\neg Q$ $P \to Q$ $\therefore \neg P$	$(\neg Q \land (P \to Q)) \to \neg P$	Modus Tollens
③ $P \to Q$ $Q \to R$ $\therefore P \to R$	$((P \to Q) \land (Q \to R)) \to (P \to R)$	Hypothetical syllogism
④ $P \lor Q$ $\neg P$ $\therefore Q$	$\neg P \to Q$	Disjunctive syllogism
⑤ P $\therefore P \lor Q$	$P \to (P \lor Q)$	Addition
⑥ $P \land Q$ $\therefore P$	$(P \land Q) \to P$	simplification
⑦ $P \lor Q$ $\neg P \lor R$ $\therefore Q \lor R$	$(P \lor Q) \to (\neg P \lor R)$	Resolution
⑧ P Q $\therefore P \land Q$	$P \land Q \to P \land Q$	Conjunction

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AS

⑤ Show that : "If I play football then I am tired the next day." "I will take rest if I am tired." "If did not take rest." Will lead to conclusion I didnot play football.

↳ soln:

P : I play football.

Q : I am tired.

R : I will take rest.

$$\begin{array}{l} P \rightarrow Q \\ Q \rightarrow R \\ \neg R \\ \hline \therefore \neg P \end{array}$$

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03

~~Q) Hypothesis:~~

~~i) P→Q~~

~~ii) Q→R~~

~~iii)~~

Q. Show that "if the interest rate drops, the housing market will improve", "The federal discount rate will drop or the housing market will not improve,", "Interest rate will drop". Prove that the federal discount rate will drop.

Soln.

P: The interest rate drops.

Q: The housing market will improve.

R: The federal discount rate will drop.

Hypothesis:

i) $P \to Q$

ii) $R \vee \neg Q$

iii) P

$\therefore R$

S.N.	step:	Reason:
1	$P \to Q$	Given Hypothesis
2	$\neg P \vee Q$	Implication on 1.
3	$R \vee \neg Q$	Given Hypothesis
4	$\neg P \vee R$	Resolution from 2 and 3
5	P	Given Hypothesis
6	R	from 4 and 5

(Q.5) → soln;

Hypothesis: i) $P \to Q$
ii) $Q \to R$
iii) $\neg R$

$\therefore \neg P$

S.N.	steps	Reasons.
1	$P \to Q$	Given Hypothesis.
2.	$Q \to R$	Given Hypothesis.
3.	$P \to R$	Hypothetical syllogism from 1 and 2
4.	$\neg R$	Given Hypothes.
5.	$\neg P$	Modus Tollens from 3 and 4

(Q) 1 "If clinton doesnot live in france, then he doesnot speak french", "Clinton doesnot drive car", "If Clinton lives in france, then he rides a motorcycle", "Either clinton speaks french or he drives car". Prove that clinton drives motorcycle.

↳ soln,

P : Clinton lives in france

Q : He speaks french

R : Clinton drives car

S : He rides a motorcycle.

Hypothesis:

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i) $\neg P \to \neg Q$

ii) $\neg R$

iii) $P \to S$

iv) $Q \vee R$

Conclusion $\therefore$ S

S.N.	steps	Reasons.
1.	$\neg R$	Given Hypothesis
2.	$Q \vee R$	Given Hypothesis
3.	Q	from 1 and 2 (Disjunctive syllogism)
4.	$\neg P \to \neg Q$	Given Hypothesis
5.	P	from 3 and 4 (Modus tollens).
6.	$P \to S$	Given Hypothesis
7.	S	from 5 and 6 (Modus ponens)

Q. If it is not raining, then the road is not wet.

$\overbrace{\text{it is not raining}}^{\neg P}$, then $\underbrace{\text{the road is not}}_{\neg q}$ wet.

$$\neg P \to \neg Q.$$

Q If it is raining, then the road is wet, and if the road is wet then it is raining.

$$(P \to Q) \wedge (Q \to P)$$

Q It is not the case that both it is raining and the road is wet.

$$\neg(P \wedge Q)$$

Q. If the road is wet, then it is raining, but if it is not raining then the road is not wet.

08-27 A:

$$(Q \rightarrow P) \vee (\neg P \rightarrow \neg Q)$$

* Predicate Logic / first Order Logic (FOL):

↳ Consider the following

P: "All men are mortal"

Q: "Ram is man"

R: "Ram is mortal"

→ No rule of propositional logic will allow us to conclude the truth of R.

→ Therefore, we need more powerful type of logic called first order logic. or predicate logic

↳ To understand predicate logic, we need to understand

a) subject

b) Predicates

c) Quantifiers

d) Domain

Example:

"Jose is tall"

subject: Jose → predicate: tall

Notation: Tall(Jose)

a) Subject:

↳ The subject is what (or whom) the sentence is about.

b) Predicates:

↳ Refers to the property that subject of statement can have.

↳ Consider the statement.

x is greater than 3

$\underbrace{x}_{\text{subject}}$ is $\underbrace{\text{greater than 3}}_{\text{predicate}}$

we can denote the statement as

P(x) :

P(6) → True
P(5) → True
P(2) → False

Q
↳ let Q(x) denotes the statement "The word x' contain the letter a"

What are the truth value of Q(asmita) and Q(logic)

soln:
Q(asmita) : True
Q(logic) : False

Q let c(x,y) denotes the statement "x is the capital of y"
what are the truth value of c(kathmandu, Nepal) → True
c(Texas, USA) → False

Q Consider the following statement
P(x) : x is greater than 10
Domain : All positive natural number
Can we say the above statement is true for all values of x within our domain?
↳ No, P(x) is true for <u>some values</u> of x.

8) Consider the statement

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AS

$Q(x)$: " $x < x+1$ "

Domain : All positive natural number.

↳ $Q(x)$ is true for <u>all values</u> of x in given domain.

<u>081-09-16</u>

In predicate logic, predicates are used alongside quantifiers to express the extent to which a predicate is true over a range of elements. Using quantifiers to create such proposition is called quantifications. There are two types of quantifiers in predicate logic:-

1) Universal quantifier ($\forall_x$) } imp in license also
2) Existential quantifier ($\exists_x$)

1) Universal quantifier ($\forall_x$) :

"Every cat drinks milk"

x_1
x_2 x_3
Domain of cat.

Above statement is equivalent to

x_1 drinks milk $\wedge$ x_2 drinks milk $\wedge$ x_3 drinks milk.

let,

Milk(x) : "x drinks milk" → It can be read as: for all Milk(x), or for every x, Milk(x).

$\boxed{\forall_x \cdot Milk(x)}$

↳ The universal quantification of $P(x)$ is : " $P(x)$ for all value of x in the domain " = $\forall_x P(x)$

↳ we can also read $\forall_x P(x)$ as :- <u>for all P(x)</u> or for every x, P(x)

09-11
AI

Q All student of BECE takes course on AI.

let,

$A(x)$ = "x takes course on AI

$\Rightarrow \boxed{\forall x \cdot A(x)}$

Q Let $Q(x)$ be the statement "x less than 5". What is the truth value of quantification where domain is all real number.

↳ $\boxed{\forall x \cdot Q(x)}$

let us consider, $x = 6$.

$6 < 5$ (False) ⇒ counter example

$\therefore \forall x \cdot Q(x) = $ false.

Truth value of

↳ An element for which $Q(x)$ is false is called counter example of $\forall x \cdot Q(x)$.

Q What is the truth value of $\forall x \cdot P(x)$ where,
$P(x)$ = "$x^2 < 20$" and the domain consist of positive number not exceeding 4.

Domain = $\{1, 2, 3, 4\}$

. for all values in domain, $P(x)$ is true.

$\therefore$ The truth value of $\forall x \cdot P(x)$ = True.

09-16
AI

2) Existential quantifier ($\exists x$) :

"Some lions drinks milk"

Domain of lion.

Above statement can be written as :

x_1 drinks milk $\vee$ x_2 drinks milk $\vee$ x_3 drinks milk.

Let,

Milk(x) : "x drinks milk"

$\boxed{\exists_x \text{Milk}(x)}$

Read as,

for some x, Milk(x) or there is at least one x, such that Milk(x) or there exist an x in domain such that Milk(x).

↳ The existential quantification of P(x) is : "P(x) for some value of x in the domain" = $\exists_x P(x)$.

Q) Some student of BECE takes course on Data mining.

Let

D(x) = " x takes course on Data Mining"

$\Rightarrow \boxed{\exists_x \cdot D(x)}$

BECE.

Q) Let P(x) denotes the statement "x > 3". What is the truth value of quantification $\exists_x P(x)$ where the domain contain all real number.

for x = 4,

4 > 3 (True)

$\therefore$ The truth value of $\exists_x \cdot P(x)$ = True

Universal – all true gives true

Existential – one true gives true.

09-16 A2

(5) What is the truth value of $\exists_x \cdot P(x)$ where $P(x)$ is a statement "$x^2+1 > 20$" and domain is all positive number not exceeding 4.

↳ Domain = {1, 2, 3, 4}

For all values of x, $p(x)$ is false.

∴ The truth value of $\exists_x \cdot P(x)$ = False.

(8). Express the statement "Every student in BECE has studied calculus." Using predicates, and quantification.

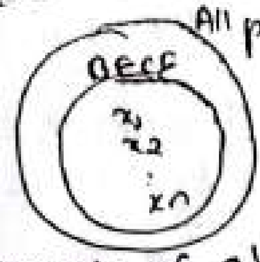
let.

$C(x)$: "x ~~...~~ has studied calculus"

$\Rightarrow$ $\boxed{\forall_x C(x)}$

(8) Domain: all people within class.

Statement: There is a person x, if person x is in the class (BECE), then x has studied calculus.



Domain of all people

↳ let.

$S(x)$: "x is in the class (BECE)"

$C(x)$: "x has studied calculus"

$\forall_x [S(x) \rightarrow C(x)]$

for domain: all people within college

$\forall_x [S(x) \wedge C(x)]$

(Q) Express the statement "some student in this class has visited Ilam" using predicates and quantifiers. Domain: BECE student within class.

09-10
A7

Statement becomes,

There is a student x in this class that has visited Ilam.

let,

I(x) : "x has visited Ilam"

$\Rightarrow \boxed{\exists_x I(x)}$

(Q) Domain: all people.

Statement becomes,

There is a person x, such that person x is in the class and x has visited Ilam.

let,

S(x) : "x is in the class"

I(x) : "x has visited Ilam"

$\Rightarrow \boxed{\exists x \, [S(x) \land I(x)]}$

* Negation of Quantification

(*) Every student in BECE has taken Data Mining. (Domain: All BECE student)

let

D(x) : "x has taken Data Mining"

$\Rightarrow \boxed{\forall_x \cdot D(x)}$

The negation of above statement is

"There is some student in BECE who has not taken Data Mining".

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AJ

$\Rightarrow$ $\boxed{\exists x \cdot \neg D(x)}$

(*) "There is a student in class who has long hair".

let,

$H(x)$: "x has long hair"

$\Rightarrow$ $\boxed{\exists x \cdot H(x)}$

The negation of above statement is:

Every student in class has not long hair.

$\Rightarrow$ $\boxed{\forall x \cdot \neg H(x)}$

(*) "Express the statement "Every student in this this class has visited Jhapa or kathmandu.

↳ let

$J(x)$: "x has visited Jhapa"

$K(x)$: "x has visited Kathmandu"

$\Rightarrow$ $\boxed{\forall x \cdot [J(x) \vee K(x)]}$

The negation of above statement is

There is some student in this class who has not visited Jhapa or kathmandu.

$\Rightarrow$ $\exists x \cdot [\neg [J(x) \vee K(x)]]$

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A1

(1) Let $P(x)$ be statement "x can speak Russian" $Q(x)$ be the statement "x knows the computer language C++" (Domain: all student at your college).

a) There is student at your college who can speak Russian and who knows C++.

$$\exists x\, [P(x) \land Q(x)]$$

b) There is student at your college who can speak Russian but who does not know C++.

$$\exists x\, [P(x) \land \neg Q(x)]$$

c) Every student at your college either can speak Russian or know C++.

$$\forall x\, [P(x) \lor Q(x)]$$

d) No student at your college can speak Russian or knows C++.

$$\forall x\, [\neg P(x) \land \neg Q(x)]$$

Nested Quantifiers:

Consider the following statement.

"The sum of any two positive real number is positive."

This can be stated as:-

"for every x and for every y, if $x>0$ and $y>0$, then $x+y>0$"

let.

$P(x,y)$: "$(x>0) \land (y>0) \longrightarrow (x+y)>0$

The given statement says that the sum of any two positive real number is positive, so we need to quantification

$\therefore \forall_x \forall_y [P(x,y)]$

Q. The product of two -ve number is positive.

let.

$P(x,y)$: "$(x<0) \land (y<0) \rightarrow (x*y) >0$"

$\therefore \forall_x \forall_y [P(x,y)]$

Q. If person is a student and is a computer science major, then this person takes a course in mathematics.

let

$S(x)$: "x is a student"

$C(x)$: "x is a computer science major"

$T(x,m)$: "x takes a course in m"

$\forall_x \; [(S(x) \land C(x)) \longrightarrow \exists_m T(x,m)]$

09-18
 A3

Q Everybody loves somebody.

~~Statement~~:

for every person x, there exist y such that x loves y.

$L(x, y)$: "x loves y"

$\forall_x \exists_y L(x, y)$

Q Everybody loves everybody.

$\forall_x \forall_y L(x, y)$.

Q Somebody loves somebody.

$\exists_x \exists_y L(x, y)$

Q Someone is loved by everybody.

$\exists_x \forall_y L(x, y)$

Q If person is female and is a parent, then this person is someone's mother.

$F(x)$: "x is female"

$P(x)$: "x is parent"

$M(x, y)$: "x is somebody's mother"

$\forall_x [(F(x) \wedge P(x)) \rightarrow \exists_y M(x, y)]$

(Q) Every gardener likes the sun. ~~vvf on~~

~~$\forall x \forall y$ ...~~

~~$\forall x [g(x) \Rightarrow s(x)]$~~

$\forall x$ gardener$(x) \rightarrow$ likes $(x$, sun$)$.

(Q) You can fool some of the peopale ~~of~~ all of the time. There exist some people who can be fooled all of the time.

~~$\forall x$ fool(x)~~

~~$\exists x \forall x$ fool(x) fool $(x$, somepeople$)$~~

$\exists x \forall t$ person $(x) \wedge$ time $(t) \rightarrow$ can_fool(x, t)

(Q) You can fool all of the people some ~~time~~ of the time.

~~There exist some~~ for every people, there exist some time when all people can be fooled.

(Q) All purple mushrooms are poisonous.

~~$\forall x\, M(x) \rightarrow P(x)$~~

$\forall x$ ~~...~~ [mushroom$(x) \wedge$ purple$(x)] \rightarrow$ poisonous (x)

(Q) No purple mushroom is poisonous.

$\forall x$ [mushroom$(x) \wedge$ purple$(x)] \rightarrow \neg$poisonous (x)

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(Q) Aakash is tall.
↳ Tall (Aakash)

(Q) Aakash is not tall.
$\neg$Tall (Aakash)

(Q) There are exactly two purple mushroom.

(Q) ~~ϕ~~ x is above y if and only if x is ~~oo~~ on directly on top of y or there is pile of one or more other objects directly on top of one another starting with x and ending with y.

(Q) Everyone at NCIT is smart.

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AI

* Resolution theorem proving:

Rules:

1. ~~Correct~~ Convert the english statements into FOL.
2. Convert FOPL to CNF.
3. Apply negation to what's need to be proven
4. Draw resolution graph.

Rules to convert FOL to CNF

① $X \rightarrow Y \Rightarrow \neg X \vee Y$

② $X \wedge Y \rightarrow Z$

$\neg(X \wedge Y) \vee Z$

$\neg X \vee \neg Y \vee Z$

③ $\forall a\, dog(a) \Rightarrow dog(a)$

VVI

Q. Consider the following set of facts.

The humidity is high or the sky is cloudy

If the sky is cloudy, then it will rain.

If the humidity is high then it is hot

It is not hot.

Prove that it will rain.

converting into FOL

Step-I: ①

P : ~~Humidity~~ The humidity is high

Q : The sky is cloudy.

$P \vee Q$

(2)

R : It will rain

$Q \rightarrow R$

(3)

S : It is hot

$P \rightarrow S$

(4) $\neg S$

Step-II: Converting FOL to CNF.

(2) $Q \rightarrow R \Rightarrow \neg Q \vee R$

(3) $P \rightarrow S \Rightarrow \neg P \vee S$

(1) $P \vee Q \Rightarrow P \vee Q$

(4) $\neg S \Rightarrow \neg S$.

Step-III: Negating the statement to be proven.

It will rain $\Rightarrow$ It will not rain

$\neg R$

Step-IV: Drawing Resolution Graph.

$\neg R$ $\quad$ $\neg Q \vee R$

$\neg Q$ $\quad$ $P \vee Q$

P $\quad$ $\neg P \vee S$

S $\quad$ $\neg S$

ϕ

Hence the statement we took i.e. $\neg R$ is false. So, R will be true i.e. it will rain.

proved

V.V.I (8)

09-22
AF

Mother(Lulu, fifi)

Alive(Lulu)

$\forall x \forall y$. Mother(x,y) $\rightarrow$ parent(x,y)

$\forall x \forall y$ (parent(x,y) $\land$ Alive(x)) $\rightarrow$ older(x,y)

Prove. older(Lulu, fifi)

~~Sstep~~ Step-I: Converting into FOL

All statements are in FOL.

Step-II: Converting FOL to CNF

(a) Mother(Lulu, fifi)

(b) Alive(Lulu)

(c) $\forall x \forall y$ Mother(x,y) $\rightarrow$ parent(x,y)

$\Rightarrow$ $\neg$Mother(x,y) $\lor$ Parent(x,y)

(d) $\forall x \forall y$ (parent(x,y) $\land$ Alive(x)) $\rightarrow$ older(x,y)

$\Rightarrow$ $\neg$parent(x,y) $\lor$ $\neg$Alive(x) $\lor$ older(x,y)

Step-III: Negating statement to be proven

older(Lulu, fifi) $\Rightarrow$ $\neg$older(Lulu, Fifi)

Step-IV: Drawing Resolution Graph

Crd $\rightarrow$

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$\neg older(Lulu, Fifi)$ $\quad$ $\neg parent(x,y) \lor \neg Alive(x) \lor older(x,y)$

$\{x/Lulu, y/Fifi\}$

Lulu, Fifi $\quad$ Lulu

$\neg parent(Lulu, Fifi) \lor \neg Alive(Lulu)$ $\quad$ $Alive(Lulu)$

$\cancel{\neg Mother(x,y) \lor Parent(x,y)}$

$\cancel{\neg Mother(x,y)}$

$\neg parent(Lulu, Fifi)$ $\quad$ $\neg Mother(x,y) \lor Parent(x,y)$

$\{x/Lulu, y/Fifi\}$

$\neg Mother(Lulu, Fifi)$ $\quad$ $Mother(Lulu, Fifi)$

ϕ

* Reasoning Under Uncertainity :

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AI

~~Lec~~-

Uncertainity occurs when we don't have complete information.

Eg:- ① We don't know if it will rain tomorrow

② We cannot predict the exact outcome of dice roll

↳ We often need to make decision despite this uncertainity

↳ Suppose that we are trying to determine if a patient has pneumonia. We observe the following symptoms.

i) Patient has a cough.

ii) Patient has a fever.

iii) Patient has difficulty in breathing.

Now, we would like to determine how likely the patient has pneumonia given these symptoms

Even though symptoms are common in pneumonia, we are not 100% certain that the patient has pneumonia. In this case, we are dealing with uncertainity.

↳ Propositional and predicate logic cannot handle this uncertainity because

1) Computational complexity

↳ Listing all the antecedents and consequences in the agent's environment is too big.

2) Lack of domain knowledge

↳ We cannot list every possibility because we do not know enough about the domain.

3) Paragmatic factors

↳ Use of predicate logic may be impractical.

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* Probability:

↳ Probability deals with uncertainity by assigning numerical values to something (range 0 to 1)

↳ Probability is Just a number indicating our belief in a statement. It is often taken to be our degree of willingness to particular outcome

0 → no belief

1 → high confidence.

* Prior Probability:

↳ This ais an unconditional probability without taking any evidence into consideration.

Eg:- Assume that public health experts have examined 1 lakh people at random and found that 10 of them have measles. Now, we pick a person at random. What is the prior probability that person has measles?

Ans $P(H) = \frac{10}{10000} = 0.0001$.

$P(\neg H) = 1 - P(H) = 0.9999$

* Posterior Probability:

↳ $P(H/E)$ is the probability that hypothes 'H' is true given that Evidence 'E' is observed

Eg:- suppose we see red spots (E) on person's body and we consider the hypothesis that they have the measles (H).

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AF

↳ $P(E/\neg H)$ is the probability that Evidence 'E' will be observed even if hypothes 'H' is not true.

*** Baye's Rule:**

$$P(H/E) = \frac{P(E/H) \cdot P(H)}{P(E)}$$

$$P(E) = P(E/H) \cdot P(H) + P(E/\neg H) \cdot P(\neg H)$$

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*** Bayesian Network:**

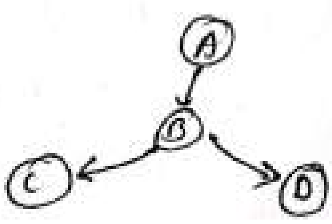
↳ Also known as belief networks ~~called~~ or bayes nets, belongs to the family of probabilistic graphical models. These graphical structures are used to represent knowledge about ~~an~~ an uncertain domain. In particular, each node in the graph represents a random variable while the edges between the nodes represents the probabilistic dependencies among the corresponding random variables.

↳ Bayesian network help us reason with uncertainity

↳ They are used in many application. Eg:- spam filtering, speech recognition, robotics, diagnostic system, etc.

↳ A bayesian network is made up of:

1) Directed Acyclic graph



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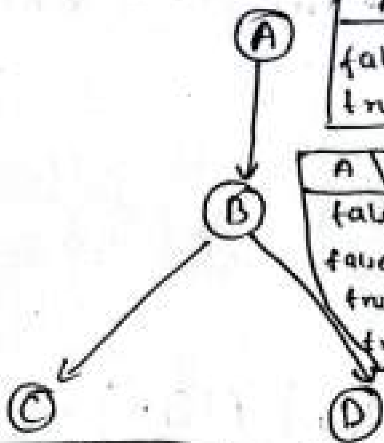
2) A set of tables for each node in the graph.



A	P(A)
false	0.6
true	0.4

A	B	P(B/A)
false	false	0.01
false	true	0.99
true	false	0.7
true	true	0.3

B	C	P(C/B)
false	false	0.02
false	true	0.98
true	false	0.05
true	true	0.95

B	D	P(D/B)
false	false	0.4
false	true	0.96
true	false	0.9
true	true	0.1

↳ Each node in the graph is random variable.

↳ A node x is parent of anothe node y if there is arrow from node x to node y. Eg:- A is parent of B (in fig)

↳ Informally an arrow from node x to node y means x has direct influence on y.

↳ Each node x_i has conditional probability distribution $P(x_i / parent(x_i))$ that quantifies the effects of parent on the node.

↳ These parameters are probabilities in conditional probability table

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AI

(Q)

Rain $P(R) = 0.4$

cause

effect. wet grass $P(W/R) = 0.9$

$P(W/!R) = 0.2$

What is the probability that rain is the cause?

$$P(R/W) = \frac{P(W/R) * P(R)}{P(W)}$$

Now,

$$P(W) = P(W/R) * P(R) + P(W/!R) * P(!R)$$

$$= 0.9 * 0.4 + 0.2 * 0.6$$

$$= 0.48$$

$$\therefore P(R/W) = \frac{0.9 * 0.4}{0.48} = 0.75 \quad \underline{\text{Ans}}$$

(Q) Consider $P(<30)$: probability of person randomly selected from local bar environment be born within 30 miles of ~~[illegible]~~ manchester.

While watching a game of champion league football in a cafe, you observe someone who is actually supporting manchester united in a game. Using Baye's rule, calculate the probability that they were actually born within 30 miles of manchester. Assume that the probability that randomly selected person in bar is born within 30 miles of manchester is $1/29$

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AS

The chance that person born within 30 miles of manchester actually supports manchester united is $7/19$.

The probability that the person not born within 30 miles of manchester support manchester united is $1/19$

↳ soln;

$$P(<30) = \frac{1}{29}$$

$$P(M/<30) = \frac{7}{19}$$

$$P(M/!<30) = \frac{1}{19}$$

$$P(<30/M) = ?$$

Now,

$$P(<30/M) = \frac{P(M/<30) * P(<30)}{P(M)}$$

$$P(M) = P(M/<30) * P(<30) + P(M/!<30) * P(!<30)$$

$$= \frac{7}{19} * \frac{1}{29} + \frac{1}{19} * \frac{28}{29}$$

$$= \frac{35}{551}$$

$$\therefore P(<30/M) = \frac{7/19 * \frac{1}{29}}{\frac{35}{551}} = \frac{1}{5} = 0.2 \text{ Ans}$$

(9)

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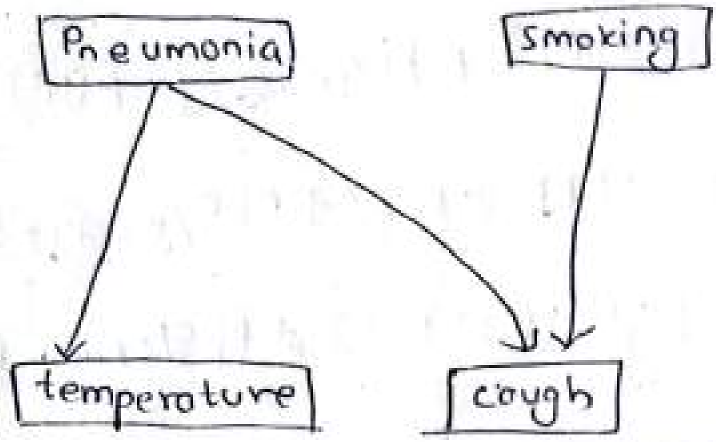
A1

Pneumonia	
true	0.1
false	0.9



Smoking	
true	0.2
false	0.8

Pneumonia	Temperature	
	Yes	NO
Yes	0.9	0.1
No	0.2	0.8

Pneumonia	Smoking	Cough	
		true	false
true	Yes	0.95	0.05
true	no	0.8	0.2
false	Yes	0.6	0.4
false	no	0.05	0.95

find the probability.

1) P[cough / pneumonia and smoking]

= 0.95.

2) P[smoking / cough]

$$= \frac{P[cough/smoking] * P(smoking)}{\cancel{P[cough/smoking] * P(smoking) + P[cough/!smoking] * P(!smoking)} \; P[cough]}$$

$$= \frac{(0.95 + 0.6) * 0.2}{(0.95 + 0.6) * 0.2 + (0.8 + 0.05) * 0.8}$$

= 0.3131

$$= \frac{\{P[C/P \text{ and } S] * P[P] + P[C/!P \text{ and } S] * P[!P]\} * P(S)}{P[C/P \text{ and } S] * P[P] * P[S] + P[C/P \text{ and } !S] * P[P] * P[!S] + P[C/!P \text{ and } S] * P[!P] * P[S] + P[C/!P \text{ and } !S] * P[!P] * P[!S]}$$

$$= \frac{(0.95 * 0.1 + 0.6 * 0.9) * 0.2}{0.95 * 0.1 * 0.2 + 0.8 * 0.1 * 0.8 + 0.6 \times 0.9 \times 0.2 + 0.05 * 0.9 * 0.8}$$

$$= 0.5594 \approx \left(\frac{127}{227}\right)$$ Ans

08-09-25

* Markov Network

↳ The ~~deen~~ canonical probabilistic model of temporal or sequential data is called Markov Model or Markov Network.

↳ "future is independent of the past given the present."

↳ They model a process that proceeds in steps (time, sequence, trials, etc) like series of probability trees.

Definition:

Discrete random variables $x_1 \ldots x_n$ forms a discrete time markov chain if their joint distribution respects the following graphical pattern.

$$x_1 \longrightarrow x_2 \longrightarrow x_3 \longrightarrow \cdots\ x_n.$$

Mathematically,

$$P(x_t / x_1, x_2, \dots x_{t-1}) = P(x_t / x_{t-1})$$

$$P(x_1, \dots x_n) = P(x_1) \cdot P\left(\tfrac{x_2}{x_1}\right) P\left(\tfrac{x_3}{x_2}\right) \cdots P\left(\tfrac{x_n}{x_{n-1}}\right)$$

$$= P(x_1) \cdot P\left(\tfrac{x_2}{x_1}\right) \cdot P\left(\tfrac{x_3}{x_1, x_2}\right) \cdots P\left(\tfrac{x_n}{x_1, \dots x_{n-1}}\right)$$

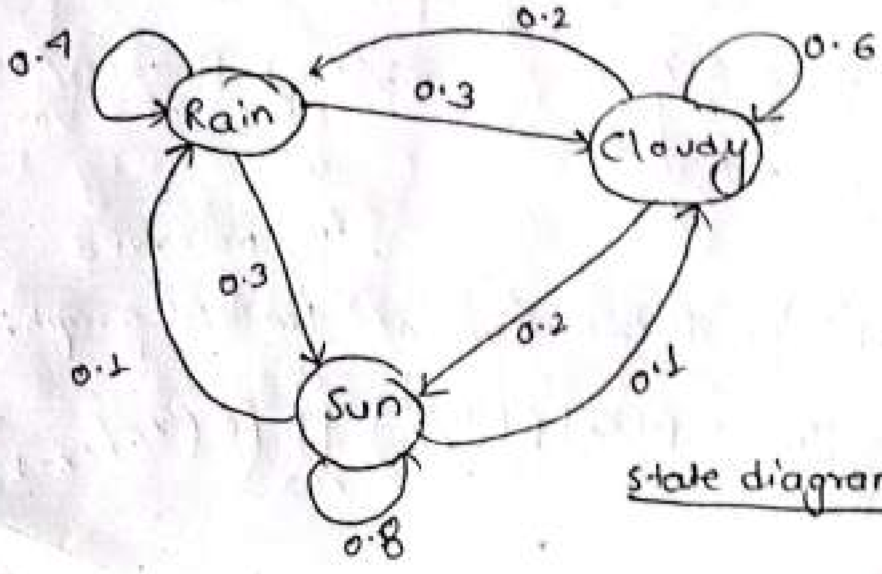
Markov Model Example:

state1 : Rain
state2 : cloudy
state3 : Sun

} three states of weather.

		Tomorrow(a)		
		Rain	Cloudy	Sun
Today	Rain	0.4	0.3	0.3
	cloudy	0.2	0.6	0.2
	Sun	0.1	0.1	0.8

⇐ we obtain the transition matrix with long term of observation.



State diagram

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AS

Summary of Markov Model:

↳ They are combination of probabilities in matrix operation.
↳ They model a process that proceeds in steps (time, sequence, trials, etc) like series of probability trees.
↳ The model can be in one state at each step.
↳ When the next step occurs, the process can be in the same state or move to another state.
↳ Movement betn states are defined by probabilities.
↳ We can then find the probability of being in any given state.

VVI for short note

* Hidden Markov Model: [HMM]

↳ In hidden markov model, is a stochastic finite automation where each states generate an observation.

Let x_t = hidden states/variables

y_t = observations

m = possible states,

then, $x_t \in \{1, 2, \ldots m\}$ and $y_t \in \{1, 2, \ldots m\}$.

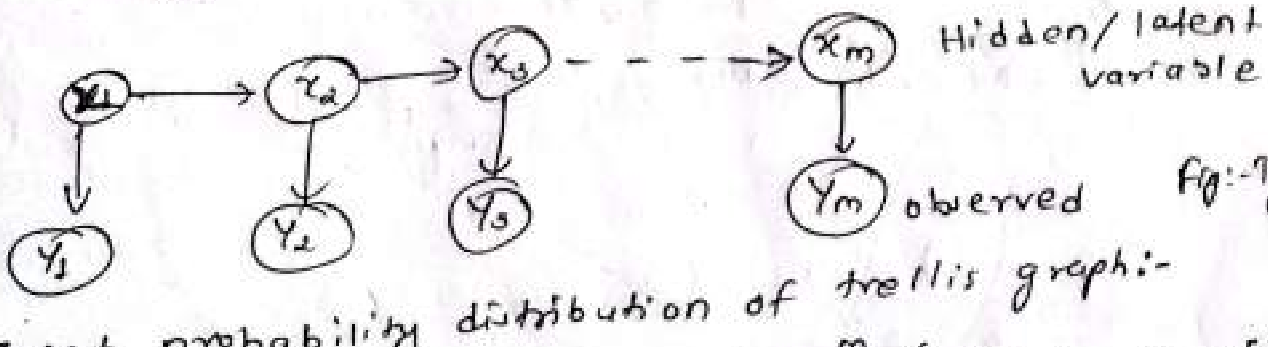
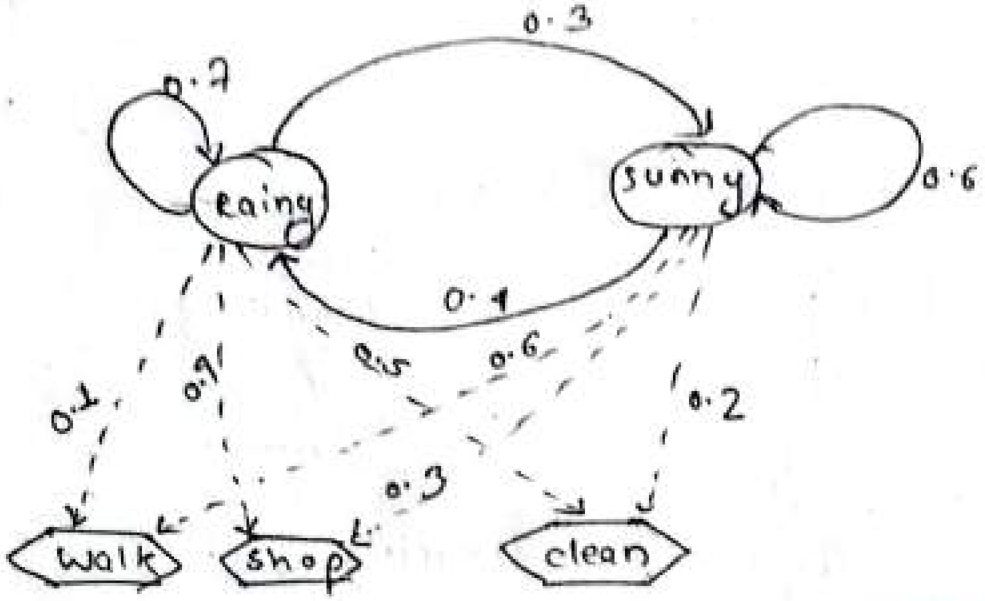


fig:- Trellis graph

The joint probability distribution of trellis graph:-

$$p(x_1, \ldots x_m, y_1 \ldots y_m) = p(x_1)\, p(y_1/x_1) \prod_{k=2}^{m} \left(p(x_k/x_{k-1})\; p(y_k/x_k) \right)$$

Example:



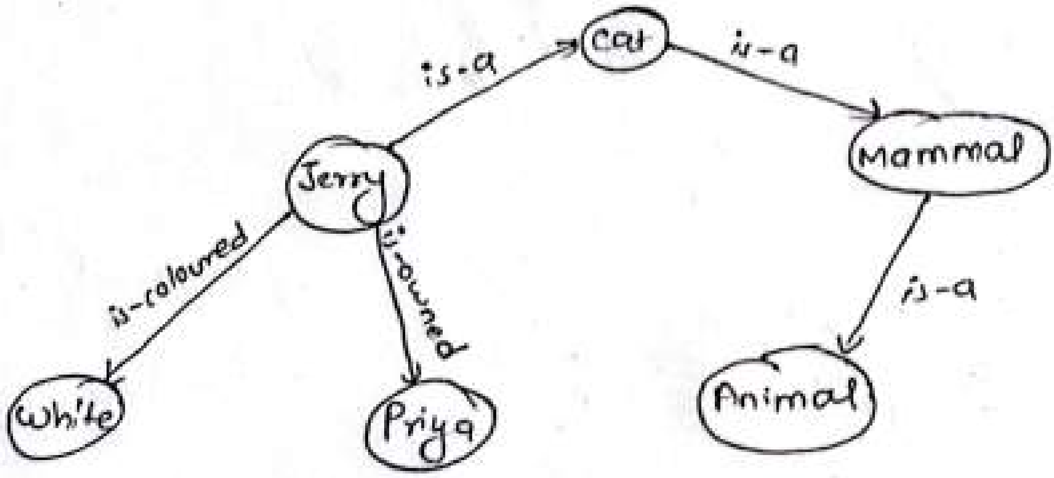
× Other approaches to knowledge representation

- Knowledge representation is critical for AI system.
- It structure knowledge in machine readable form for reasoning and problem solving.
- Common approaches to knowledge representation includes:
  1. → Semantic Network
  → Frames
  → Ontologies

1) Semantic Network:
- Graph based representation of knowledge.
- Node represents concepts or entities.
- Edge represents relationships betn concepts (eg.- is-a, has-a)
- inheritance allows concepts to share properties.

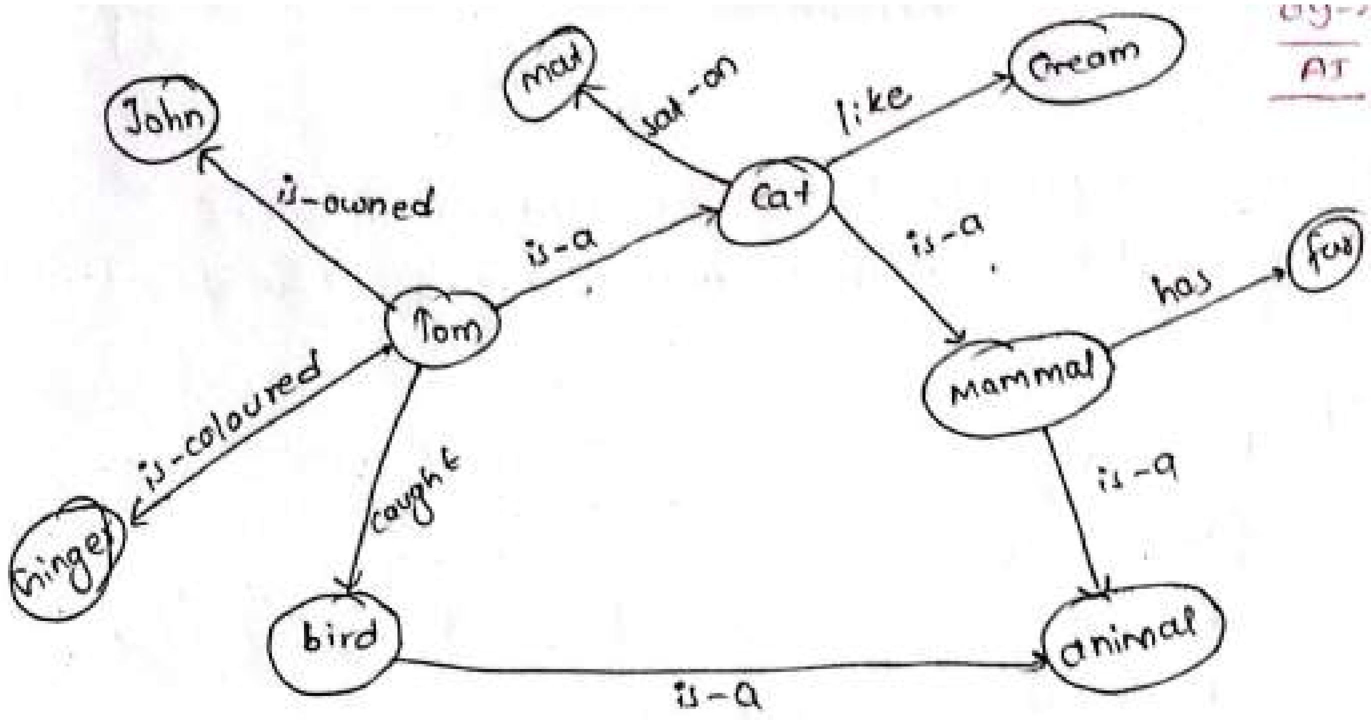
Q) Represent the following knowledge in semantic network
- Jerry is a cat.
- Jerry is a mamal.
- Jerry is owned by priya
- Jerry is white coloured
- All mammals are animal.

Ans



(9) Represent the following information in semantic network.

- Tom is a cat
- Tom caught a bird
- Tom is owned by John
- Tom is ginger in colour
- Cats like cream
- The cats sat on a mat.
- A cat is a mammal
- A bird is an animal.
- All mammals are animal
- Mammals have fur.



Advantages of semantic network

1) Intutive and flexible for relational knowledge
2) Easy to visualize and understand
3) Inherits properties from parent concepts.

Disadvantages:

1) Lack of formal rule.
2) Difficult to represent complex relationships
3) limited reasoning capabilities.

2)
*Frames:

↳ structured representation using slots and values.
↳ A frame is like a data structure or object with attributes (Slots).

Example:-

frame Dog
[Type: Animal
legs: 4
Fur: Yes
Habitat: Domestic
Sound: Bark]

Advantages:

1) Structured and detailed representation of objects
2) Support default values and inheritance.
3) Easy to extend and Update

Disadvantages:

1) More rigid structure compared to semantic network.
2) Complexity increases with deeply nested inheritance.
3) Can become complex with large no. of frames.

3) Ontological Representation:

*Ontology:

↳ formal representation of domain's concepts and relationship.
↳ Defines class, instance and relation betn concepts.
↳ supports logical reasoning.

eg:-

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class : Animal

has_a : leg
has_a : heart
is_a : mammal
is_a : bird

class : mammal

has_a : fur
gives_birth : yes

class : ~~a~~ bird

has_wings : yes
lays-eggs : yes.

Advantages:

1) Rich formalism support automated reasoning.
2) Interoparability approach systems.
3) Highly expressive for complex knowledge domains.

Disadvantages:

1) complex and resource intensive to build and maintain.
2) Ambiguity in domain interpretation
3) Scalability challenges as ontobogy grows.

Chapter-5

Machine Learning

What is machine learning?

HW Explain supervised and unsupervised learning vvI

081-10-03

* K-Nearest Neighbour (KNN).

Q) Consider the ~~above~~ following data points and find class label of last data point using KNN algorithm. (let k = 5).

sepal length	sepal width	species (class)
5.3	3.7	setosa
5.1	3.8	setosa
5.4	3.4	setosa
7.2	3.0	verginica
7.4	2.8	verginica
5.8	2.7	verginica
6.1	2.8	versicolor
6.3	2.3	versicolor
5.5	2.4	versicolor
5.2	3.1	?

Distance from 1st data

$$\sqrt{(5.3-5.2)^2 + (3.7-3.1)^2}$$

$$= 0.6[illegible]1$$

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Distance from 2nd data:

$$\sqrt{(5.1-5.2)^2+(3.8-3.1)^2} = 0.71$$

Distance from 3rd data.

$$\sqrt{(5.4-5.2)^2+(3.4-3.1)^2} = 0.36$$

Distance from 4th data

$$\sqrt{(7.2-5.2)^2+(3.0-3.1)^2} = 2.00$$

Distance from 5th data

$$\sqrt{(7.4-5.2)^2+(2.8-3.1)^2} = 2.22$$

Distance from 6th data.

$$\sqrt{(5.8-5.2)^2+(2.7-3.1)^2} = 0.72$$

Distance from 7th data

$$\sqrt{(6.1-5.2)^2+(2.8-3.1)^2} = 0.95$$

Distance from 8th data.

$$\sqrt{(6.3-5.2)^2+(2.3-3.1)^2} = 1.36$$

Distance from 9th data

$$\sqrt{(5.5-5.2)^2+(2.4-3.1)^2} = 0.76$$

The 5 nearest neighbours are:

setosa, setosa, setosa, verginica, versicolor.

∴ The majority is setosa, so the assigned class for last data point is setosa.

* Evaluating classifier :

↳ We need to be familiar with terminology that we come across while evaluating a classifier. We have +ve examples (example of main class of interest) and -ve examples (all other examples).

↳ Given two classes for example : the positive examples may be buy computer = yes while the negative examples are buy computer = no. suppose, we used our classifier on test set of labelled examples. P is the no. of +ve examples and N is the no. of -ve examples.

TP (True Positive)
↳ These refer to the +ve examples that were correctly labelled by the classifier.

TN (True Negative)
↳ These are the -ve example that were correctly labelled by the classifier.

FP (false Positive)
↳ These are the -ve examples that were incorrectly labelled as positive.

FN (false Negative)
↳ These are the +ve examples that were mislabelled as negative.

↳ These terms are summarized in confusion matrix as shown in figure below :-

CH

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AI

Actual class/ predicted class	C1	-C1
C1	TP	FN
-C1	FP	TN

not needed in exam

081-10-07

*Performance Measure

* Accuracy / Recognition rate:
  ↳ The accuracy of classifier on given test set is the percentage of test set examples that are correctly classified by the classifier.

$$\text{Accuracy} = \frac{TP + TN}{P + N} \quad (\text{in } \%)$$

* Error Rate / Misclassification rate:
  ↳ The error rate of classifier is the percentage of test set examples that are incorrectly classified by the classifier.

$$\text{Error Rate} = \frac{FP + FN}{P + N}$$

or,

$$1 - \text{Accuracy}.$$

* Sensitivity / True positive rate (recall):
↳ The sensitivity of classifier on a given test set is the portion of positive examples that are correctly classified.

$$\text{sensitivity} = \frac{TP}{P} \quad (P = TP + FP)$$

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* Specificity / True Negative rate

↳ The specificity of classifier on given test set is the portion of negative examples that are correctly identified

$$\text{Specificity} = \frac{TN}{N}$$

$$F_1\text{-score} = 2 \times \frac{\text{precision} * \text{recall}}{\text{precision} + \text{recall}}$$

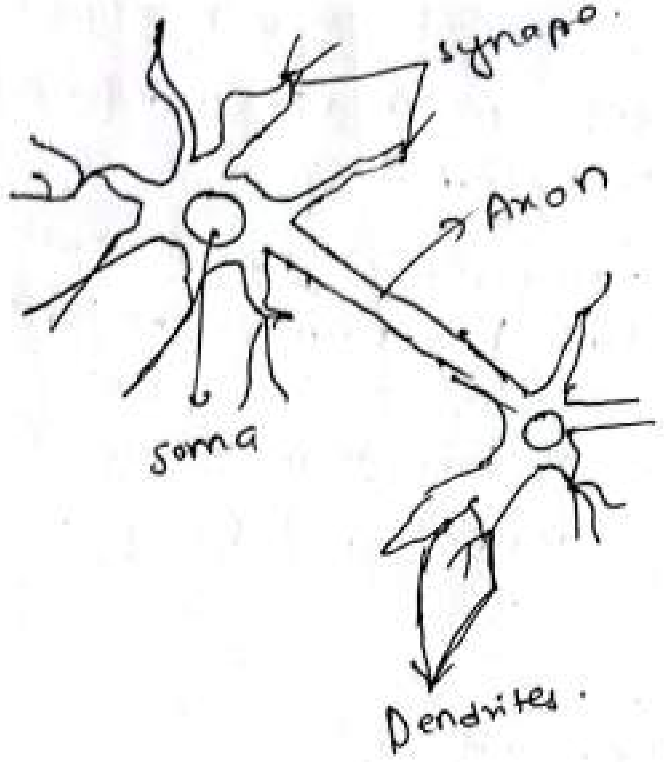
$$\text{precision} = \frac{TP}{TP + FP}$$

*Artificial Neural Network:

* Biological neural network:

↳ A neural network can be defined as model of reasoning based on the human brain. The brain consists of densely intsterconnected set of nerve cells or basic information processing units called neurons.

↳ The human brain incorporates nearly 10 billion neurons and 60 trillion connections, synapses between them. By using multiple neurons simultaneously, the brain can perform its function much faster than the fastest computers in existence today.



Each neuron has very simple structure, but an army of such elements creates a tremendous processing power.

Neuron:
fundamental f×nal unit of nervous system

Soma:
Cell body that contain nucleus.

Dendrites:
A number of fibres input

Axon:
single long fibre with many branches o/p

Synapse:
Junction of dendrites & axon. Each neuron form synapse with 10 to 100000 other neurons.

* Artificial Neural Network:

- A computing system made up of no. of simple highly interconnected processing elements, which process information by dynamic state response to external i/ps.
- Artificial neural network is composed of no. of nodes or units connected by links. Each link has numeric weight associated with it
- Weights are the primary means of long-term storage in neural network and learning usually takes place by updating weights.
- Each unit has:
  1) a set of i/p links from other units
  2) set of o/p links to other units.
  3) a current activation level
- Means of computing the activation level at the next time ~~state~~ step given its input weights.

Analogy btw Biological and Neural Network:

Soma → Neuron

Dendrite → i/p

Axon → o/p

synapse → weight.

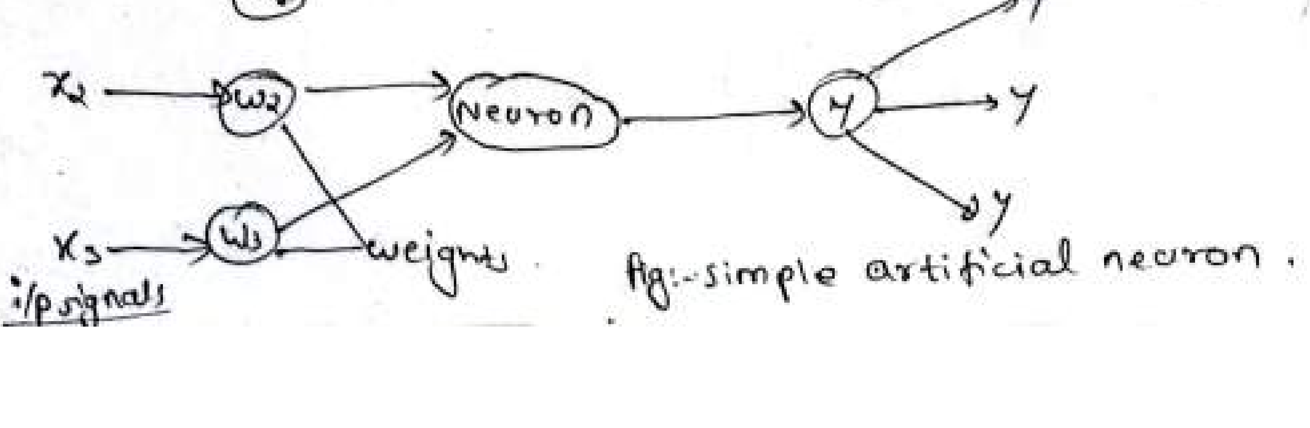


fig:- simple artificial neuron.

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AI

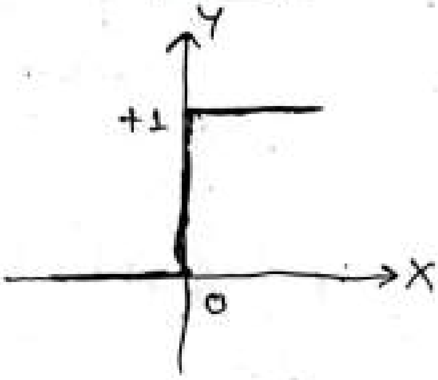
↳ The neuron computes the weighted sum of the i/p signals and compares the result with threshold value, θ (bias). If the net i/p is less than threshold, the neuron o/p is -1. But if the net i/p is greater than or equal to threshold, the neuron becomes activated and its o/p attains a value $+1$.

↳ The neuron uses the following transfer fxn (activation fxn)

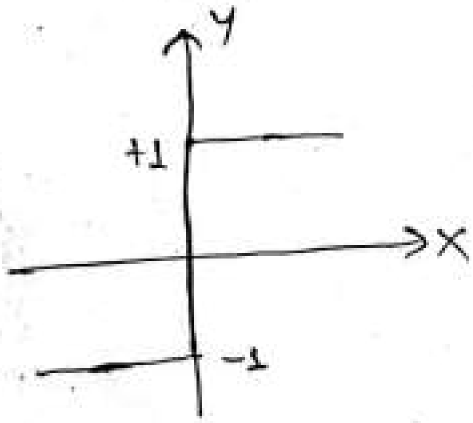
$$X = \sum_{i=1}^{n} x_i w_i \qquad Y = \begin{cases} +1, & \text{if } X \geq \theta \\ -1, & \text{if } X < \theta \end{cases}$$

The above used activation fxn is called sign fxn.

step fxn



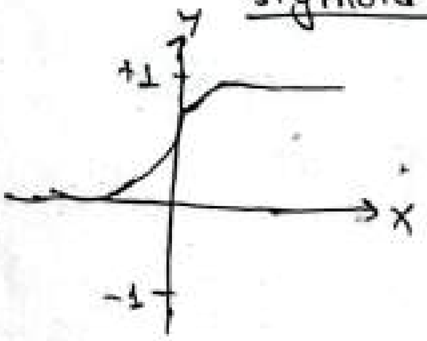
sign fxn



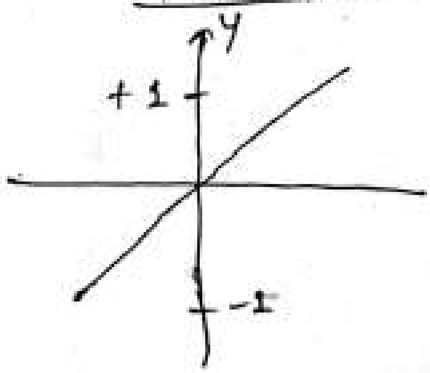
$$Y^{step} = \begin{cases} 1 & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$$

$$Y^{sign} = \begin{cases} +1 & \text{if } x \geq 0 \\ -1 & \text{if } x < 0 \end{cases}$$

sigmoid fxn



linear fxn.



$$Y^{sigmoid} = \frac{1}{1+e^{-x}}$$

$$Y^{linear} = x$$

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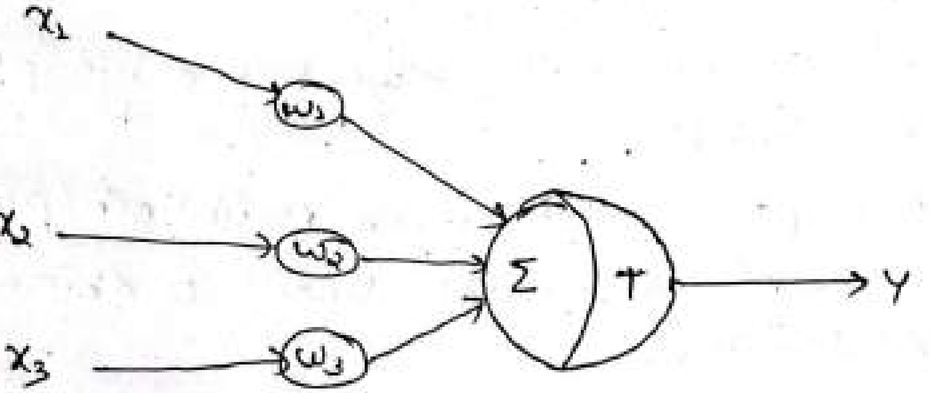
AI

× Mcuhlloch/pit Neuron :

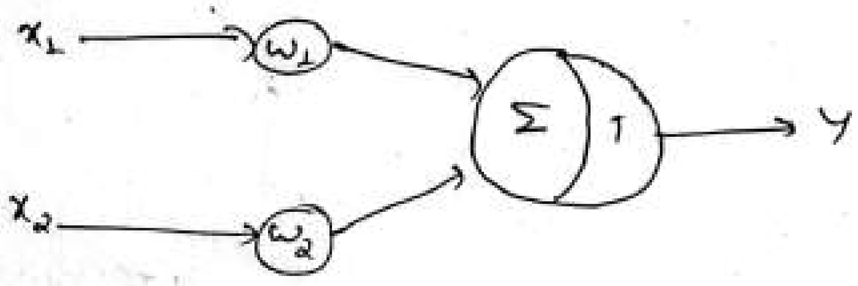
- One of the first neuron models to be implemented.
- Its o/p is 1 or 0.
- Each i/p is weighted with weights in the range -1 to +1.
- It has threshold value (bias), T



The neuron fires if the inequality is true.

$$x_1 w_1 + x_2 w_2 + x_3 w_3 > T$$

Q) Design or gate implementation using Mcuhlloch/pit neuron.



$w_1 = w_2 = 0.7$
$T = 0.4$ } suppose

if $x_1 = x_2 = 0$, $Y = 0$

if $x_1 = 0$ and $x_2 = 1$, $Y = 1$

if $x_1 = 1$ and $x_2 = 0$, $Y = 1$

if $x_1 = 1$ and $x_2 = 1$, $Y = 1$

* ~~Perceptr~~ Perceptron:

↳ In 1958, Frank Rosenblatt introduced a training algorithm that provided the first procedure for training a simple artificial neutral networ (ANN) called a perceptron.

↳ The perceptron is the simplest form of the Neural network.

↳ It consists of single neuron with adjustable synaptic weights and a hard limiter.

↳ The operation of perceptron is based on mcuhlloch/pit neuron model. The model consists of linear combiner followed by hard limiter.

↳ The weighted sum of i/ps is applied to the hard limiter which provides an o/p equal to +1 if its i/p is positive and -1 if its i/p is -ve.

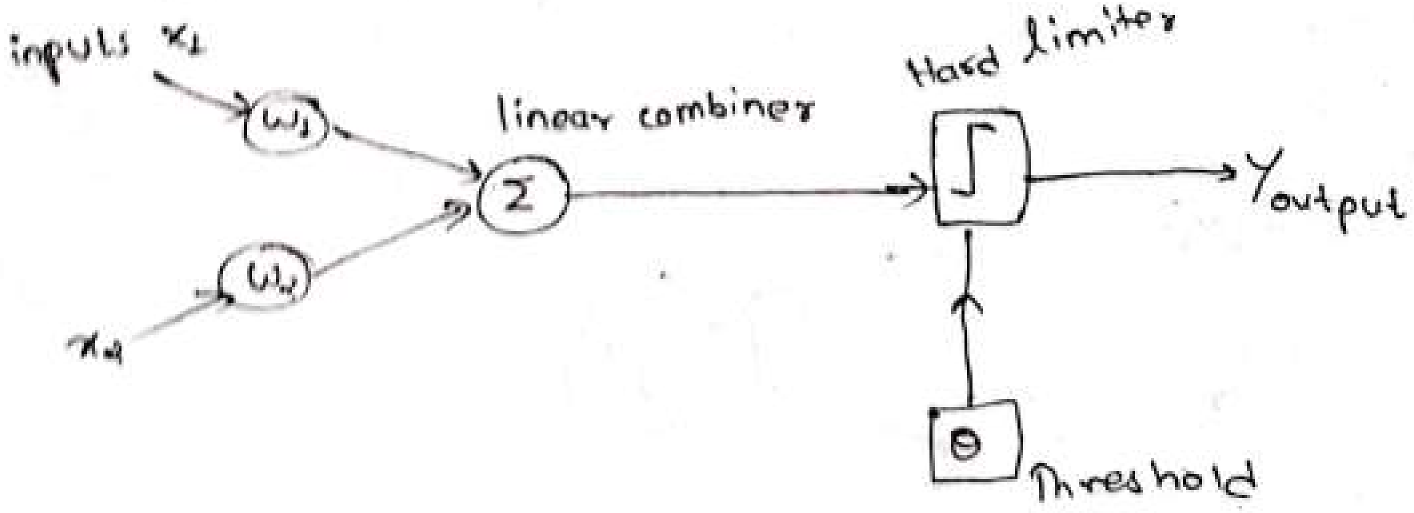


fig:- single layer two input perceptron.

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AI

perceptron learning rule:

$$w_i(p+1) = w_i(p) + \alpha * x_i(p) * e(p)$$

where, $p = 1, 2, 3$

α is learning rate, a positive constant less than unity

* Perceptron Training Algorithm:

Step-1: Initialization

↳ set initial weight $w_1, w_2, \dots w_n$ and threshold θ to random number in the range [-0.5 to 0.5]

Step-2: Activation

↳ Activate the perceptron by applying i/ps $x_1(p), x_2(p) \dots x_n(p)$ and desired o/p $Y_d(p)$

↳ Calculate the actual o/p $Y(p)$ at iteration $p=1$

$$Y(p) = \text{sign} \sum_{i=1}^{n} [x_i w_i - \theta]$$

activation fxn

where, n is the no. of perceptron i/ps and sign is the activation fxn.

↳ Calculate error

$$e(p) = Y_d(p) - Y(p)$$

Step-3: Weight update or training

↳ Update the weights of perceptron $\boxed{w_i(p+1) = w_i(p) + \Delta w_i(p)}$

where, $\Delta w_i(p)$ is weight correction at iteration p.

The weight correction is computed by formula;

$$\Delta w_i(p) = \alpha * x_i(p) * e(p)$$

Step-4 : Iteration

↳ Increase iteration p by 1, go back to step 2 and repeat the process until convergence (until weights are same continuously).

*** The learning rate (α).**

↳ The performance of neuron heavily depends on the value of the learning rate (α).

↳ If it is too large, the system will not converge.

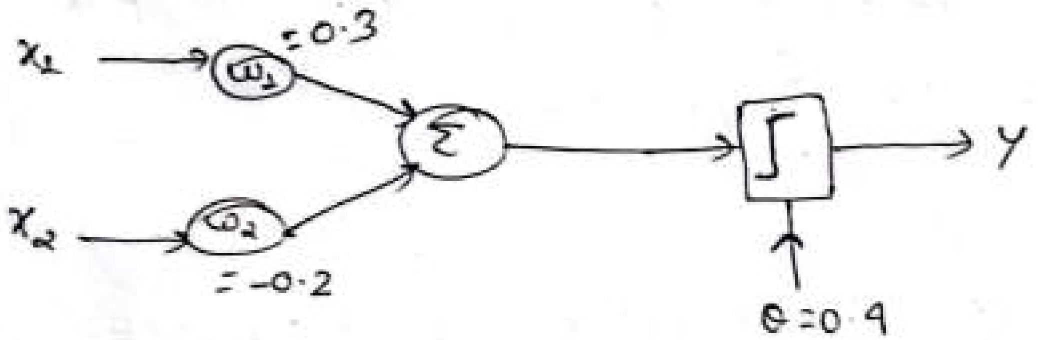
↳ If it is too small, the convergence will take long.

↳ α is selected based on hit and trial (trial and error) method. We usually keep value of $\alpha = 0.1$

Q. Realize ~~OR~~ AND gate using perceptron:

let us consider $\alpha = 0.2$.

(1)



x_1	x_2	y
0	0	0
0	1	0
1	0	0
1	1	1

Let us consider the activation fxn be step fxn

(2)

epoch 1

x_1	x_2	$Y_d(P)$	Y_{est}	Error
0	0	0	$step(0*0.3+0*(-0.2)-0.4)=0$	$0-0=0$
0	1	0	$step(0*(0.3)+1*(-0.2)-0.4)=0$	$0-0=0$
1	0	0	$step(1*0.3+0*(-0.2)-0.4)=0$	$0-0=0$
1	1	1	$step(1*0.3+(1*(-0.2)-0.4)=0$	$1-0=1$

table ...

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AI

w_1	w_2	status
0.3	-0.2	No change
0.3	-0.2	No change
0.3	-0.2	No change
0.5	0	Change.

At i/p (1,1)

$\Delta w_1 = \alpha * x_1 * e(t)$
$= 0.2 * 1 * 1$
$= 0.2$

$w_1 = w_1 + \Delta w_1$
$= 0.3 + 0.2 = 0.5$

$\Delta w_2 = \alpha * x_2 * e(t)$
$= 0.2 * 1 * 1$
$= 0.2$

$w_2 = w_2 + \Delta w_2$
$= -0.2 + 0.2 = 0.$

epoch 2

x_1	x_2	$Y_d(p)$	Yest	Error	w_1	w_2	Status.
0	0	0	step(0×0.5 + 0×0 − 0.4) = 0	0	0.5	0	No change
0	1	0	step(0×0.5 + 1×0 − 0.4) = 0	0	0.5	0	No change
1	0	0	step(1×0.5 + 0×0 − 0.4) = 1	-1	0.3	0	Change.
1	1	1	step(1×0.3 + 1×0 − 0.4) = 0	1	0.5	0.2	Change.

At i/p (1,0)

$\Delta w_1 = \alpha * x_1 * e(t)$
$= 0.2 * 1 * (-1)$
$= -0.2$

$w_1 = w_1 + \Delta w_1$
$= 0.5 - 0.2$
$= 0.3$

$\Delta w_2 = \alpha * x_2 * e(t)$
$= 0.2 * 0 * (-1)$
$= 0$

$w_2 = w_2 + \Delta w_2$
$= 0 + 0$
$= 0$

At i/p (1,1)

$\Delta w_1 = \alpha * x_1 * e(t)$
$= 0.2 * 1 * 1$
$= 0.2$

$w_1 = w_1 + \Delta w_1$
$= 0.3 + 0.2$
$= 0.5$

$\Delta w_2 = \alpha * x_2 * e(t)$
$= 0.2 * 1 * 1$
$= 0.2$

$w_2 = w_2 + \Delta w_2$
$= 0 + 0.2$
$= 0.2$

epoch 3

x_1	x_2	$Y_d(P)$	Yest	Error	w_1	w_2	status
0	0	0	step(0*0.5+0*0.2-0.9)=0	0	0.5	0.2	No change
0	1	0	step(0*0.5+1*0.2-0.9)=0	0	0.5	0.2	No change
1	0	0	step(1*0.5+0*0.2-0.9)=1	0-1=-1	0.3	0.2	change
1	1	1	step(1*0.3+1*0.2-0.9)=1	0	0.3	0.2	No change

At i/p (1,0)

$\Delta w_1 = \alpha * x_1 * e(t)$
$= 0.2 * 1 * (-1)$
$= -0.2$

$w_1 = w_1 + \Delta w_1$
$= 0.5 - 0.2$
$= 0.3$

$\Delta w_2 = \alpha * x_2 * e(t)$
$= 0.2 * 0 * (-1)$
$= 0$

$w_2 = w_2 + \Delta w_2$
$= 0.2 + 0$
$= 0.2$

epoch 4

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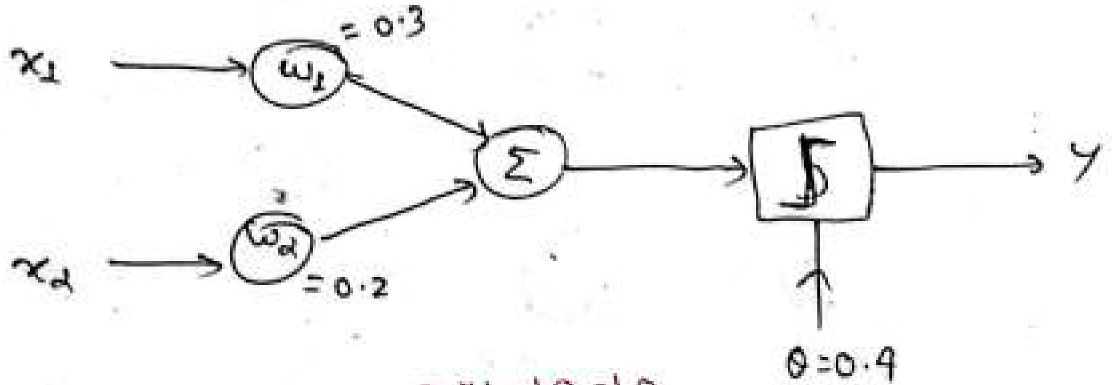
AI

x_1	x_2	$Y_d(p)$	Y_{est}	Error	w_1	w_2	status
0	0	0	step $(0\times0.3+0\times0.2-0.4)=0$	0	0.3	0.2	No change.
0	1	0	step $(0\times0.3+1\times0.2-0.4)=0$	0	0.3	0.2	No change
1	0	0	step $(1\times0.3+0\times0.2-0.4)=0$	0	0.3	0.2	No change
1	1	1	step $(1\times0.3+1\times0.2-0.4)=1$	0	0.3	0.2	No change



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* Multilayer Perceptron:

- Multilayer perceptron is a feed forward neural network with one or more hidden layers.
- The network consists of an i/p layer of source neurons, at least one middle or hidden layer of computational neurons and an o/p layer of computational neurons.
- The i/p signals are propagated in a forward direction on layer by layer basis.
- The hidden layer hides its desired o/p. Neuron in the hidden layer cannot be observed through the i/p o/p behavior of the network.

↳ Perceptron can be improved if placed in multilayer network.

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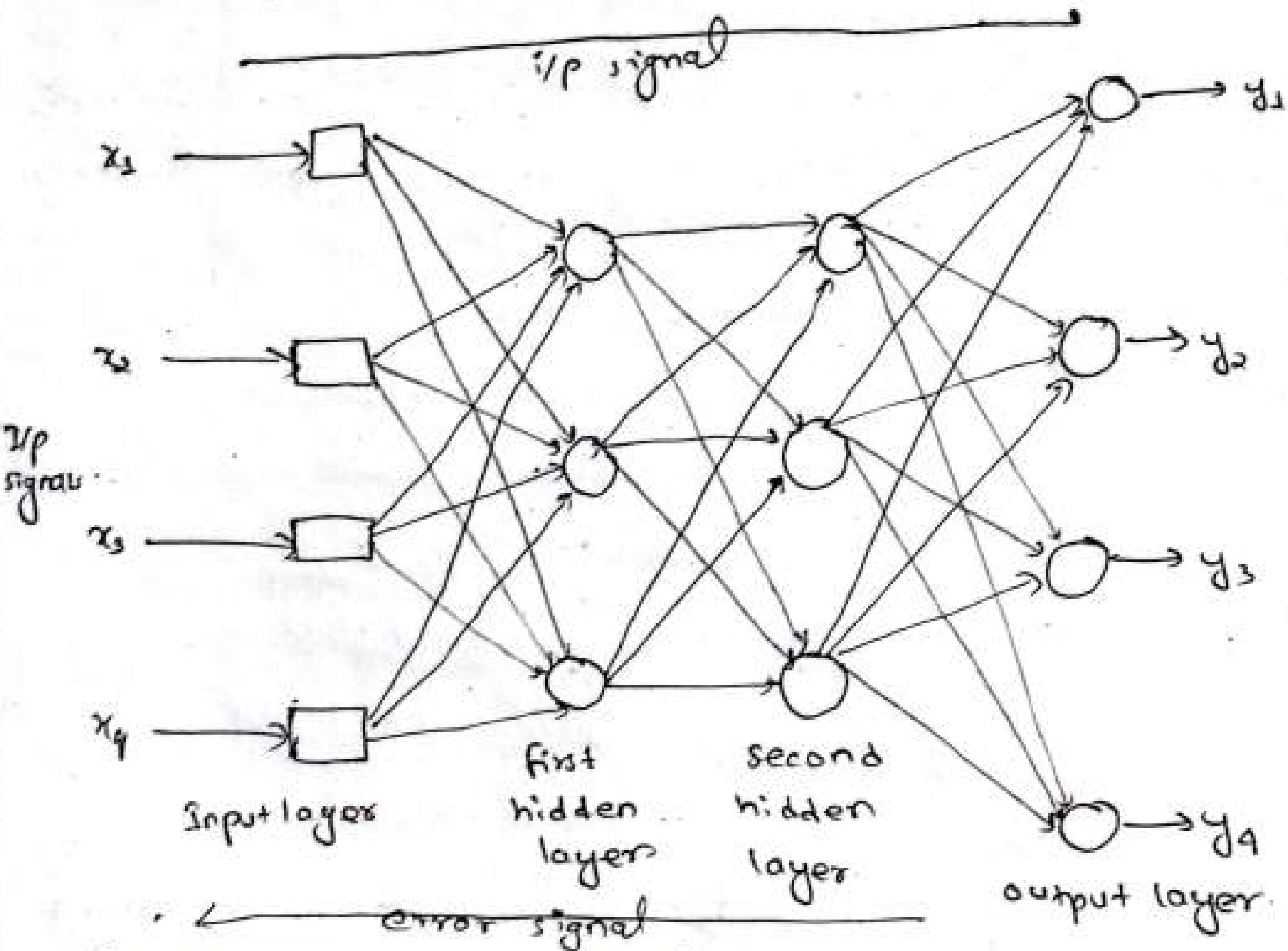
A2

Multilayer Neural Network:



× Backpropagation Neural Network:

↳ Learning in multilayer network proceeds in the same way as per perceptron.

↳ Training set of i/p patterns is presented to the network.

↳ The network computes its o/p pattern and if there is error or in other word difference betn actual o/p and desired o/p the weights are adjusted to reduce the error.

↳ In back propagation neural network, the algorithm has two phase:

1) A training i/p pattern is presented to the i/p layer. The network propagates the i/p from layer to layer until the o/p ~~layer~~ pattern is generated by the o/p layer.

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2) If the pattern is different from the desired o/p, an error is calculated and then propagated backword through the network from o/p layer to i/p layer. The weights are modified as the error is propagated.

* Back propagation training algorithm:

Step 1: Initialization:

Set all the weights and threshold levels of the network to random numbers uniformly distributed inside the small range: $\left(-\frac{2.4}{F_i}, +\frac{2.4}{F_i}\right)$

where,

F_i is total no. of i/ps of neuron i in the network.

The weight initialization is done on neuron by neuron basis.

Step-2: Activation:

Activate the back propagation neural network by applying i/p $x_1(p), x_2(p), \ldots x_n(p)$ and desired o/p $y_{d,1}(p), y_{d,2}(p), \ldots y_{d,n}(p)$.

a) Calculate the actual o/p of the neuron in the hidden layers.

$$y_j(p) = \text{sigmoid}\left[\sum_{i=1}^{n} x_i(p) \cdot w_{ij}(p) - \theta_j\right]$$

n is the no. of i/ps of neuron j in the hidden layer

b) Calculate the actual o/p of the neuron in the o/p layer

$$y_k(p) = \text{sigmoid}\left[\sum_{j=1}^{m} x_{jk}(p) \cdot w_{jk}(p) - \theta_k\right]$$

m is the no. of i/p of neuron k in the o/p layer.

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AI

Step-3: Weight Training :

Update the weights in the back propagation network, propagating the error (backward) associated with the o/p neurons.

a) Calculate the error gradient for the neuron in the o/p layer.

$$\delta_k(P) = Y_k(P) \cdot [1 - Y_k(P)] \cdot e_k(P)$$

where,

$$e_k(P) = Y_{d,k}(P) - Y_k(P)$$

calculate the weight correction :-

$$\Delta w_{jk}(P) = \alpha \cdot Y_j(P) \cdot \delta_k(P)$$

Update the weight at o/p neuron.

$$w_{jk}(P+1) = w_{jk}(P) + \Delta w_{jk}(P)$$

b) Calculate the error gradient for the neuron in the hidden layer

$$\delta_j(P) = Y_j(P) \cdot [1 - Y_j(P)] \cdot \sum_{K=1}^{l} \delta_k(P) \cdot w_{jk}(P)$$

calculate the weight correction.

$$\Delta w_{ij}(P) = \alpha \times x_i(P) \cdot \delta_j(P)$$

Update the weights at the hidden layer.

$$w_{ij}(P+1) = w_{ij}(P) + \Delta w_{ij}(P)$$

Step-7: Iteration:

Increase iteration p by 1. Go back to step 2 and repeat the process until convergence.

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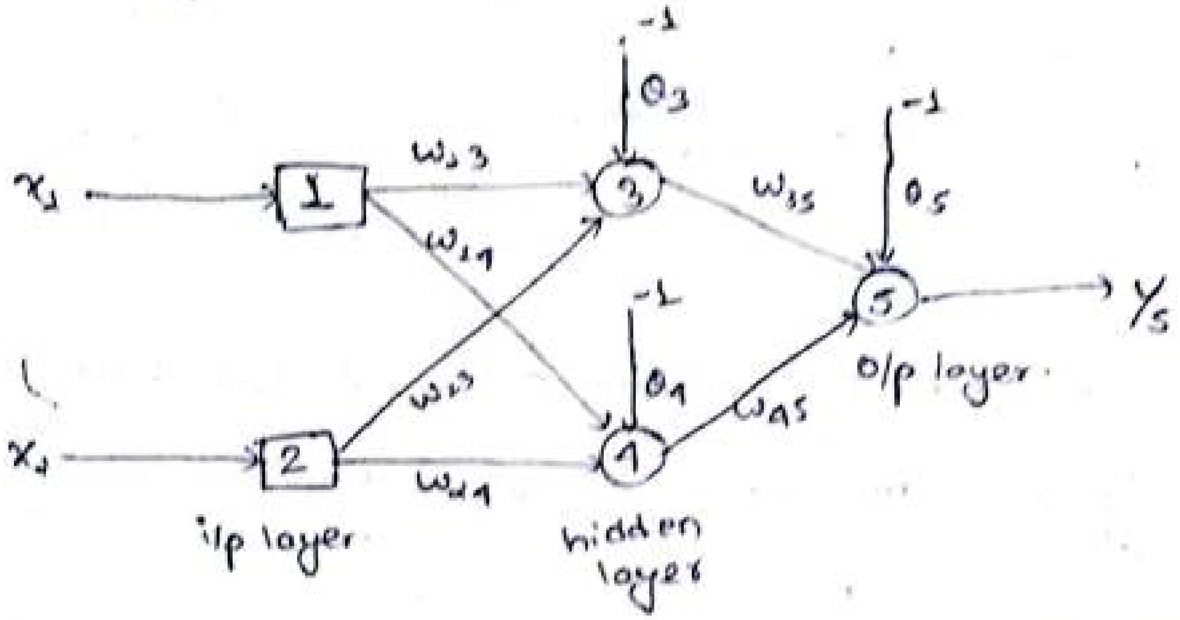
A7

Eg:
~~Design~~ Realize XOR operation using back propagation algorithm.



$w_{13} = 0.5$, $w_{14} = 0.9$, $w_{23} = 0.4$, $w_{24} = 1.0$, $w_{35} = -1.2$

$w_{45} = 1.1$, $\theta_3 = 0.8$, $\theta_4 = -0.1$, $\theta_5 = 0.3$

Activation function : sigmoid

↳ soln: Step-1:

We consider the training set where i/p x_1 and x_2 are equal to 1 and desired o/p $Y_{d,5} = 0$.

Step-2:

The actual o/p of neuron 3 and neuron 4 at hidden layer are calculated as.

$$Y_3 = \text{sigmoid}\left[x_1 w_{13} + x_2 w_{23} - \theta_3\right]$$

$$= \frac{1}{1+e^{-[(1*0.5+1*0.4)-0.8]}}$$

$$= 0.52450$$

$$Y_4 = \text{sigmoid}\,(x_2 \cdot w_{24} + x_1 w_{14} - \theta_4)$$

$$= \frac{1}{1+e^{-[(1\times 1.0+1*0.9)-(-0.1)]}}$$

$$= 0.8808$$

Now, the actual o/p at the neuron 5 is calculated as.

$$Y_5 = \text{sigmoid}\,(y_3 * w_{35} + Y_4 * w_{45} - \theta_5)$$

$$= \frac{1}{1+e^{-\{(0.5250*(-1.2)+0.8808*1.1)-0.3\}}}$$

$$= 0.5097$$

$$e = Y_{d,5} - Y_5$$

$$= 0 - 0.5097$$

$$= -0.5097$$

The next step is to update weight at neuron 5.

$$\delta_5 = Y_5(1-Y_5)\,e$$

$$= 0.5097\,(1-0.5097) * (-0.5097)$$

$$= -0.1274$$

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A1

Assume $\alpha = 0.1$

$$\Delta w_{35} = \alpha \cdot y_3 * \delta_5$$
$$= 0.1 * 0.5250 * (-0.1277)$$
$$= -0.0067$$

$$\Delta w_{45} = \alpha \cdot Y_4 * \delta_5$$
$$= 0.1 * 0.8808 * (-0.1277)$$
$$= -0.0112$$

$$\Delta \theta_5 = \alpha \cdot (-1) \cdot \delta_5$$
$$= 0.1 * (-1) * (-0.1277)$$
$$= -0.0127$$

At hidden layer;

$$\delta_3 = Y_3(1 - Y_3) * \delta_5 * w_{35}$$
$$= 0.5250 * (1 - 0.5250) * (-0.1277) * (-1.42)$$
$$= 0.0381$$

$$\delta_4 = Y_4 (1 - Y_4) \cdot \delta_5 \cdot w_{45}$$
$$= 0.8808 (1 - 0.8808) * (-0.1277) * (1.1)$$
$$= -0.0147$$

$$\Delta w_{13} = \alpha * x_1 * \delta_3$$
$$= 0.1 * 1 * 0.0381$$
$$= 0.0038$$

$$\Delta w_{23} = \alpha * x_2 * \delta_3$$
$$= 0.1 * 1 * 0.0381$$
$$= 0.0038$$

$\Delta\theta_3 = \alpha * (-1) * \delta_3 = 0.1 * (-1) * 0.0381 = -0.0038$

$\Delta w_{14} = \alpha * x_1 * \delta_4 = 0.1 * 1 * (-0.0147) = -0.0015$

$\Delta w_{24} = \alpha * x_2 * \delta_4 = 0.1 * 1 * (-0.0147) = -0.0015$

$\Delta\theta_4 = \alpha * (-1) * \delta_4 = 0.1 * (-1) * (-0.0147) = 0.0015$

Next, we update weights and threshold.

$w_{13} = w_{13} + \Delta w_{13} = 0.50381$

$w_{14} = w_{14} + \Delta w_{14} = 0.8985$

$w_{23} = w_{23} + \Delta w_{23} = 0.40381$

$w_{24} = w_{24} + \Delta w_{24} = 0.9985$

$w_{35} = w_{35} + \Delta w_{35} = -1.2067$

$w_{45} = w_{45} + \Delta w_{45} = 1.0888$

$\theta_3 = \theta_3 + \Delta\theta_3 = 0.7962$

$\theta_4 = \theta_4 + \Delta\theta_4 = -0.098$

$\theta_5 = \theta_5 + \Delta\theta_5 = 0.287$

weights and threshold are updated and the process is continued until convergence.

Chapter-7

MYCIN → 1st expert system.

Expert system:

↳ Expert systems are computer programs built for commercial applications using the programming techniques of AI which are developed for problem solving.

↳ Built for variety of purposes including medical diagnosis, electronic fault finding, mineral prospecting, computer system configuration, etc.

↳ The process of building expert system is called knowledge engineering and is done by knowledge engineers.

* Basic Terminologies:

* Knowledge Engineer:

↳ is a human with background of computer science and AI who knows how to buid an expert system.

↳ Decide how to represent knowledge and helps programmer how to write code

↳ Creates a relation with human expert to elicit knowledge.

* Knowledge Engineering:

↳ It is the process of gathering of knowledge from human experts or other sources

* Expert

↳ Evaluates the expert systems and gives reports to knowledge engineer

Features of expert system:

1) facility for non-expert person to solve problems that require some expertise.
2) Speedy solution
3) Reliable solution
4) Cost reduction.
5) Power to manage without human experts.
6) Wider areas of knowledge.

Use of expert system is specially recommended when:

- human experts are difficult to find.
- human experts are expensive.
- problems are incompletely defined.
- there is lack of knowledge among all those who need it.
- the problem is rapidly changing.

* Architecture of expert system: VVI

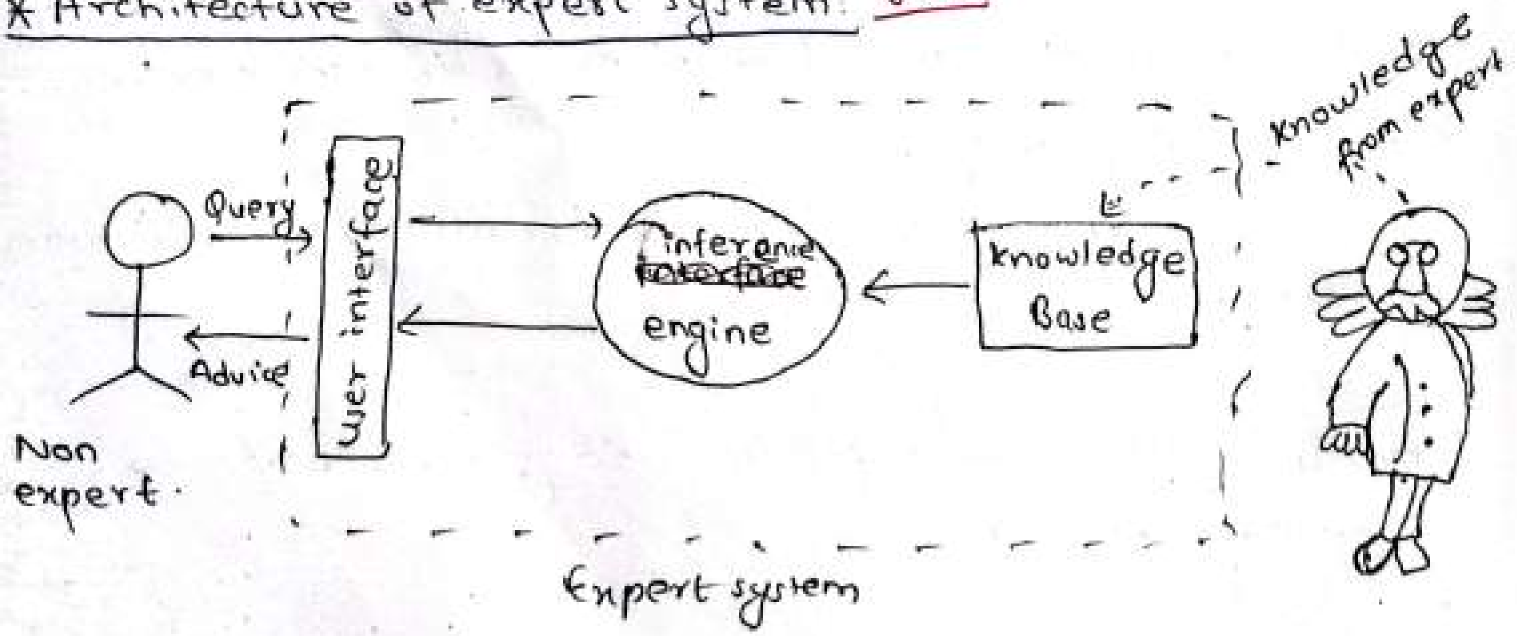

fig:- Architecture of Expert system

* Knowledge Base:

↳ It contains domain specific and high quality knowledge. Knowledge is required to exhibit intelligence. The success of expert system majorly depends upon the collection of highly accurate and precise knowledge.

What is knowledge?

↳ The data is collection of facts. The information is organized as data and facts about the task domain.
Data, information and past experience combined is called knowledge.

Components of knowledge base

↳ The knowledge base ~~as~~ stores both the factual & heuristic knowledge.

factual knowledge

↳ It is the information widely accepted by the knowledge engineers and scholars in the task domain.

Heuristic knowledge

↳ It is about practice, accurate judgement, one's ability to evaluate and guess.

* Inference Engine

↳ Use of efficient procedures and rules by the inference engine is essential in deducting a correct solution.

1) In case of knowledge-based expert system

↳ The inference engine acquires and manipulates the knowledge from the knowledge base to arrive at a particular solution.

2) In case of rule-based expert system.

↳ Applies rules repeatedly to the facts.

↳ Adds new knowledge into the knowledge base if required.

↳ Resolves rule conflict when multiple rules are applicable to a particular case.

* User Interface:

↳ It provides interactions betn user of the expert system and the expert system itself. It generally uses NLP. The user of expert system need not to be necessarily an expert in artificial intelligence.

↳ It explains how the expert system has arrived at a particular solution. The explanation may appear in following form.

- Natural language displayed on screen.
- verbal narration in natural language.

Advantages of expert system:

1) Increased availability.

↳ Expert system is mass production of expertise which is available in any suitable computers.

2) Reduced cost:

↳ Expert system cost much cheaper than real human expert.

3) Reduced danger:

↳ Can be used in environment that might be dangerous to human.

4) Performance: 10-13/14

↳ Unlike human experts which might retire, quit, and die, expert system's knowledge last indefinitely.

5) Multiple expertise:

↳ Knowledge of many expert system could be made work simultaneously and continuously on a problem

6) Increased reliability:

↳ Expert system could provide second option or break a tie in case of disagreement among human expert.

7) Steady, unemotional, complete response at all time

↳ Important in real time and emergency situations when human experts cannot operate on its peak due to stress or some sentimental influences.

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* Inference Mechanisms:

↳ refers to the processes of strategies used by an expert system to derive new facts or conclusions from the existing knowledge base.

1) Forward Chaining:

↳ It is a data-driven ~~system~~ inference mechanism.

↳ It starts with available facts and applies rules to derive new facts or conclusions.

Process:

i) start with known facts in the knowledge base

ii) search for rules whose cond's (antecedents) are satisfied by facts.

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iii) Apply the rule to derive new facts (consequences).

iv) Repeat the process until no more facts can be derived or the goal is reached.

Eg:-

IF it is raining THEN the ground will be wet.

If we know that "it is raining", we can infer that "the ground is wet".

~~[illegible]~~

2) Backward chaining:

↳ It is a goal-driven inference mechanism.

↳ starts with a goal or hypothesis and works backward to determine if the facts support it.

Process:

i) Start with a goal or hypothesis that needs to be proven.

ii) Search for rules that have this goal as their conclusion.

iii) Check if the conditions (antecedents) of those rules are satisfied by existing facts.

iv) If not, recursively attempt to prove the $cond^n s$.

Eg:

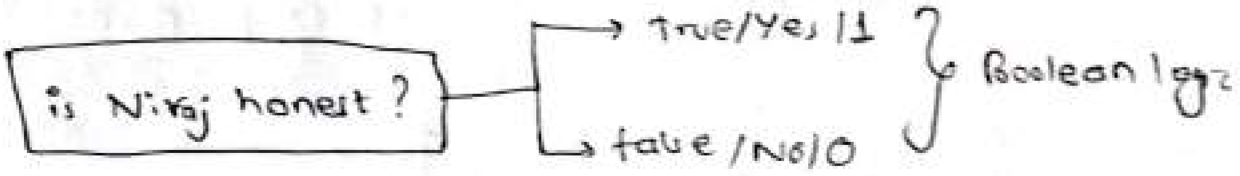
Goal: Is the ground wet?

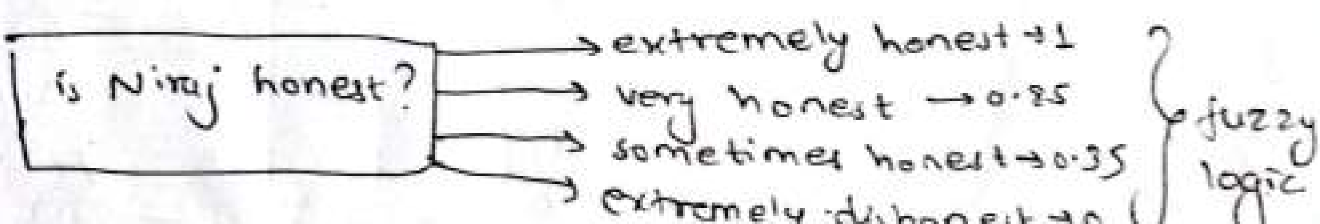
Find a rule that concludes "ground is wet", such as "IF it is raining THEN the ground will be wet". check if the $cond^n$ (it is raining) is true.

Chapter - 6.

Fuzzy Logic

- The words fuzzy refers to things which are not clear or vague. Any event process or function that is changing continuously cannot be defeined as either true or false.
- In real world many times, we encounter a situation when we cannot determine whether the state is true or false.
- In boolean system truth value, 1.0 represent true value and 0 represent false value. But in fuzzy system, there is no logic for truth and absolute false value. But in fuzzy logic, there is intermediate value too present which is partially false.





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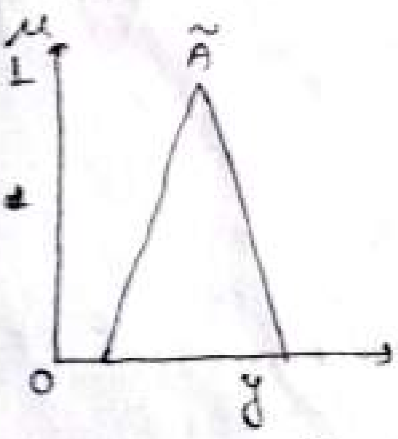
A?

* Key difference of fuzzy logic and classical crisp logic:

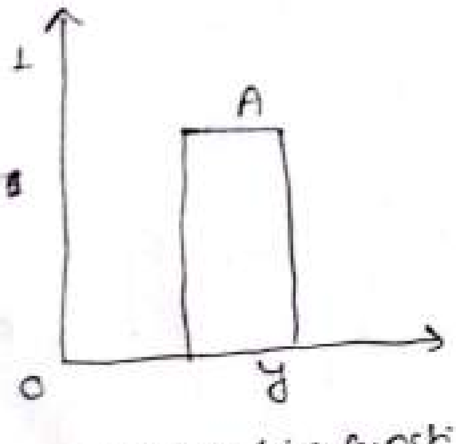
	crisp logic	fuzzy logic
values	0 or 1	continuous 0 to 1.
Basis	precise, rigid rules.	approximation
Example	is it hot?	How hot it is?

* fuzzy logic set theory:

↳ fuzzy sets can be considered as an extension and gross oversimplication of classical sets. It can be understood in the context of membership. Basically, it allows partial membership which means that it contain elements that have varying degree of membership in set.



membership function of fuzzy ~~logic~~ set Ã.



membership function of classical set A

Eg:- Fuzzy set: "warm tempr"

$\mu(20) = 0.2$ (partially warm)
$\mu(25) = 0.8$ (warmer)
$\mu(30°C) = 1.0$ (fully warm).

A fuzzy set $\tilde{A}$ ~~exists~~ in the universe of information U 15/10-14
can be defined as set of ordered pairs and it can
be represented mathematically as,

$$\tilde{A} = \{(y, \mu\tilde{A}(y)) \mid y \in U\}$$

Here,

$\mu\tilde{A}(y)$ = degree of membership of y.

$\tilde{A}$ assumes value (0 to 1).

Membership fxn:

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↳ Membership fxn define the degree of membership for each elements. Common types includes :-

Triangular Membership fxn → Defined by 3 parameters (a, b, c)

$$\mu(x) = \begin{cases} 0 & x < a \text{ or } x > c \\ \dfrac{x-a}{b-a} & a \leq x \leq b \\ \dfrac{c-x}{c-b} & b \leq x \leq c \end{cases}$$

Example:-

For the fuzzy set, warm tempr with parameters $(a, b, c) = (\underset{a}{15}, \underset{b}{25}, \underset{c}{35})$

find.

$\mu(20°C) = ?$, $\mu(30) = ?$

$$\mu(30) = \frac{35-30}{35-25} = \frac{1}{2} = 0.5$$

$$\mu(20) = \frac{20-15}{25-15} = \frac{1}{2} = 0.5$$

Gaussian Member fun:

A smooth bell shaped curve

$$\mu(x) = e^{\frac{-(x-c)^2}{2\sigma^2}}$$

where, c is centroid and σ is the spread.

Fuzzy set Operations:

① union (OR):

$$\mu_{A\cup B}(x) = \max(\mu_A(x), \mu_B(x))$$

Eg:

$A = \{0.4, 0.6, 0.8\}$, $B = \{0.5, 0.3, 0.9\}$

$\therefore \mu_{A\cup B} = \{0.5, 0.6, 0.9\}$

② AND (Intersection):

$$\mu_{A\cap B}(x) = \min(\mu_A(x), \mu_B(x))$$

Eg:

$A = \{0.4, 0.6, 0.8\}$, $B = \{0.5, 0.3, 0.9\}$

$\therefore \mu_{A\cap B} = \{0.4, 0.3, 0.8\}$

③ Complement (NOT)

$$\mu_{A'}(x) = 1 - \mu_A(x)$$

Eg:

$A = \{0.4, 0.6, 0.8\}$

$\therefore \mu_{A'} = \{0.6, 0.4, 0.2\}$

Trapezoidal membership fun.

$$mu(x) = \begin{cases} \frac{b-x}{b-a} & \text{if } a<x<b \\ 1 & \text{if } b<x<c \\ \frac{d-x}{d-c} & \text{if } c<x<d \\ 0 & \text{otherwise} \end{cases}$$

*Operation of fuzzy System: VV.I

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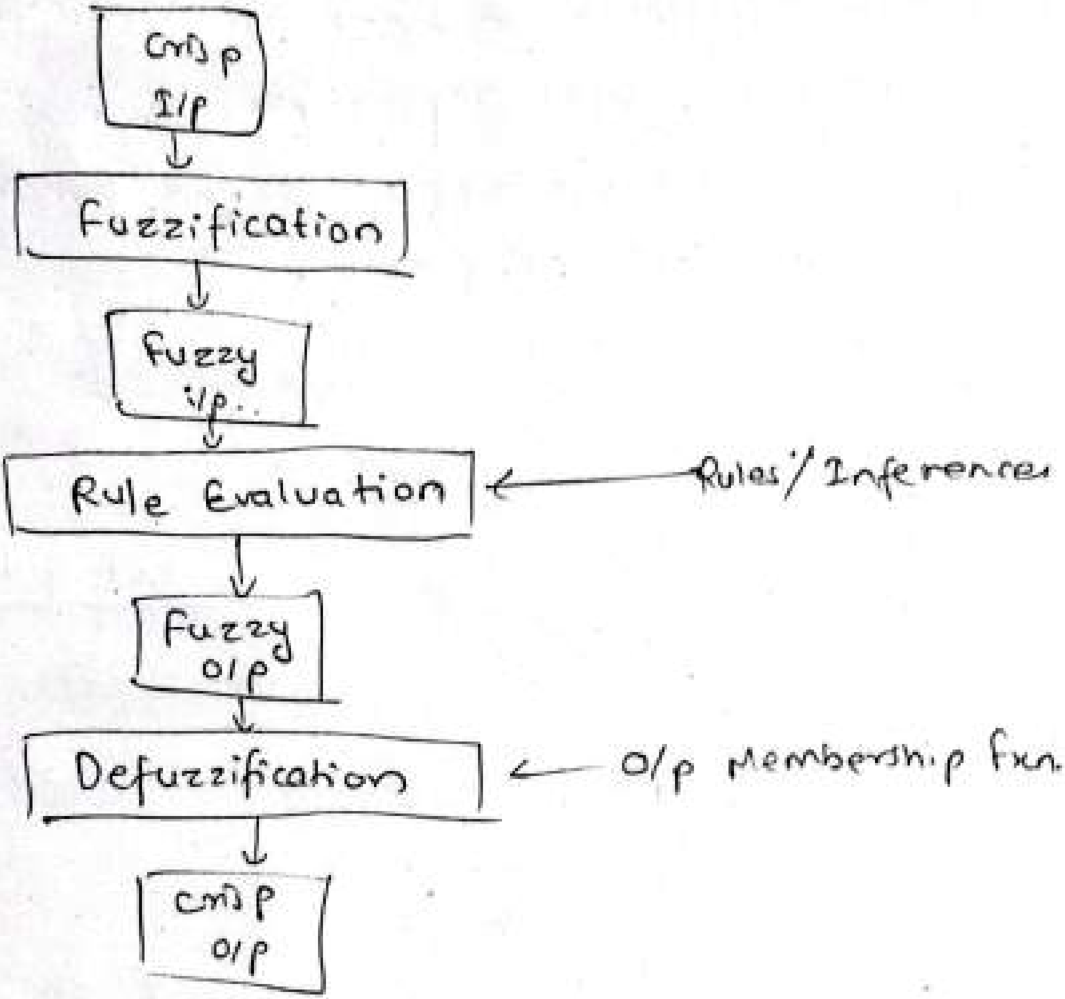

*Fuzzification:

- is the process of translating a precise value (like length of holiday) into a fuzzy concept like long.
- fuzzy concept such as long is represented using a membership fxn that shows how strongly something belongs to that category.
- Membership value ranges from 0 to 1.
- <u>for eg</u>: A hodiday of two week is not long (membership =0), while a holiday of four week is fully long (membership =1). Between two and four weeks, the membership value increases gradually. Eg:- 3 weeks holiday is long with membership value is 0.5.

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↳ This approach enables fuzzy logic to handle concepts with gradual transitions like long or hot. For instance, hotness might be divided into categories like warm, hot, very hot, each represented by its own membership fxn. These overlapping ranges make fuzzy reasoning, flexible, and adaptable.

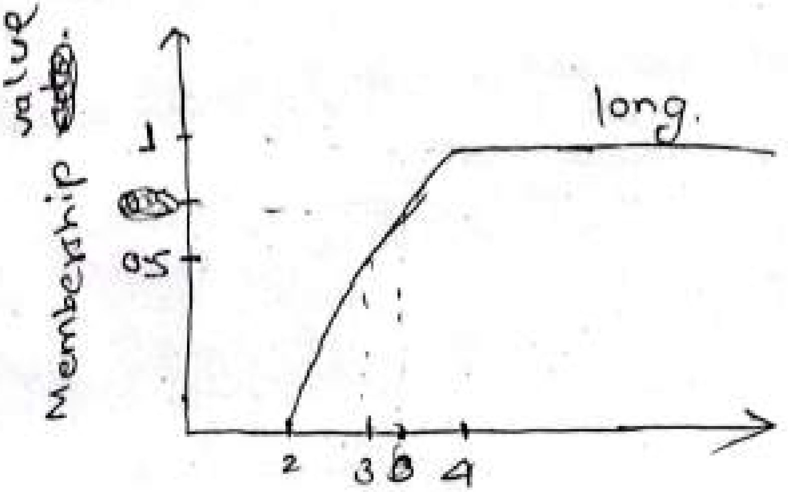

Fig:- Fuzzy concept 'long related to length of holidays in weeks'

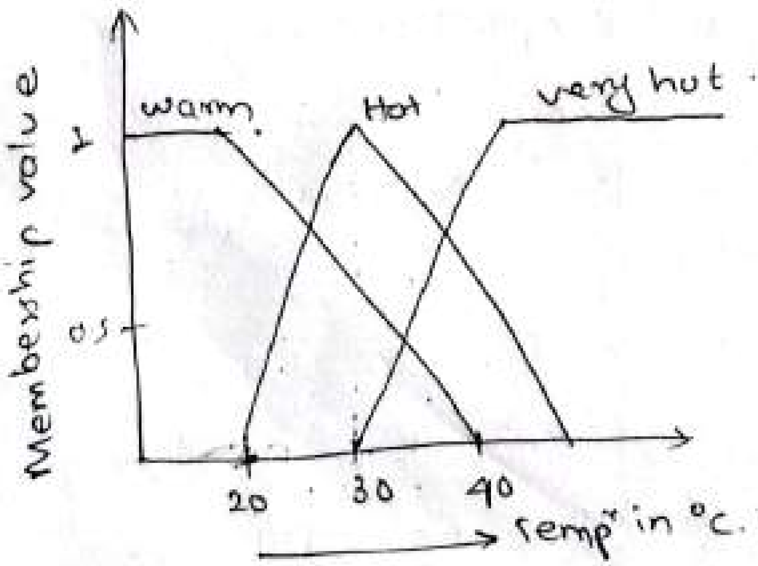

Fig:- Membership fxns for warm, hot, and very hot

* Rule Evaluation / fuzzy Inference:

↳ fuzzy inference applies fuzzy logic to make decisions based on rules. It follows the following general steps

a) fuzzification : convert real-world values into membership values for fuzzy condⁿ.

b) Evaluate rules:

- for a rule (eg: if location is expensive and holiday is long, then spending money is high), determine the minimum membership values of the condⁿ
- Eg:- If location is expensive has membership value 0.9 and holiday is long has membership value 0.7, the outcome spending money is high gets the minimum value, so the membership value for spending money is high = 0.7.

c) Apply rule weight:

- Each rule has a weight (default = 1). Multiply the rule weight by membership value of the outcome.
- Eg:- 0.7 * 1 = 0.7

~~d) Defuzzification~~ d) Combine results:

~~* Defuzzification:~~

- Multiple rules may point to the same value (eg: spending money is high). Take the max. value of membership from all relevant rules.
- Eg:- if there is another rule (if holiday is exotic then spending money is high), may assign membership value of

0.9. Combining these values, membership for spending money is high becomes 0.9 (max). 10-25 AI

* Defuzzification:

↳ is the process of turning fuzzy results into single clear number that can be used practically.

↳ Imagine we use a fuzzy rule system to decide how much money should someone take on holiday. The system doesnot give us one exact number but instead says:

- spending mongey is low with confidence of 0.3
- spending money is medium with confidence of 0.5
- spending money is high with confidence of 0.7

→ These ~~conditions~~ confidence ~~label~~ levels (0.3, 0.5, 0.7) are called membership values. They tell us how strongly is fuzzy categrory applies.

→ Each category (low, medium, and high) is tied to specific spending amount

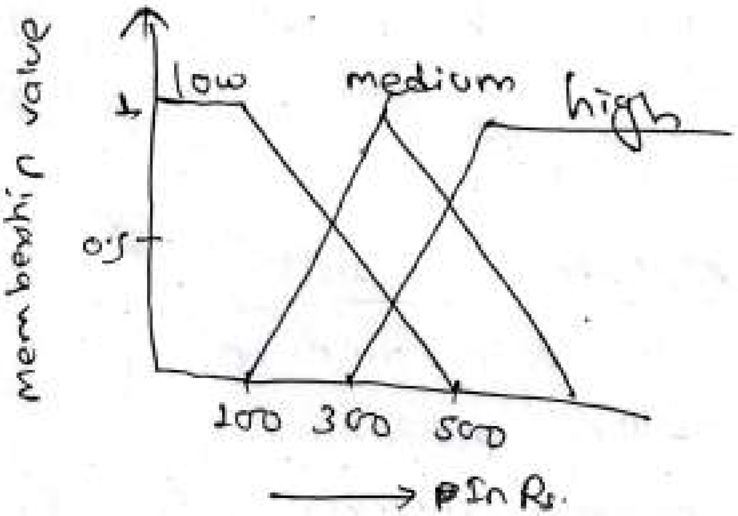

fig:- membership fn for low, median, and high-

↳ The expression for defuzzified value is :-

$$\frac{(HV\_low \times MV\_low + HV\_med \times MV\_med + HV\_high \times MV\_high)}{(MV\_low + MV\_med + MV\_high)}$$

for above eg:

$$= \frac{100 \times 0.3 + 300 \times 0.5 + 500 \times 0.7}{0.3 + 0.5 + 0.7}$$

$$= 353.33$$